$$\begin{pmatrix} N_B \\ T \\ m_D \\ x \end{pmatrix} = \text{Find}(N_B, T, m_D, x) \qquad \begin{pmatrix} N_B \\ T \end{pmatrix} = \begin{pmatrix} 413.05 \\ 129.08 \end{pmatrix} \text{N} \qquad m_D = 25.6 \text{ kg}$$

$$x = 0.052 \text{ m}$$

Since $x = 51.6\,\text{mm} < \frac{b}{2} = 125\,\text{mm}$ our assumption is correct $\qquad m_D = 25.6$ kg

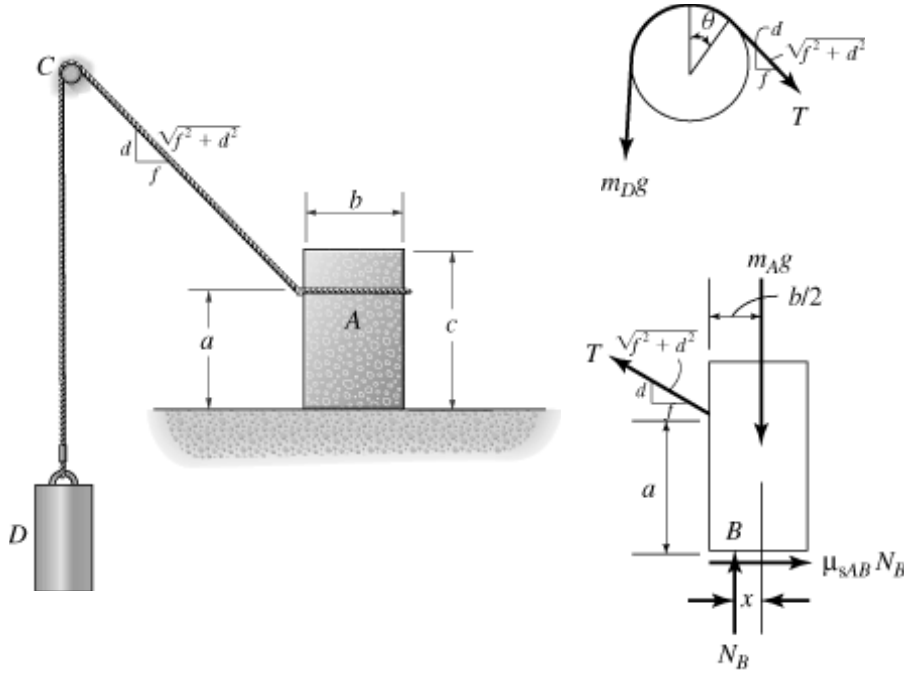
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Problem 8-106

Block A rests on the surface for which the coefficient of friction is μ_{sAB}. If the mass of the suspended cylinder is m_D, determine the smallest mass m_A of block A so that it does not slip or tip. The coefficient of static friction between the cord and the fixed peg at C is μ_{sC}.

Units Used:

$g = 9.81 \frac{\text{m}}{\text{s}^2}$



Given:

$\mu_{sAB} = 0.25$

$m_D = 4$ kg

$\mu_{sC} = 0.3$

$a = 0.3$ m

$b = 0.25$ m

$c = 0.4$ m

$d = 3$

$f = 4$

Solution: Assume that slipping is the crtitical motion $\qquad \beta = \pi - \text{atan}\left(\frac{f}{d}\right)$

The initial guesses: $N_B = 100$ N $\qquad T = 50$ N $\qquad m_A = 1$ kg $\qquad x = 10$ mm

Given $\qquad m_D g = T e^{\mu_{sC}\beta} \qquad \left(\frac{d}{\sqrt{f^2+d^2}}\right) T - m_A g + N_B = 0$

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$$\left(\frac{-f}{\sqrt{f^2+d^2}}\right)T+\mu_{sAB}N_B=0 \qquad \left(\frac{f}{\sqrt{f^2+d^2}}\right)Ta-\left(\frac{d}{\sqrt{f^2+d^2}}\right)T\frac{b}{2}-N_Bx=0$$

$$\begin{pmatrix}N_B\\T\\m_A\\x\end{pmatrix}=\text{Find}\left(N_B,T,m_A,x\right) \qquad \begin{pmatrix}N_B\\T\end{pmatrix}=\begin{pmatrix}64.63\\20.20\end{pmatrix}\text{ N} \qquad m_A=7.82\text{ kg}$$

$$x=0.052\text{ m}$$

Since $x=51.6\,\text{mm}<\frac{b}{2}=125\,\text{mm}$ our assumption is correct $\qquad m_A=7.82\text{ kg}$

Problem 8-107

The collar bearing uniformly supports an axial force **P**. If the coefficient of static friction is μ_s, determine the torque **M** required to overcome friction.

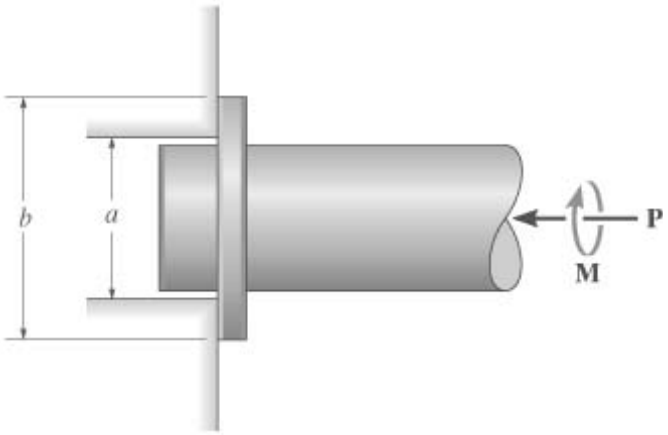
Given:

$a = 2$ in

$b = 3$ in

$P = 800$ lb

$\mu_s = 0.3$



Solution:

$$M=\frac{a}{b}\mu_sP\left[\frac{\left(\frac{b}{2}\right)^3-\left(\frac{a}{2}\right)^3}{\left(\frac{b}{2}\right)^2-\left(\frac{a}{2}\right)^2}\right] \qquad M=\frac{a}{b}\,\mu_s\,P\left[\frac{1}{2}\left(\frac{a^2+ab+b^2}{a+b}\right)\right] \qquad M=304.00\,\text{lb}\cdot\text{in}$$

Problem 8-108

The collar bearing uniformly supports an axial force **P**. If a torque **M** is applied to the shaft and causes it to rotate at constant velocity, determine the coefficient of kinetic friction at the surface of contact.

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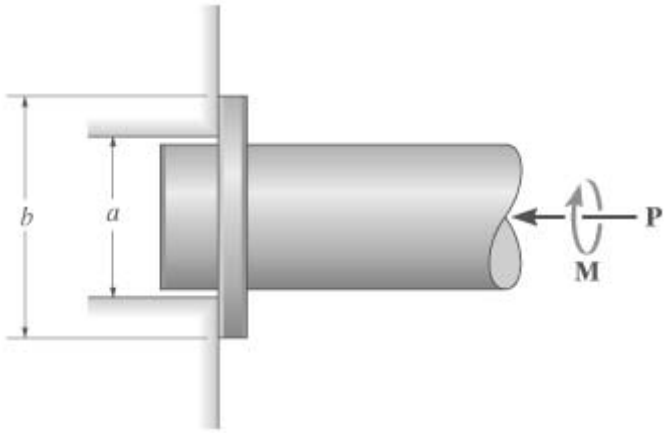
Given:

$a = 2$ in

$b = 3$ in

$P = 500$ lb

$M = 3$ lb ft



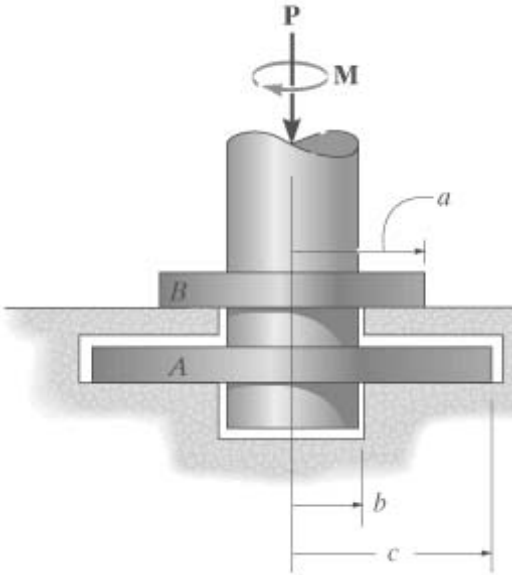
Solution:

$$M = \frac{a}{b}\mu_k P\left[\frac{\left(\frac{b}{2}\right)^3 - \left(\frac{a}{2}\right)^3}{\left(\frac{b}{2}\right)^2 - \left(\frac{a}{2}\right)^2}\right] = \left(\frac{a\mu_k P}{2b}\right)\left(\frac{b^3 - a^3}{b^2 - a^2}\right)$$

$$\mu_k = \frac{2Mb\left(b^2 - a^2\right)}{aP\left(b^3 - a^3\right)} \qquad \mu_k = 0.0568$$

Problem 8-109

The *double-collar bearing* is subjected to an axial force **P**. Assuming that collar *A* supports *kP* and collar *B* supports $(1 - k)P$, both with a uniform distribution of pressure, determine the maximum frictional moment *M* that may be resisted by the bearing.



Units Used:

$\text{kN} = 10^3$ N

Given:

$P = 4$ kN

$a = 20$ mm

$b = 10$ mm

$c = 30$ mm

$\mu_s = 0.2$

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$k = 0.75$

Solution:

$$M = \frac{2}{3}\mu_s\left[\frac{c^3 - b^3}{c^2 - b^2}kP + \frac{a^3 - b^3}{a^2 - b^2}(1 - k)P\right]$$

$M = 16.1\,\text{N}\cdot\text{m}$

Problem 8-110

The annular ring bearing is subjected to a thrust P. If the coefficient of static friction is μ_s, determine the torque **M** that must be applied to overcome friction.

Given:

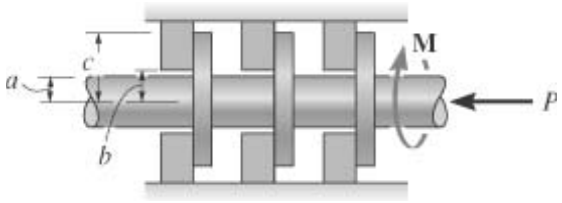
$P = 800$ lb

$\mu_s = 0.35$

$a = 0.75$ in

$b = 1$ in

$c = 2$ in



Solution:

$$M = \frac{2}{3}\mu_s P\left(\frac{b^3 - c^3}{b^2 - c^2}\right)$$

$M = 36.3\,\text{lb}\cdot\text{ft}$

Problem 8-111

The floor-polishing machine rotates at a constant angular velocity. If it has weight W, determine the couple forces **F** the operator must apply to the handles to hold the machine stationary. The coefficient of kinetic friction between the floor and brush is μ_k. Assume the brush exerts a uniform pressure on the floor.

Given:

$W = 80$ lb

$\mu_k = 0.3$

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$a = 1.5$ ft

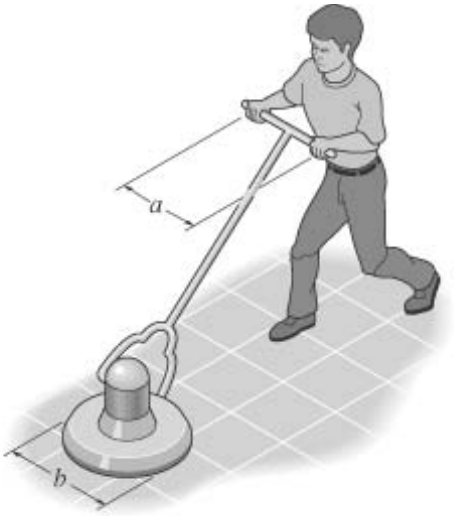
$b = 2$ ft

Solution:

$$F\,a = \frac{2}{3}\mu_k W\left(\frac{b}{2}\right)$$

$$F = \frac{1}{3}\,\mu_k\,W\left(\frac{b}{a}\right)$$

$$F = 10.7\,\text{lb}$$



Problem 8-112

The plate clutch consists of a flat plate *A* that slides over the rotating shaft *S*. The shaft is fixed to the driving plate gear *B*. If the gear *C*, which is in mesh with *B*, is subjected to a torque *M*, determine the smallest force *P*, that must be applied via the control arm, to stop the rotation. The coefficient of static friction between the plates *A* and *D* is μ_s. Assume the bearing pressure between *A* and *D* to be uniform.

Given:

$M = 0.8$ N·m

$\mu_s = 0.4$

$a = 150$ mm

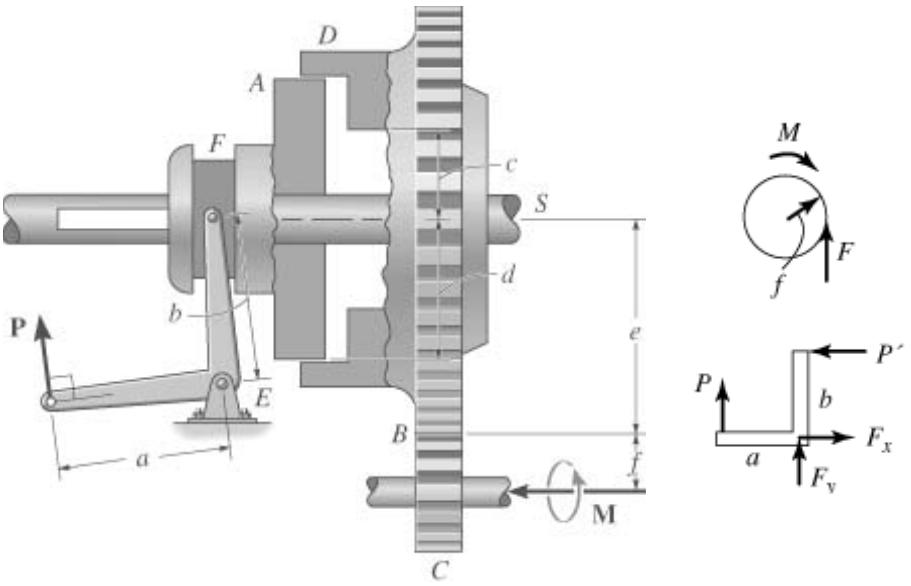
$b = 200$ mm

$c = 100$ mm

$d = 125$ mm

$e = 150$ mm

$f = 30$ mm



Solution:

$$F = \frac{M}{f} \qquad F = 26.667\ \text{N}$$

$$M_2 = F\,e \qquad M_2 = 4.00\,\text{N·m}$$

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$$M_2 = \frac{2}{3}\mu_s P'\left(\frac{d^3 - c^3}{d^2 - c^2}\right) \qquad P' = \frac{3M_2}{2\mu_s}\left(\frac{d^2 - c^2}{d^3 - c^3}\right) \qquad P' = 88.5 \text{ N}$$

$$P'\,b - P\,a = 0 \qquad P = P'\left(\frac{b}{a}\right) \qquad P = 118 \text{ N}$$

Problem 8-113

The shaft of diameter b is held in the hole such that the normal pressure acting around the shaft varies linearly with its depth as shown. Determine the frictional torque that must be overcome to rotate the shaft.

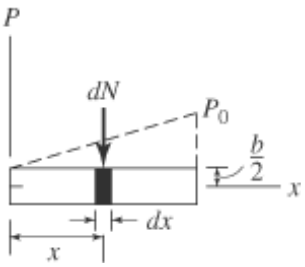
Given:

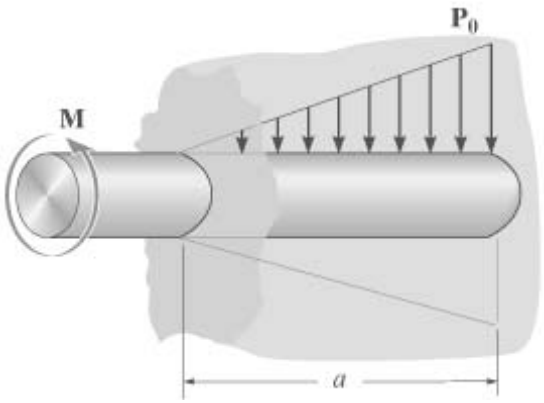
$a = 6$ in

$p_0 = 60 \dfrac{\text{lb}}{\text{in}^2}$

$b = 4$ in

$\mu_s = 0.2$





Solution:

$$N = \int_0^a \left(\frac{x}{a}\right) p_0 2\pi\left(\frac{b}{2}\right) \mathrm{d}x \qquad T = \mu_s N\left(\frac{b}{2}\right) \qquad T = 905\,\text{lb}\cdot\text{in}$$

Problem 8-114

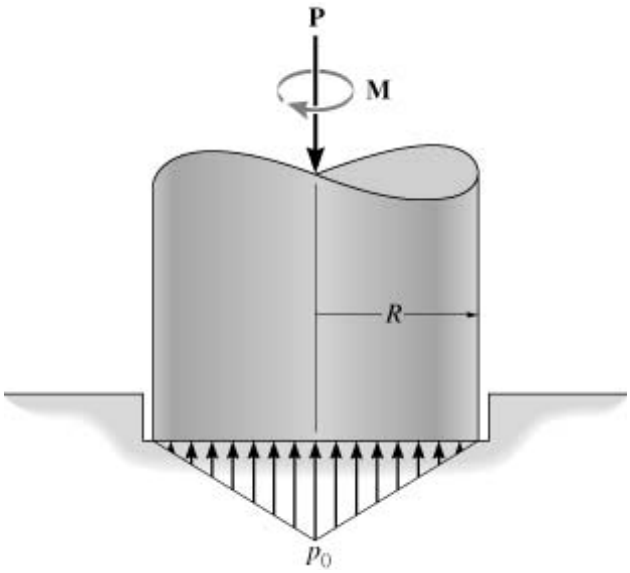
Because of wearing at the edges, the pivot bearing is subjected to a conical pressure distribution at its surface of contact. Determine the torque **M** required to overcome friction and turn the shaft, which supports an axial force **P**. The coefficient of static friction is μ_s. For the solution, it is necessary to determine the peak pressure p_0 in terms of P and the bearing radius R.

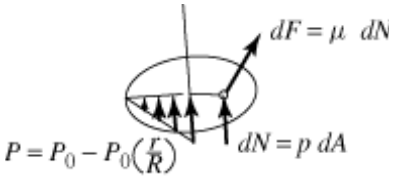
Solution:

$$P = \int_0^{2\pi}\int_0^R \left(p_0 - p_0\frac{r}{R}\right) r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{\pi p_0 R^2}{3} \qquad p_0 = \frac{3P}{\pi R^2}$$

$$\mathrm{d}M = r\mathrm{d}F = r\mu\mathrm{d}N = r\mu p\mathrm{d}A = r\mu p\,r\mathrm{d}\theta\mathrm{d}r$$

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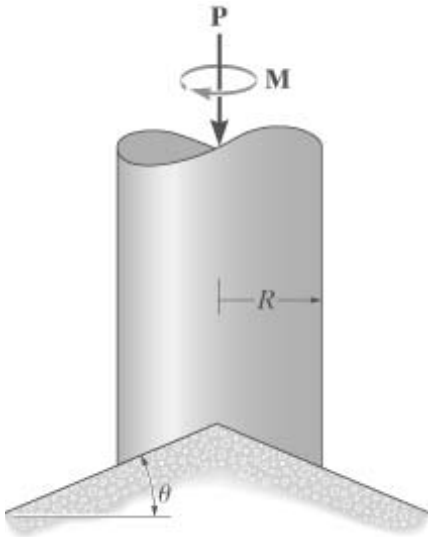


$$M = \int_0^R \mu\left(p_0 - \frac{p_0}{R}r\right) r^2\,\mathrm{d}r \int_0^{2\pi} 1\,\mathrm{d}\theta = \frac{\pi}{6}\mu p_0 R^3 = \frac{\pi}{6}\mu\left(\frac{3P}{\pi R^2}\right)R^3 = \frac{\mu P R}{2}$$

Thus, $M = \dfrac{\mu P R}{2}$

Problem 8-115

The conical bearing is subjected to a constant pressure distribution at its surface of contact. If the coefficient of static friction is μ_s, determine the torque **M** required to overcome friction if the shaft supports an axial force **P**.



Solution:

The differential Area (shaded)

$$\mathrm{d}A = 2\pi \mathrm{r}\left(\frac{\mathrm{d}r}{\cos(\theta)}\right) = \frac{2\pi r \mathrm{d}r}{\cos(\theta)}$$

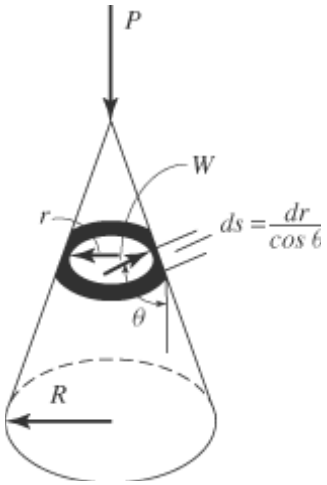
$$P = \int p\cos(\theta)\,\mathrm{d}A = \int p\cos(\theta)\left(\frac{2\pi r}{\cos(\theta)}\right)\mathrm{d}r = 2\pi p\int_0^R r\,\mathrm{d}r = \pi p R^2 \qquad p = \frac{P}{\pi R^2}$$

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$$dN = p\,dA = \frac{P}{\pi R^2}\left(\frac{2\pi r\,dr}{\cos(\theta)}\right) = \frac{2P}{R^2\cos(\theta)}r\,dr$$

$$M = \int r\,dF = \int r\mu_S\,dN = \left(\frac{2\mu_S P}{R^2\cos(\theta)}\right)\int_0^R r^2\,dr$$

$$M = \left(\frac{2\mu_S P}{R^2\cos(\theta)}\right)\frac{R^3}{3} = \frac{2\mu_S P R}{3\cos(\theta)}$$



Problem 8-116

The tractor is used to push the pipe of weight W. To do this it must overcome the frictional forces at the ground, caused by sand. Assuming that the sand exerts a pressure on the bottom of the pipe as shown, and the coefficient of static friction between the pipe and the sand is μ_s, determine the force required to push the pipe forward. Also, determine the peak pressure p_0.

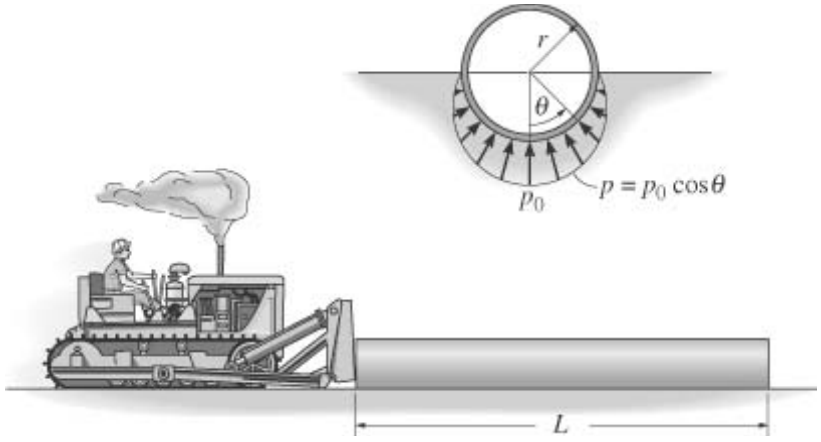
Given:

$W = 1500$ lb

$\mu_S = 0.3$

$L = 12$ ft

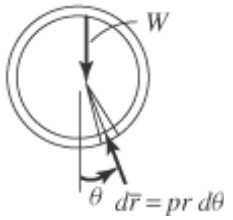
$r = 15$ in



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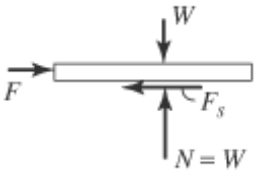
Solution:

$$+\uparrow \Sigma F_y = 0; \quad 2L\int_0^{\frac{\pi}{2}} p_0 \cos(\theta) r \, d\theta \cos(\theta) - W = 0$$



$$W = 2 p_0 L r \int_0^{\frac{\pi}{2}} \cos^2\theta \, d\theta = p_0 L r\left(\frac{\pi}{2}\right)$$

$$p_0 = 2\,\frac{W}{r\,L\,\pi} \qquad p_0 = 0.442\,\frac{\text{lb}}{\text{in}^2}$$



$$\xrightarrow{+} \Sigma F_x = 0; \quad F = \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \mu_s p_0 L r \cos(\theta) \, d\theta \qquad F = 573\,\text{lb}$$

Problem 8-117

Assuming that the variation of pressure at the bottom of the pivot bearing is defined as $p = p_0(R_2/r)$, determine the torque **M** needed to overcome friction if the shaft is subjected to an axial force **P.** The coefficient of static friction is μ_s. For the solution, it is necessary to determine p_0 in terms of P and the bearing dimensions R_1 and R_2.

Solution:

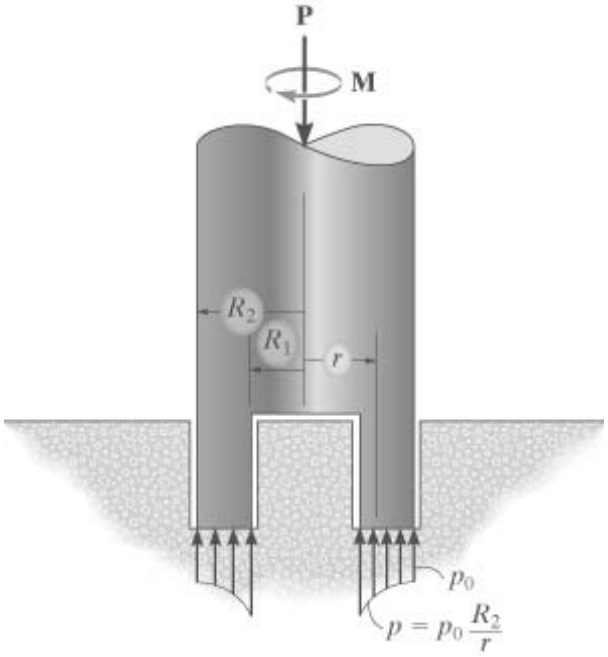
$$P = \int_0^{2\pi} \int_{R_1}^{R_2} p\, r \, dr \, d\theta$$

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$$P = \int_0^{2\pi} \int_{R_1}^{R_2} p_0\left(\frac{R_2}{r}\right) r \, dr \, d\theta$$

$$P = 2\pi p_0 R_2\left(R_2 - R_1\right)$$

$$p_0 = \frac{P}{2\pi R_2\left(R_2 - R_1\right)}$$



$$M = \int_A r \, dF = \int_0^{2\pi} \int_{R_1}^{R_2} r \mu_s \, p_0 \, r \, dr \, d\theta$$

$$M = \int_0^{2\pi} \int_0^{2\pi} \mu_s \, p_0\left(\frac{R_2}{r}\right) r^2 \, dr \, d\theta = \pi \mu_s \, p_0 R_2\left({R_2}^2 - {R_1}^2\right)$$

$$M = \pi \mu_s \left[\frac{P}{2\pi R_2\left(R_2 - R_1\right)}\right] R_2\left({R_2}^2 - {R_1}^2\right) \qquad M = \frac{\mu_s P\left(R_2 + R_1\right)}{2}$$

Problem 8-118

A disk having an outer diameter a fits loosely over a fixed shaft having a diameter b. If the coefficient of static friction between the disk and the shaft is μ_s, determine the smallest vertical force **P**, acting on the rim, which must be applied to the disk to cause it to slip over the shaft. The disk weighs W.

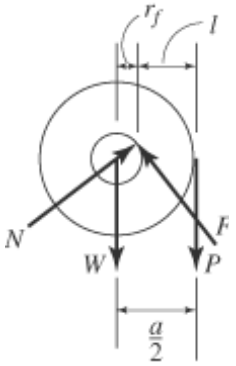
Given:

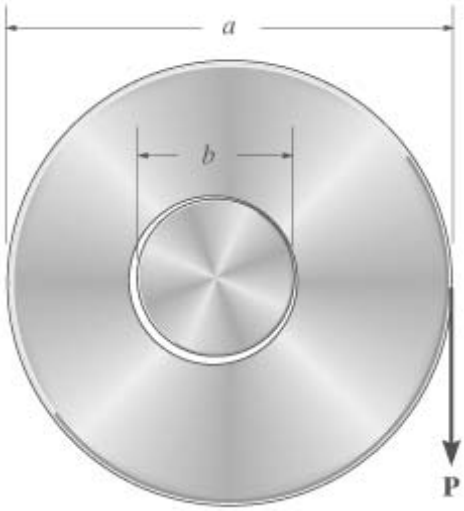
$a = 8$ in

$b = 3$ in

$\mu_s = 0.15$

$W = 10$ lb





Solution:

$\phi_s = \operatorname{atan}\left(\mu_s\right)$

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$$r_f = \left(\frac{b}{2}\right)\sin(\phi_s)$$

$$l = \left[\left(\frac{a}{2}\right) - r_f\right]$$

$\Sigma M_0 = 0;$

$$W r_f - l P = 0 \qquad P = W\left(\frac{r_f}{l}\right) \qquad P = 0.59\,\text{lb}$$

Problem 8-119

The pulley has a radius r and fits loosely on the shaft of diameter d. If the loadings acting on the belt cause the pulley to rotate with constant angular velocity, determine the frictional force between the shaft and the pulley and compute the coefficient of kinetic friction. The pulley has weight W.

Given:

$r = 3$ in

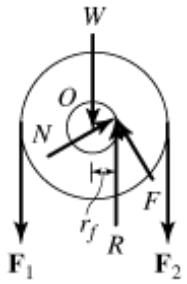
$d = 0.5$ in

$W = 18$ lb

$F_1 = 5$ lb

$F_2 = 5.5$ lb





Solution:

$+\uparrow \Sigma F_y = 0;$ $\quad R - W - F_1 - F_2 = 0$ $\quad R = W + F_1 + F_2$ $\quad R = 28.5\,\text{lb}$

$\circlearrowleft + \Sigma M_O = 0;$ $\quad -F_2 r + F_1 r + R r_f = 0$ $\quad r_f = r\left(\frac{F_2 - F_1}{R}\right)$ $\quad r_f = 0.05263\,\text{in}$

$r_f = \frac{d}{2}\sin(\phi_k)$ $\quad \phi_k = \text{asin}\left(2\frac{r_f}{d}\right)$ $\quad \phi_k = 12.15\,\text{deg}$

$\mu = \tan(\phi_k)$ $\quad \mu = 0.215$

Also,

$\circlearrowleft + \Sigma M_O = 0;$ $\quad -F_2 r + F_1 r + F\left(\frac{d}{2}\right) = 0$ $\quad F = 2r\left(\frac{F_2 - F_1}{d}\right)$ $\quad F = 6\,\text{lb}$

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Problem 8-120

The pulley has a radius r and fits loosely on the shaft of diameter d. If the loadings acting on the belt cause the pulley to rotate with constant angular velocity, determine the frictional force between the shaft and the pulley and compute the coefficient of kinetic friction. Neglect the weight of the pulley.

Given:

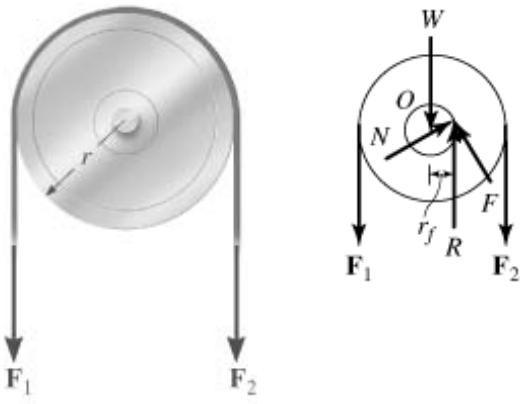
$r = 3$ in

$d = 0.5$ in

$W = 0$ lb

$F_1 = 5$ lb

$F_2 = 5.5$ lb



Solution:

$+\uparrow \Sigma F_y = 0;\quad R - W - F_1 - F_2 = 0 \qquad R = W + F_1 + F_2 \qquad R = 10.5\,\text{lb}$

$\curvearrowleft + \Sigma M_O = 0;\quad -F_2 r + F_1 r + R r_f = 0 \qquad r_f = r\left(\frac{F_2 - F_1}{R}\right) \qquad r_f = 0.14286\,\text{in}$

$r_f = \frac{d}{2}\sin(\phi_k) \qquad \phi_k = \operatorname{asin}\left(2\frac{r_f}{d}\right) \qquad \phi_k = 34.85\,\text{deg}$

$\mu = \tan(\phi_k) \qquad \mu = 0.696$

Also,

$\curvearrowleft + \Sigma M_O = 0;\quad -F_2 r + F_1 r + F\frac{d}{2} = 0 \qquad F = 2r\left(\frac{F_2 - F_1}{d}\right) \qquad F = 6\,\text{lb}$

Problem 8-121

A pulley of mass M has radius a and the axle has a diameter D. If the coefficient of kinetic friction between the axle and the pulley is μ_k determine the vertical force **P** on the rope required to lift the block of mass M_B at constant velocity.

Given:

$a = 120$ mm

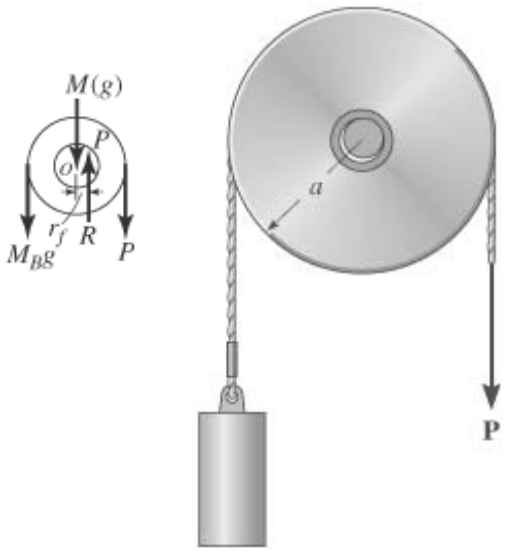
$M = 5$ kg

$D = 40$ mm

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$\mu_k = 0.15$

$M_B = 80$ kg



Solution:

$$\phi_k = \operatorname{atan}(\mu_k)$$

$$r_f = \left(\frac{D}{2}\right)\sin(\phi_k)$$

$\Sigma M_p = 0;$

$$M_B g(a + r_f) + M g r_f - P(a - r_f) = 0$$

$$P = \frac{M_B g(a + r_f) + M g r_f}{a - r_f} \qquad P = 826 \text{ N}$$

Problem 8-122

A pulley of mass M has radius a and the axle has a diameter D. If the coefficient of kinetic friction between the axle and the pulley is μ_k determine the force **P** on the rope required to lift the block of mass M_B at constant velocity. Apply the force P horizontally to the right (not as shown in the figure).

Given:

$a = 120$ mm

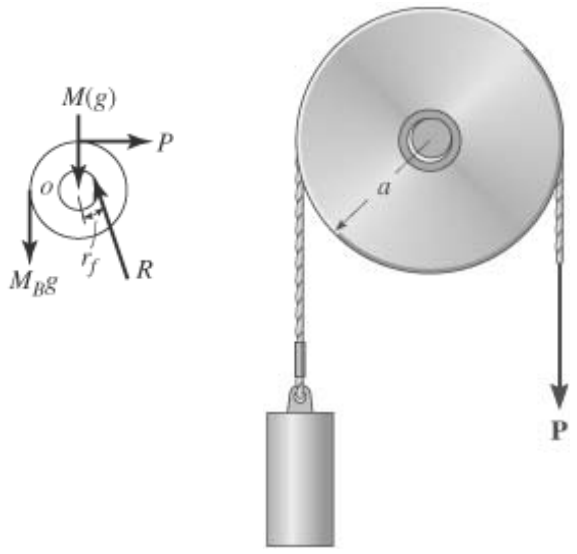
$M = 5$ kg

$D = 40$ mm

$\mu_k = 0.15$

$M_B = 80$ kg

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

$$\phi_k = \operatorname{atan}(\mu_k)$$

$$r_f = \frac{D}{2}\sin(\phi_k)$$

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Guesses

$P = 1\ \text{N} \qquad R = 1\ \text{N} \qquad \alpha = 1\ \text{deg}$

Given

$R\cos(\alpha) - M_B g - M g = 0$

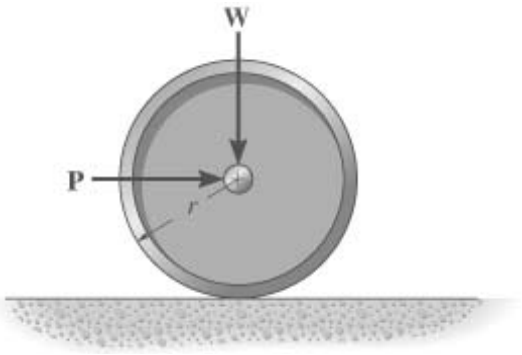
$P - R\sin(\alpha) = 0$

$M_B g\, a - P\, a + R\, r_f = 0$

$\begin{pmatrix} P \\ R \\ \alpha \end{pmatrix} = \text{Find}(P, R, \alpha) \qquad P = 814\ \text{N}$

Problem 8-123

A wheel on a freight car carries a load *W*. If the axle of the car has a diameter *D*, determine the horizontal force P that must be applied to the axle to rotate the wheel. The coefficient of kinetic friction is μ_k.



Units Used:

$\text{kip} = 1000\ \text{lb}$

Given:

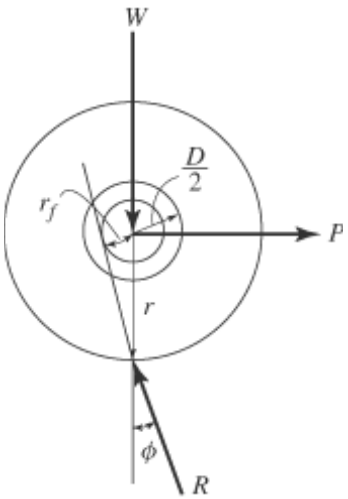
$W = 20\ \text{kip}$

$D = 2\ \text{in}$

$\mu_k = 0.05$

$r = 16\ \text{in}$

Solution:



$\Sigma F_x = 0; \qquad P - R\sin(\phi) = 0$

$\Sigma F_y = 0; \qquad R\cos(\phi) - W = 0$

Thus, $\qquad P = W\tan(\phi)$

$\phi_k = \text{atan}(\mu_k) \qquad \phi_k = 2.86\,\text{deg}$

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$r_f = \left(\frac{D}{2}\right)\sin(\phi_k)$ $r_f = 0.04994\,\text{in}$

$\phi = \text{asin}\left(\frac{r_f}{r}\right)$ $\phi = 0.1788\,\text{deg}$

$P = W\tan(\phi)$

$P = 62.4\,\text{lb}$

Problem 8-124

The trailer has a total weight W and center of gravity at G which is directly over its axle. If the axle has a diameter D, the radius of the wheel is r, and the coefficient of kinetic friction at the bearing is μ_k, determine the horizontal force **P** needed to pull the trailer.

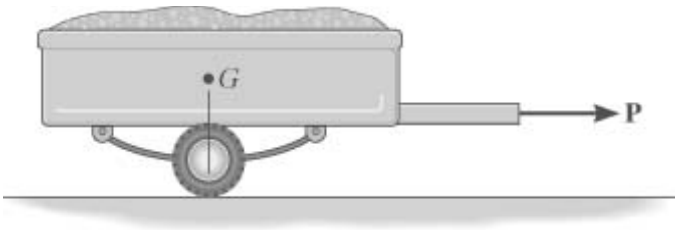
Given:

$W = 850\ \text{lb}$

$r = 1.5\ \text{ft}$

$\mu_k = 0.08$

$D = 1\ \text{in}$

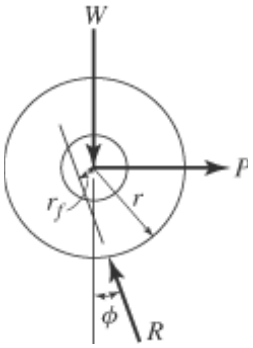


Solution:

$\Sigma F_x = 0;$ $P - R\sin(\phi) = 0$

$\Sigma F_y = 0;$ $R\cos(\phi) - W = 0$

Thus, $P = W\tan(\phi)$



$\phi_k = \text{atan}(\mu_k)$ $\phi_k = 4.57\,\text{deg}$

$r_f = \left(\frac{D}{2}\right)\sin(\phi_k)$ $r_f = 0.03987\,\text{in}$

$\phi = \text{asin}\left(\frac{r_f}{r}\right)$ $\phi = 0.1269\,\text{deg}$

$P = W\tan(\phi)$ $P = 1.88\,\text{lb}$

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Problem 8-125

The collar fits *loosely* around a fixed shaft that has radius *r*. If the coefficient of kinetic friction between the shaft and the collar is μ_k, determine the force **P** on the horizontal segment of the belt so that the collar rotates counterclockwise with a constant angular velocity. Assume that the belt does not slip on the collar; rather, the collar slips on the shaft. Neglect the weight and thickness of the belt and collar. The radius, measured from the center of the collar to the mean thickness of the belt is *R*.

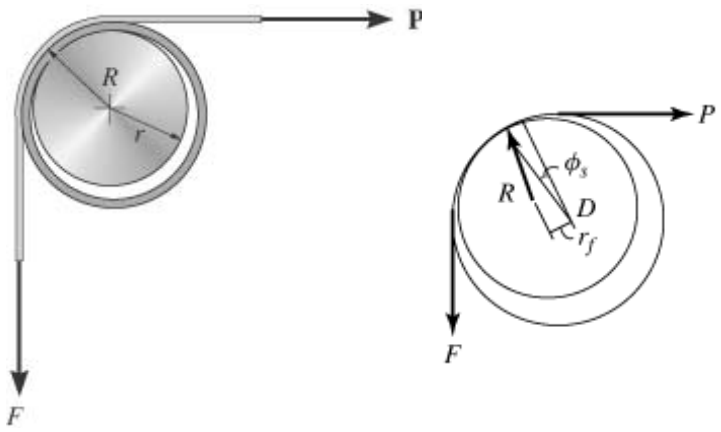
Given:

$r = 2$ in

$\mu_k = 0.3$

$R = 2.25$ in

$F = 20$ lb



Solution: $\phi_k = \operatorname{atan}(\mu_k)$ $\phi_k = 16.699\,\text{deg}$

$r_f = r\sin(\phi_k)$ $r_f = 0.5747\,\text{in}$

Equilibrium:

$+\uparrow \Sigma F_y = 0;$ $R_y - F = 0$ $R_y = F$ $R_y = 20.00\,\text{lb}$

$\xrightarrow{+} \Sigma F_x = 0;$ $P - R_x = 0$ $R_x = \text{P}$

$$R = \sqrt{R_x{}^2 + R_y{}^2} = \sqrt{P^2 + F^2}$$

Guess $P = 1$ lb

Given $-\left(\sqrt{P^2 + F^2}\right) r_f + F R - P R = 0$ $P = \text{Find}(P)$ $P = 13.79\,\text{lb}$

Problem 8-126

The collar fits *loosely* around a fixed shaft that has radius *r*. If the coefficient of kinetic friction between the shaft and the collar is μ_k, determine the force **P** on the horizontal segment

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of the belt so that the collar rotates clockwise with a constant angular velocity. Assume that the belt does not slip on the collar; rather, the collar slips on the shaft. Neglect the weight and thickness of the belt and collar. The radius, measured from the center of the collar to the mean thickness of the belt is R.

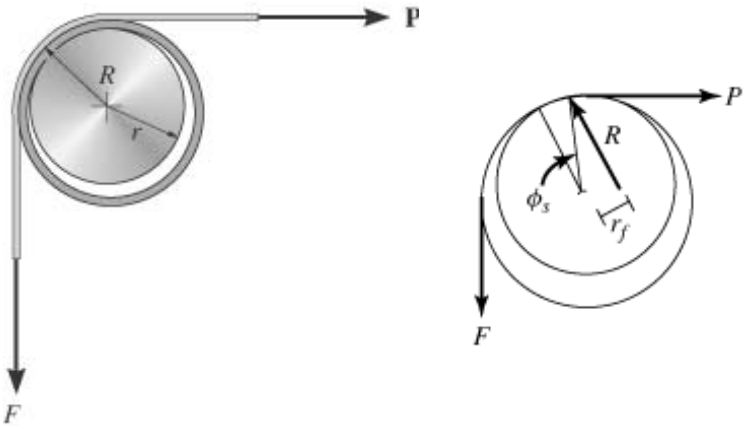
Given:

$r = 2$ in

$\mu_k = 0.3$

$R = 2.25$ in

$F = 20$ lb



Solution: $\phi_k = \text{atan}(\mu_k)$ $\phi_k = 16.699\,\text{deg}$

$r_f = r\sin(\phi_k)$ $r_f = 0.5747\,\text{in}$

Equilibrium:

$+\uparrow \Sigma F_y = 0;$ $R_y - F = 0$ $R_y = F$ $R_y = 20.00\,\text{lb}$

$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $P - R_x = 0$ $R_x = P$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{P^2 + F^2}$$

Guess $P = 1$ lb

Given $\left(\sqrt{P^2 + F^2}\right) r_f + F R - P R = 0$ $P = \text{Find}(P)$ $P = 29.00\,\text{lb}$

Problem 8-127

The connecting rod is attached to the piston by a pin at B of diameter d_1 and to the crank shaft by a bearing A of diameter d_2. If the piston is moving downwards, and the coefficient of static friction at these points is μ_s, determine the radius of the friction circle at each connection.

Given:

$d_1 = 0.75$ in

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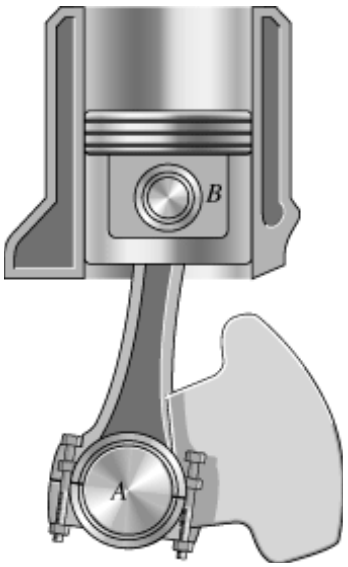
$d_2 = 2$ in

$\mu_s = 0.2$

Solution:

$r_{fA} = \frac{1}{2} d_2 \mu_s$ $r_{fA} = 0.2\,\text{in}$

$r_{fB} = \frac{1}{2} d_1 \mu_s$ $r_{fB} = 0.075\,\text{in}$



Problem 8-128

The connecting rod is attached to the piston by a pin at B of diameter d_1 and to the crank shaft by a bearing A of diameter d_2. If the piston is moving upwards, and the coefficient of static friction at these points is μ_s, determine the radius of the friction circle at each connection.

Given:

$d_1 = 20$ mm

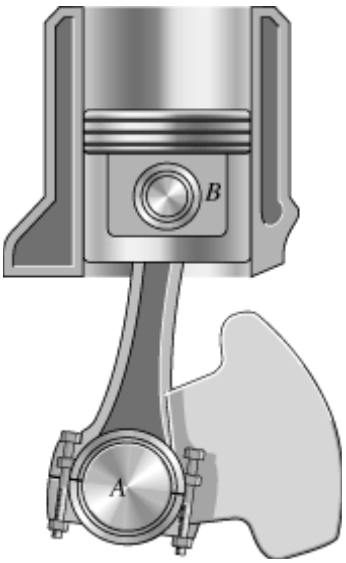
$d_2 = 50$ mm

$\mu_s = 0.3$

Solution:

$r_{fA} = \frac{1}{2} d_2 \mu_s$ $r_{fA} = 7.50\,\text{mm}$

$r_{fB} = \frac{1}{2} d_1 \mu_s$ $r_{fB} = 3\,\text{mm}$



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Problem 8-129

The lawn roller has mass M. If the arm BA is held at angle θ from the horizontal and the coefficient of rolling resistance for the roller is r, determine the force **P** needed to push the roller at constant speed. Neglect friction developed at the axle, A, and assume that the resultant force **P** acting on the handle is applied along arm BA.

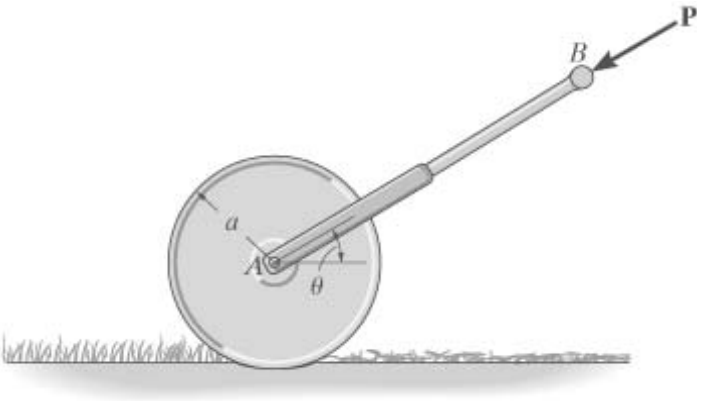
Given:

$M = 80$ kg

$\theta = 30$ deg

$a = 250$ mm

$r = 25$ mm



Solution:

$$\theta_1 = \operatorname{asin}\left(\frac{r}{a}\right)$$

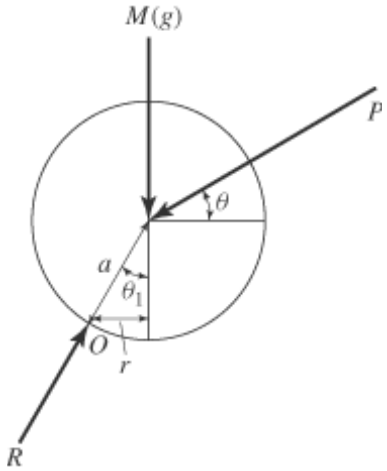
$$\theta_1 = 5.74\,\text{deg}$$

$\Sigma M_0 = 0;$

$$-rMg - P\sin(\theta)r + P\cos(\theta)a\cos(\theta_1) = 0$$

$$P = \frac{rMg}{-\sin(\theta)\,r + \cos(\theta)\,a\,\cos(\theta_1)}$$

$$P = 96.7\ \text{N}$$



Problem 8-130

The handcart has wheels with a diameter D. If a crate having a weight W is placed on the cart, determine the force **P** that must be applied to the handle to overcome the rolling resistance. The coefficient of rolling resistance is μ. Neglect the weight of the cart.

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Given:

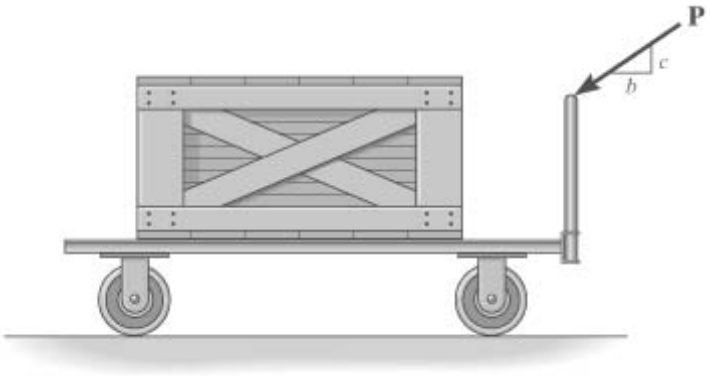
$W = 1500$ lb

$D = 6$ in

$a = 0.04$ in

$c = 3$

$b = 4$



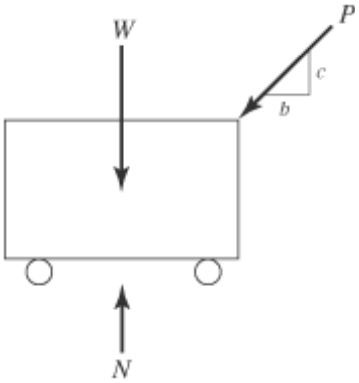
Solution:

Guesses $N = 1$ lb $P = 1$ lb

Given

$$N - W - P\left(\frac{c}{\sqrt{c^2 + b^2}}\right) = 0$$

$$\left(\frac{b}{\sqrt{b^2 + c^2}}\right)P = N\left(\frac{2a}{D}\right)$$

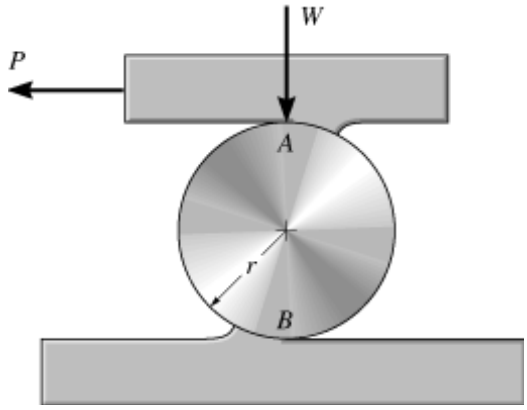


$$\begin{pmatrix} N \\ P \end{pmatrix} = \text{Find}(N, P) \qquad N = 1515\,\text{lb} \qquad P = 25.3\,\text{lb}$$

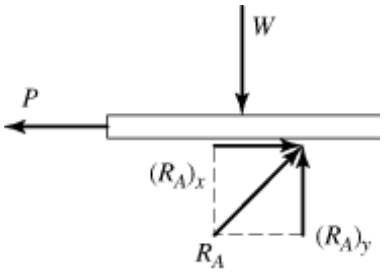
Problem 8-131

The cylinder is subjected to a load that has a weight W. If the coefficients of rolling resistance for the cylinder's top and bottom surfaces are a_A and a_B respectively, show that a force having a magnitude of $P = [W(a_A + a_B)]/2r$ is required to move the load and thereby roll the cylinder forward. Neglect the weight of the cylinder.

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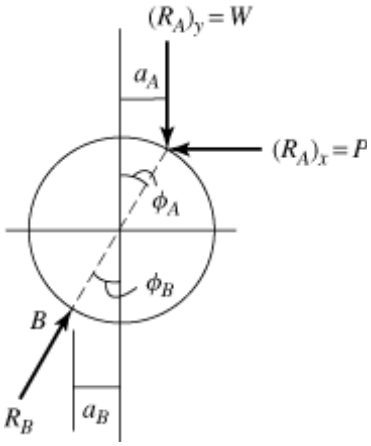
Solution:



$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad R_{Ax} - P = 0 \quad R_{Ax} = P$

$+\uparrow \Sigma F_y = 0; \quad R_{Ay} - W = 0 \quad R_{Ay} = \mathrm{P}$

$\circlearrowleft + \Sigma M_B = 0; \quad P\left(r\cos\left(\phi_A\right) + r\cos\left(\phi_B\right)\right) - W\left(a_A + a_B\right) = 0$



Since ϕ_A and ϕ_B are very small,

$\cos\left(\phi_A\right) = \cos\left(\phi_B\right) = 1$ Hence from Eq.(1)

$$P = \frac{W\left(a_A + a_B\right)}{2r} \qquad \text{(QED)}$$

Problem 8-132

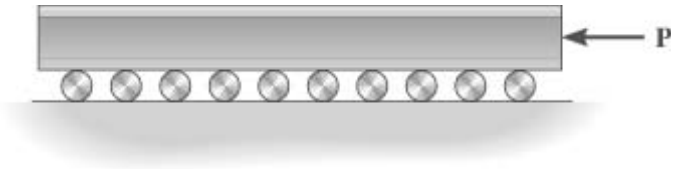
A steel beam of mass *M* is moved over a level surface using a series of rollers of diameter *D* for which the coefficient of rolling resistance is a_g at the ground and a_s at the bottom surface of the beam. Determine the horizontal force **P** needed to push the beam forward at a constant speed. *Hint:* Use the result of Prob. 8–131.

Units Used:

Mg = 1000 kg

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Given:



$M = 1.2\ \text{Mg}$

$D = 30\ \text{mm}$

$a_g = 0.4\ \text{mm}$

$a_s = 0.2\ \text{mm}$

Solution:

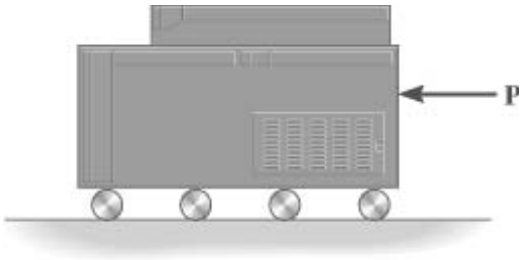
$$P = \frac{M g\left(a_g + a_s\right)}{2\left(\frac{D}{2}\right)} \qquad P = 235\,\text{N}$$

Problem 8-133

A machine of mass *M* is to be moved over a level surface using a series of rollers for which the coefficient of rolling resistance is a_g at the ground and a_m at the bottom surface of the machine. Determine the appropriate diameter of the rollers so that the machine can be pushed forward with a horizontal force **P**. *Hint:* Use the result of Prob. 8-131.

Units Used:

$\text{Mg} = 1000\ \text{kg}$



Given:

$M = 1.4\ \text{Mg}$

$a_g = 0.5\ \text{mm}$

$a_m = 0.2\ \text{mm}$

$P = 250\ \text{N}$

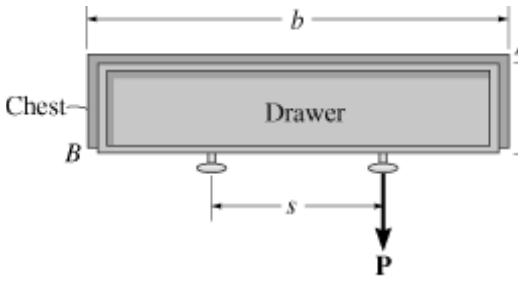
Solution:

$$P = \frac{M\ g\left(a_g + a_m\right)}{2\ r}$$

$$r = M g\left(\frac{a_g + a_m}{2\ P}\right) \qquad r = 19.2\,\text{mm} \qquad d = 2\ r \qquad d = 38.5\,\text{mm}$$

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Problem 8-134



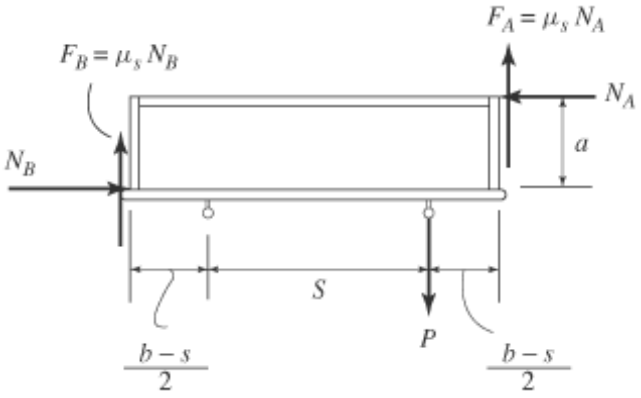
A single force **P** is applied to the handle of the drawer. If friction is neglected at the bottom and the coefficient of static friction along the sides is μ_s determine the largest spacing *s* between the symmetrically placed handles so that the drawer does not bind at the corners *A* and *B* when the force **P** is applied to one of the handles.

Given:

$\mu_s = 0.4$

$a = 0.3$ m

$b = 1.25$ m



Solution:

Equation of Equilibrium and Friction : If the drawer does not bind at corners *A* and *B*, slipping would have to occur at points *A* and *B*. Hence, $F_A = \mu N_A$ and $F_B = \mu N_B$

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad N_B - N_A = 0$

$N_A = N_B = N$

$+\uparrow \Sigma F_y = 0; \quad \mu_s N_A + \mu_s N_B - P = 0 \qquad P = 2\mu_s N$

$\circlearrowleft + \Sigma M_B = 0; \quad Na + \mu_s N b - P\left(\frac{s+b}{2}\right) = 0$

$\left[a + \mu_s\, b - 2\mu_s \left(\frac{s+b}{2}\right)\right] N = 0$

$a + \mu_s\, b - \mu_s\, (s+b) = 0 \qquad s = \frac{a}{\mu_s} \qquad s = 0.750$ m

---

Problem 8-135

The truck has mass *M* and a center of mass at *G*. Determine the greatest load it can pull if (a) the truck has rear-wheel drive while the front wheels are free to roll, and (b) the truck has

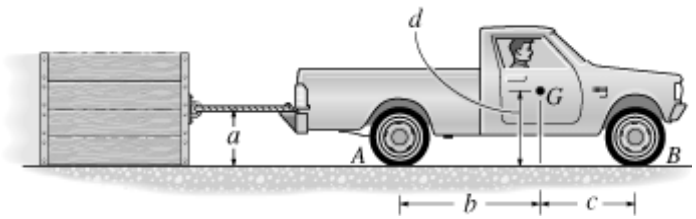
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four-wheel drive. The coefficient of static friction between the wheels and the ground is μ_{st} and between the crate and the ground, it is μ_{sc}.

Units Used:

$\text{kN} = 10^3\ \text{N}$

$\text{Mg} = 1000\ \text{kg}$



Given:

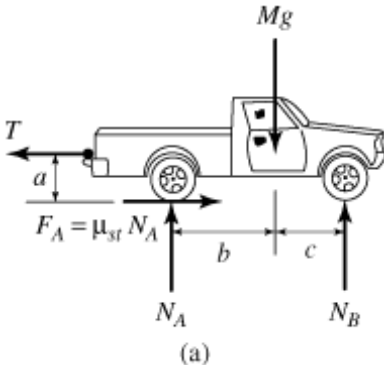
$M = 1.25\ \text{Mg}$

$\mu_{st} = 0.5$ $\quad a = 600\ \text{mm}$

$\mu_{sc} = 0.4$ $\quad b = 1.5\ \text{m}$

$c = 1\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $\quad d = 800\ \text{mm}$



(a)

Solution:

Guesses $\quad N_A = 1\ \text{N} \quad N_B = 1\ \text{N}$

$T = 1\ \text{N} \quad N_C = 1\ \text{N} \quad W = 1\ \text{N}$

(a) Rear wheel drive

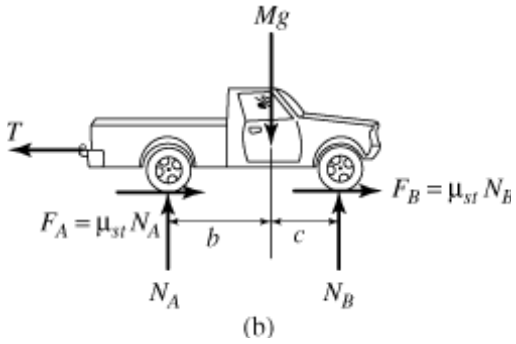
Given $\quad -T + \mu_{st} N_A = 0$

$N_A + N_B - Mg = 0$

$-Mg\,b + N_B(b + c) + T a = 0$

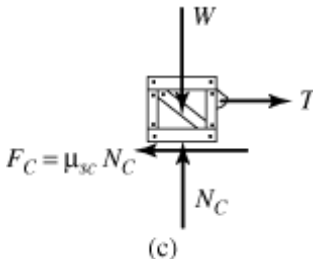
$T - \mu_{sc} N_C = 0$

$N_C - W = 0$



(b)

$$\begin{pmatrix} N_A \\ N_B \\ T \\ N_C \\ W \end{pmatrix} = \text{Find}(N_A, N_B, T, N_C, W) \qquad W = 6.97\ \text{kN}$$



(c)

(b) Four wheel drive

Given $\quad -T + \mu_{st} N_A + \mu_{st} N_B = 0$

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$$N_A + N_B - M g = 0$$

$$-M g\, b + N_B(b + c) + T a = 0$$

$$T - \mu_{sc} N_C = 0$$

$$N_C - W = 0$$

$$\begin{pmatrix} N_A \\ N_B \\ T \\ N_C \\ W \end{pmatrix} = \text{Find}\left(N_A, N_B, T, N_C, W\right) \qquad W = 15.33\,\text{kN}$$

Problem 8-136

The truck has M and a center of mass at G. The truck is traveling up an incline of angle θ. Determine the greatest load it can pull if (a) the truck has rear-wheel drive while the front wheels are free to roll, and (b) the truck has four-wheel drive. The coefficient of static friction between the wheels and the ground is μ_{st} and between the crate and the ground, it is μ_{sc}.

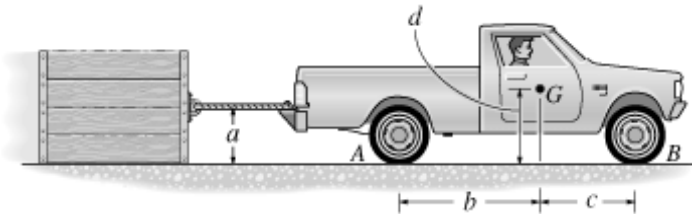
Units Used:

$\text{kN} = 10^3\ \text{N}$

$\text{Mg} = 1000\ \text{kg}$

Given:

$\theta = 10\ \text{deg}$



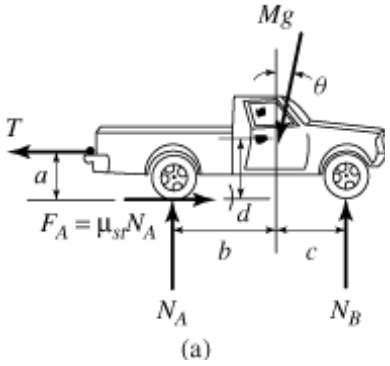
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$M = 1.25\ \text{Mg}$ $a = 600\ \text{mm}$

$\mu_{st} = 0.5$ $b = 1.5\ \text{m}$

$\mu_{sc} = 0.4$ $c = 1\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $d = 800\ \text{mm}$



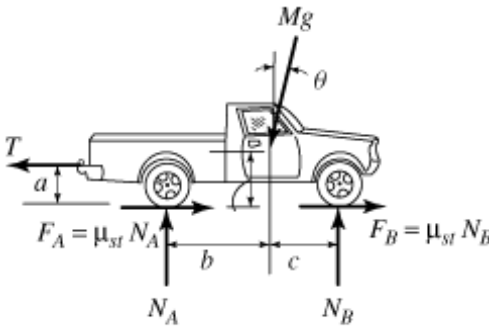
(a)

Solution:

Guesses $N_A = 1\ \text{N}$ $N_B = 1\ \text{N}$

$T = 1\ \text{N}$ $N_C = 1\ \text{N}$ $W = 1\ \text{N}$

(a) Rear wheel drive



(b)

Given

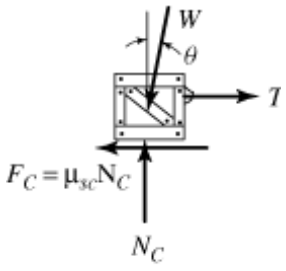
$$-T + \mu_{st} N_A - M g \sin(\theta) = 0$$

$$N_A + N_B - M g \cos(\theta) = 0$$

$$-M g b \cos(\theta) + M g d \sin(\theta) + N_B(b + c) + T a = 0$$

$$T - \mu_{sc} N_C - W \sin(\theta) = 0$$

$$N_C - W \cos(\theta) = 0$$



(c)

$$\begin{pmatrix} N_A \\ N_B \\ T \\ N_C \\ W \end{pmatrix} = \text{Find}(N_A, N_B, T, N_C, W) \qquad W = 1.25\ \text{kN}$$

(b) Four wheel drive

Given

$$-T + \mu_{st} N_A + \mu_{st} N_B - M g \sin(\theta) = 0$$

$$N_A + N_B - M g \cos(\theta) = 0$$

$$-M g b \cos(\theta) + M g d \sin(\theta) + N_B(b + c) + T a = 0$$

$$T - \mu_{sc} N_C - W \sin(\theta) = 0$$

$$N_C - W \cos(\theta) = 0$$

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$$\begin{pmatrix} N_A \\ N_B \\ T \\ N_C \\ W \end{pmatrix} = \text{Find}(N_A, N_B, T, N_C, W) \qquad W = 6.89\,\text{kN}$$

Problem 8-137

A roofer, having a mass M, walks slowly in an upright position down along the surface of a dome that has a radius of curvature ρ. If the coefficient of static friction between his shoes and the dome is μ_s determine the angle θ at which he first begins to slip.

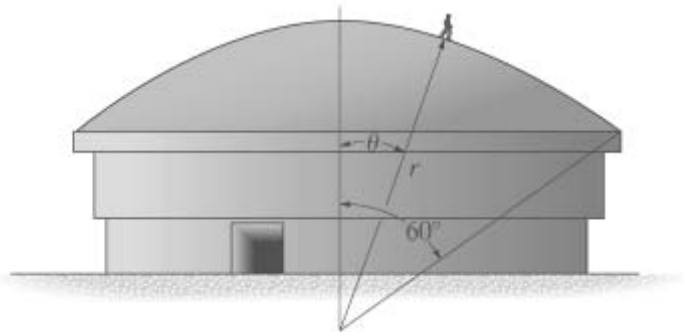
Given:

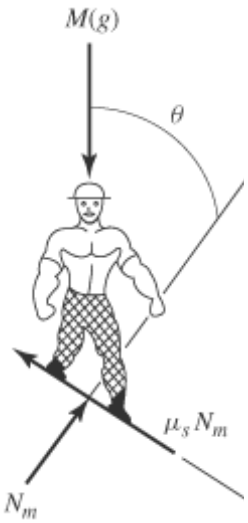
$M = 70\ \text{kg}$

$\rho = 20\ \text{m}$

$\mu_s = 0.7$

Solution:





$\Sigma F_{y'} = 0; \qquad N_m - M g \cos(\theta) = 0$

$\Sigma F_{x'} = 0; \qquad M g \sin(\theta) - \mu_s N_m = 0$

$\mu_s = \tan(\theta)$

$\theta = \text{atan}(\mu_s)$

$\theta = 35.0\,\text{deg}$

Problem 8-138

A man attempts to lift the uniform ladder of weight W to an upright position by applying a force **P** perpendicular to the ladder at rung R. Determine the coefficient of static friction between the ladder and the ground at A if the ladder begins to slip on the ground when his

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hands reach height *c*.

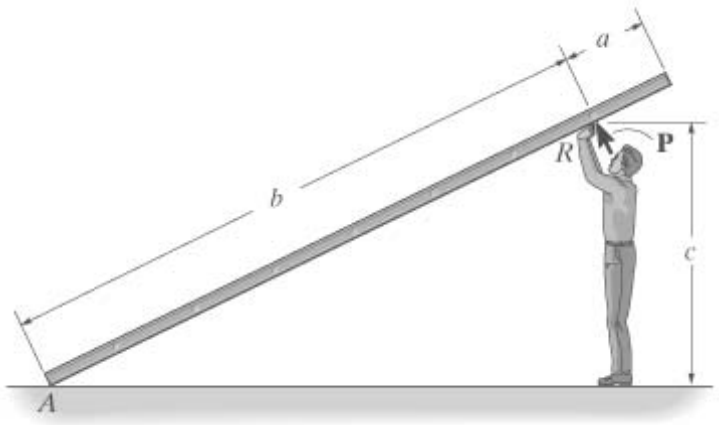
Given:

$a = 2$ ft

$b = 14$ ft

$c = 6$ ft

$W = 40$ lb

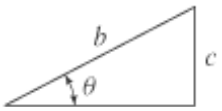


Solution:

$$\theta = \operatorname{asin}\left(\frac{c}{b}\right)$$

Initial guesses

$P = 10$ lb $\quad N_A = 100$ lb $\quad \mu_A = 100$



Given

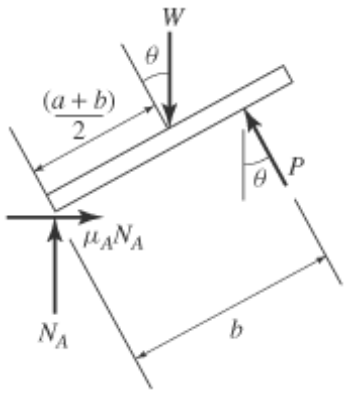
$$\Sigma F_x = 0; \quad \mu_A N_A - P\sin(\theta) = 0$$

$$\Sigma F_y = 0; \quad N_A - W + P\cos(\theta) = 0$$

$$\Sigma M_A = 0; \quad -W\left(\frac{b+a}{2}\right)\cos(\theta) + P\,b = 0$$

$$\begin{pmatrix} P \\ N_A \\ \mu_A \end{pmatrix} = \text{Find}(P, N_A, \mu_A) \qquad \begin{pmatrix} P \\ N_A \end{pmatrix} = \begin{pmatrix} 20.7 \\ 21.3 \end{pmatrix} \text{lb}$$

$\mu_A = 0.41$



Problem 8-139

Column *D* is subjected to a vertical load *W*. It is supported on two identical wedges *A* and *B* for which the coefficient of static friction at the contacting surfaces between *A* and *B* and between *B* and *C* is μ_s. Determine the force **P** needed to raise the column and the equilibrium force **P'** needed to hold wedge *A* stationary. The contacting surface between *A* and *D* is smooth.

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Units Used:

$\text{kip} = 10^3 \text{ lb}$

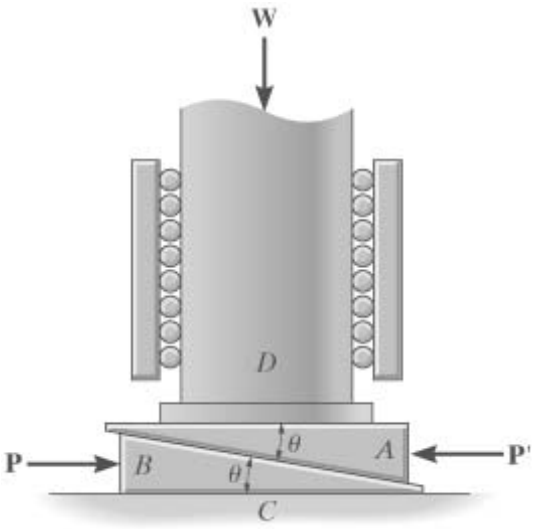
Given:

$W = 8000 \text{ lb}$

$\theta = 10 \text{ deg}$

$\mu_s = 0.4$

Solution:

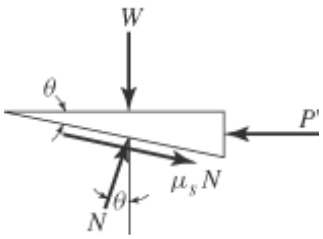


wedge *A*:

$\Sigma F_y = 0; \quad N\cos(\theta) - \mu_s N\sin(\theta) - W = 0$

$$N = \frac{W}{\cos(\theta) - \mu_s \sin(\theta)}$$

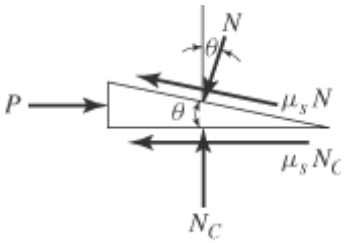
$N = 8739.8 \text{ lb}$



$\Sigma F_x = 0; \quad \mu_s N\cos(\theta) + N\sin(\theta) - P' = 0$

$P' = \mu_s N \cos(\theta) + N \sin(\theta)$

$P' = 4.96 \text{ kip}$



Wedge *B*:

$\Sigma F_y = 0; \quad N_C + \mu_s N\sin(\theta) - N\cos(\theta) = 0$

$N_C = -\mu_s N \sin(\theta) + N \cos(\theta)$

$N_C = 8000 \text{ lb}$

$\Sigma F_x = 0; \quad P - \mu_s N_C - N\sin(\theta) - \mu_s N\cos(\theta) = 0$

$P = \mu_s N_C + N \sin(\theta) + \mu_s N \cos(\theta)$

$P = 8.16 \text{ kip}$

Problem 8-140

Column *D* is subjected to a vertical load *W*. It is supported on two identical wedges *A* and *B* for which the coefficient of static friction at the contacting surfaces between *A* and *B* and between *B* and *C* is μ_s. If the forces **P** and **P'** are removed, are the wedges self-locking? The contacting surface between *A* and *D* is smooth.

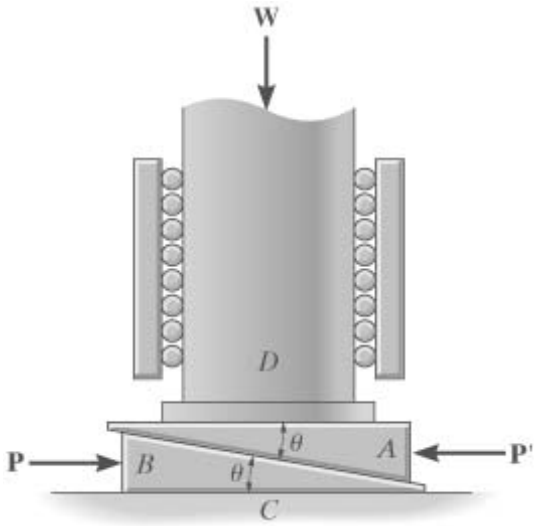
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Given:

$W = 8000$ lb

$\theta = 10$ deg

$\mu_S = 0.4$



Solution:

Wedge *A*:

$\Sigma F_y = 0;$ $N - W\cos(\theta) = 0$

$N = W\cos(\theta)$

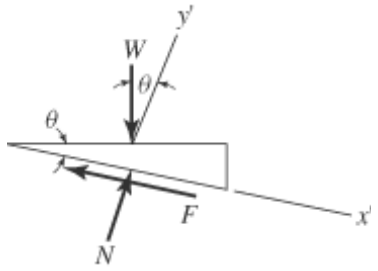
$N = 7878.5\,\text{lb}$

$\Sigma F_x = 0;$ $W\sin(\theta) - F = 0$

$F = W\sin(\theta)$

$F = 1389.2\,\text{lb}$

Friction $F_{max} = \mu_S N$



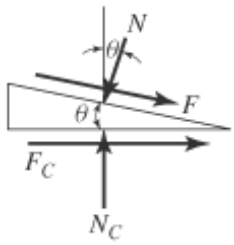
Since $F = 1389\,\text{lb} < F_{max} = 3151\,\text{lb}$ then the wedges do not slip at the contact surface *AB*.

Wedge *B*:

$\Sigma F_y = 0;$ $N_C - F\sin(\theta) - N\cos(\theta) = 0$

$N_C = F\sin(\theta) + N\cos(\theta)$

$N_C = 8000\,\text{lb}$



$\Sigma F_x = 0;$ $F_C + F\cos(\theta) - N\sin(\theta) = 0$

$F_C = -F\cos(\theta) + N\sin(\theta)$

$F_C = 0\,\text{lb}$

Friction $F_{Cmax} = \mu_S N_C$

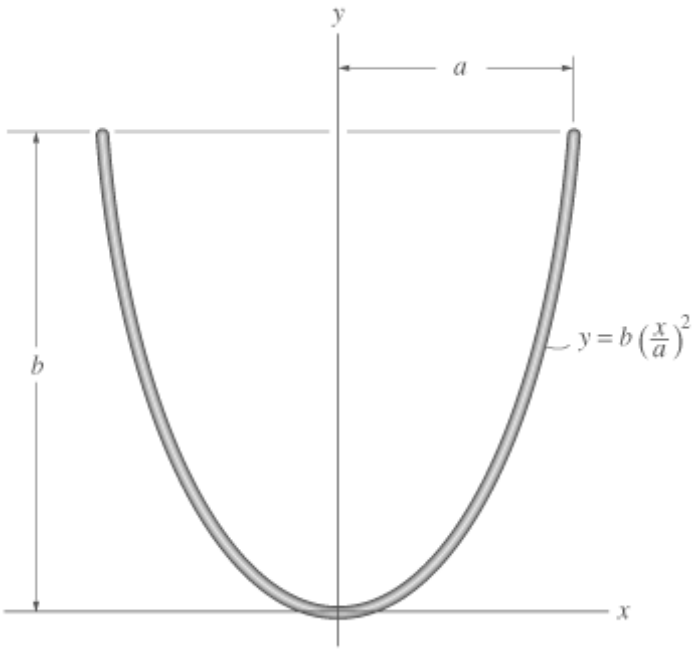
Since $F_C = 0\,\text{lb} < F_{Cmax} = 3200\,\text{lb}$ then the wedges do not slip at the contact surface BC.

Therefore the wedges are self-locking.

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Problem 9-1

Locate the center of mass of the homogeneous rod bent in the form of a parabola.



Given:

$a = 1$ m

$b = 2$ m

Solution:

$$y = b\left(\frac{x}{a}\right)^2$$

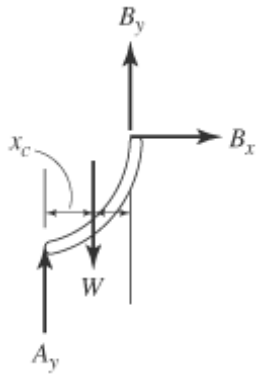
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2b}{a^2}x$$

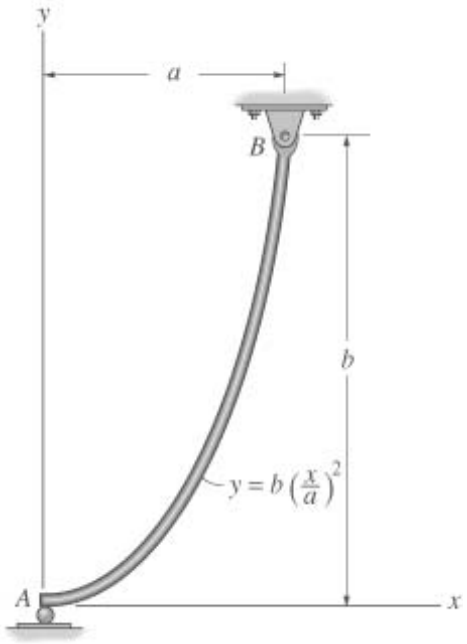
$$y_C = \frac{\displaystyle\int_0^a b\left(\frac{x}{a}\right)^2 \sqrt{1 + \left(\frac{2b}{a^2}x\right)^2}\,\mathrm{d}x}{\displaystyle\int_0^a \sqrt{1 + \left(\frac{2b}{a^2}x\right)^2}\,\mathrm{d}x}$$

$y_C = 0.912$ m $\quad$ $x_C = 0$ m

Problem 9-2

Locate the center of gravity x_c of the homogeneous rod. If the rod has a weight per unit length γ, determine the vertical reaction at A and the x and y components of reaction at the pin B.





Given:

$\gamma = 0.5 \dfrac{\text{lb}}{\text{ft}}$

$a = 1$ ft

$b = 2$ ft

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Solution:

$$y = b\left(\frac{x}{a}\right)^2 \qquad \frac{\mathrm{d}\,y}{\mathrm{d}\,x} = \frac{2\,b}{a^2}\,x$$

$$L = \int_0^a \sqrt{1 + \left(\frac{2\,b}{a^2}\,x\right)^2}\,\mathrm{d}x \qquad W = \gamma\,L \qquad W = 1.162\,\mathrm{lb}$$

$$x_C = \frac{1}{L}\int_0^a x\sqrt{1 + \left(\frac{2\,b}{a^2}\,x\right)^2}\,\mathrm{d}x \qquad x_C = 0.620\,\mathrm{ft}$$

Guesses $A_y = 1\mathrm{lb}$ $B_x = 1\mathrm{lb}$ $B_y = 1\mathrm{lb}$

Given $B_x = 0$ $A_y + B_y - W = 0$ $-A_y\,a + W\left(a - x_C\right) = 0$

$$\begin{pmatrix} B_x \\ B_y \\ A_y \end{pmatrix} = \mathrm{Find}\left(B_x, B_y, A_y\right) \qquad \begin{pmatrix} B_x \\ B_y \\ A_y \end{pmatrix} = \begin{pmatrix} 0.000 \\ 0.720 \\ 0.442 \end{pmatrix}\mathrm{lb}$$

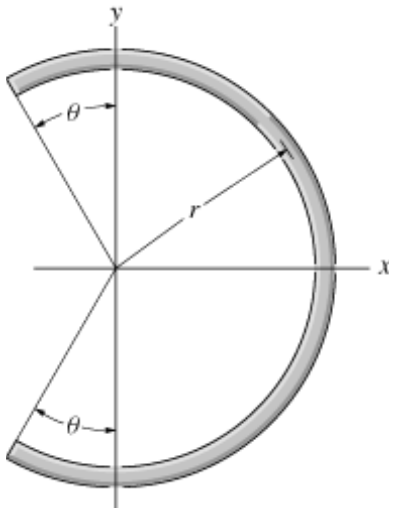
Problem 9-3

Locate the center of mass of the homogeneous rod bent into the shape of a circular arc.

Given:

$r = 300\ \mathrm{mm}$

$\theta = 30\ \mathrm{deg}$



Solution:

$y_C = 0$ Symmetry

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$$x_C = \frac{\int_{\frac{-\pi}{2}-\theta}^{\frac{\pi}{2}+\theta} r\cos(\alpha)\, r\, d\alpha}{\int_{\frac{-\pi}{2}-\theta}^{\frac{\pi}{2}+\theta} r\, d\alpha} \qquad x_C = 124.049\,\text{mm}$$

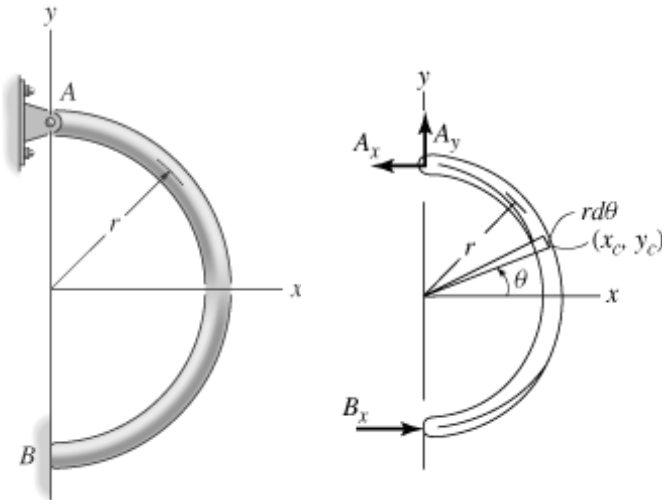
Problem 9-4

Locate the center of gravity x_c of the homogeneous rod bent in the form of a semicircular arc. The rod has a weight per unit length γ. Also, determine the horizontal reaction at the smooth support B and the x and y components of reaction at the pin A.

Given:

$$\gamma = 0.5\ \frac{\text{lb}}{\text{ft}}$$

$$r = 2\ \text{ft}$$



Solution:

$$x_C = \frac{\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} r\cos(\theta) r\, d\theta}{\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} r\, d\theta} \qquad x_C = 1.273\,\text{ft}$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad -\pi\, r\, \gamma\, x_C + B_x\,(2\, r) = 0 \qquad B_x = \frac{\pi\, r\gamma x_C}{2r} \qquad B_x = 1\,\text{lb}$$

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$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $\quad -A_x + B_x = 0$ $\quad A_x = B_x$ $\quad A_x = 1\,\text{lb}$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - \pi r\gamma = 0$ $\quad A_y = \pi r\gamma$ $\quad A_y = 3.14\,\text{lb}$

Problem 9-5

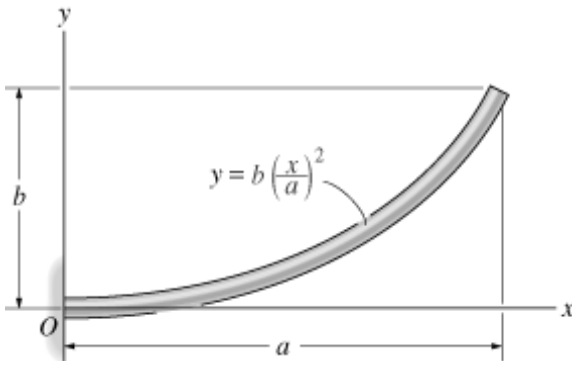
Determine the distance x_c to the center of gravity of the homogeneous rod bent into the parabolic shape. If the rod has a weight per unit length γ determine the reactions at the fixed support O.

Given:

$\gamma = 0.5\ \frac{\text{lb}}{\text{ft}}$

$a = 1\ \text{ft}$

$b = 0.5\ \text{ft}$



Solution:

$$y = b\left(\frac{x}{a}\right)^2$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2\,b\,x}{a^2}$$

$$L = \int_0^a \sqrt{1 + \left(\frac{2\,b\,x}{a^2}\right)^2}\,\mathrm{d}x \qquad L = 1.148\,\text{ft}$$

$$x_C = \frac{1}{L}\int_0^a x\sqrt{1 + \left(\frac{2\,b\,x}{a^2}\right)^2}\,\mathrm{d}x \qquad x_C = 0.531\,\text{ft}$$

$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $\quad O_x = 0$ $\quad O_x = 0\ \text{lb}$ $\quad O_x = 0\,\text{lb}$

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$+\uparrow \Sigma F_y = 0;$ $\quad O_y - \gamma L = 0$ $\quad O_y = \gamma L$ $\quad O_y = 0.574\,\text{lb}$

$\Sigma M_O = 0;$ $\quad M_O - \gamma L\, x_C = 0$ $\quad M_O = \gamma L\, x_C$ $\quad M_O = 0.305\,\text{lb}\cdot\text{ft}$

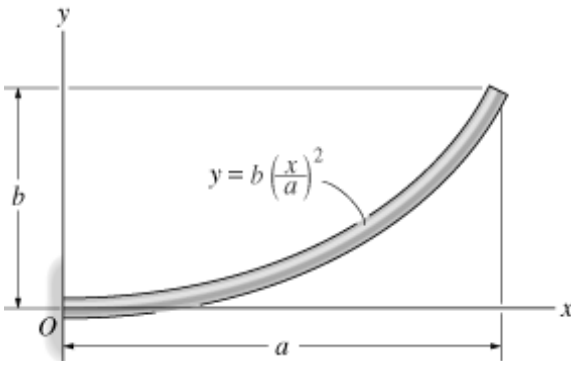
Problem 9-6

Determine the distance y_c to the center of gravity of the homogeneous rod bent into the parabolic shape.

Given:

$a = 1$ ft

$b = 0.5$ ft



Solution:

$$y = b\left(\frac{x}{a}\right)^2$$

$$\frac{dy}{dx} = \frac{2bx}{a^2}$$

$$L = \int_0^a \sqrt{1 + \left(\frac{2bx}{a^2}\right)^2}\, dx \qquad L = 1.148\,\text{ft}$$

$$y_C = \frac{1}{L}\left[\int_0^a b\left(\frac{x}{a}\right)^2 \sqrt{1 + \left(\frac{2bx}{a^2}\right)^2}\, dx\right] \qquad y_C = 0.183\,\text{ft}$$

Problem 9-7

Locate the centroid of the parabolic area.

Solution: $\quad a = \dfrac{h}{b^2}$

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$$dA = x\,dy$$

$$x_C = \frac{x}{2}$$

$$y_C = y$$

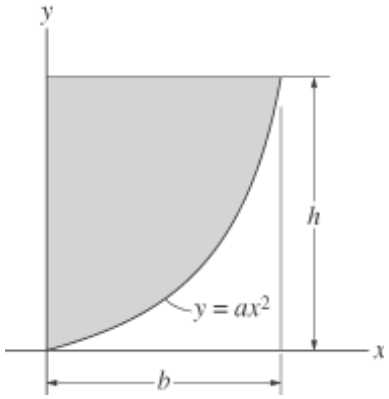
$$A = \int_0^h b\sqrt{\frac{y}{h}}\,dy = h\,b\left(\frac{y}{h}\right)^{\frac{1}{2}}$$

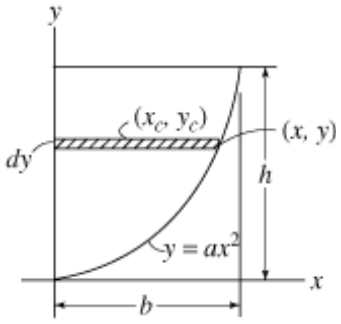
$$x_C = \frac{3}{2hb}\int_0^h \frac{1}{2}\left(b\sqrt{\frac{y}{h}}\right)^2 dy = \frac{3}{8\,h^2}\,b\,h^2$$

$$x_C = \frac{3}{8}b$$

$$y_C = \frac{3}{2hb}\int_0^h y\,b\sqrt{\frac{y}{h}}\,dy = \frac{3}{5}\,h\left(\frac{h}{h}\right)^{\frac{5}{2}}$$

$$y_C = \frac{3}{5}h$$





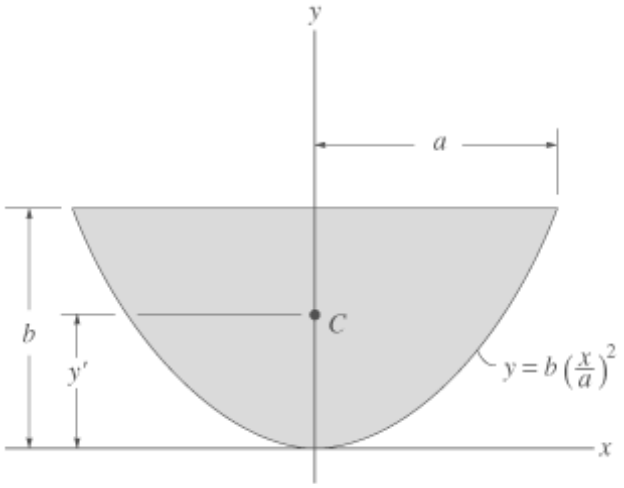
Problem 9-8

Locate the centroid y_c of the shaded area.

Given:

$a = 100$ mm

$b = 100$ mm



Solution:

$$y = b\left(\frac{x}{a}\right)^2$$

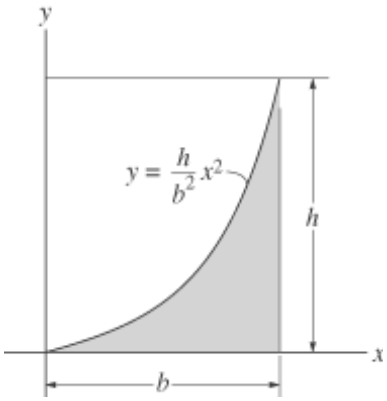
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$$x = a\sqrt{\frac{y}{b}}$$

$$y_C = \frac{\displaystyle\int_0^b y 2a\sqrt{\frac{y}{b}}\,\mathrm{d}y}{\displaystyle\int_0^b 2a\sqrt{\frac{y}{b}}\,\mathrm{d}y} \qquad y_C = 60\,\mathrm{mm}$$

Problem 9-9

Locate the centroid x_c of the shaded area.



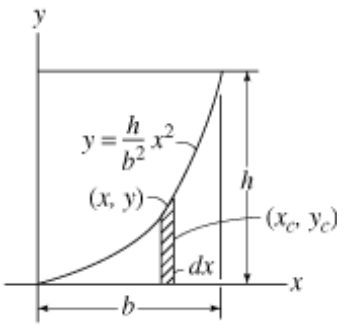
Solution:

$\mathrm{d}A = y\mathrm{d}x$

$x_C = x$

$y_C = \frac{y}{2}$

$$x_C = \frac{\displaystyle\int_0^b x\frac{h}{b^2}x^2\,\mathrm{d}x}{\displaystyle\int_0^b h\frac{x^2}{b^2}\,\mathrm{d}x} = \frac{3}{4}\,\frac{b^4}{b^3} \qquad x_C = \frac{3}{4}b$$

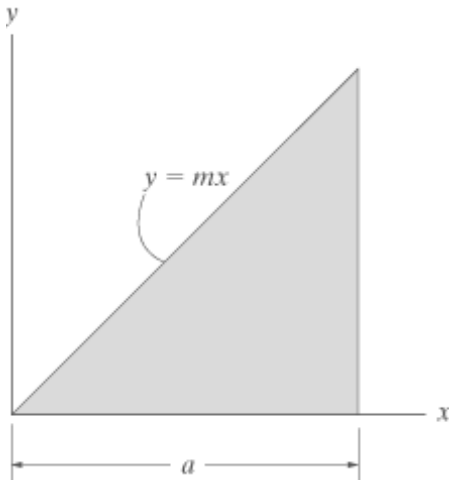


$$y_C = \frac{\displaystyle\int_0^b \frac{1}{2}\left(\frac{h}{b^2}x^2\right)^2\mathrm{d}x}{\displaystyle\int_0^b \frac{h}{b^2}x^2\,\mathrm{d}x} = \frac{3}{10}\,b^5\,\frac{h^2}{b^5\,h} \qquad y_C = \frac{3}{10}h$$

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Problem 9-10

Determine the location (x_c, y_c) of the centroid of the triangular area.



Solution:

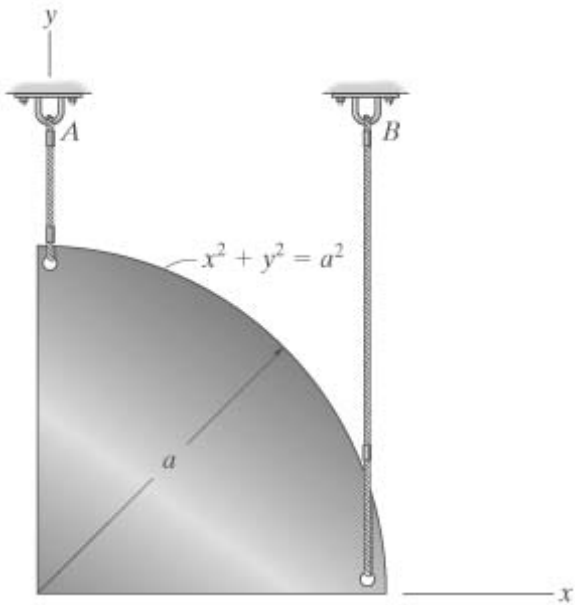
$$A = \int_0^a mx\,dx = \frac{1}{2}\,a^2 m$$

$$x_C = \frac{2}{ma^2}\int_0^a x\,mx\,dx = \frac{2}{3}\,a \qquad x_C = \frac{2}{3}a$$

$$y_C = \frac{2}{ma^2}\int_0^a \frac{1}{2}(mx)^2\,dx = \frac{1}{3}\,a\,m \qquad y_C = \frac{m}{3}a$$

Problem 9-11

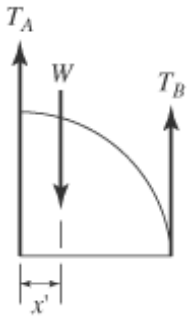
Determine the location (x_c, y_c) of the center of gravity of the quartercircular plate. Also determine the force in each of the supporting wires.The plate has a weight per unit area of γ.



Given:

$$\gamma = 5\ \frac{\text{lb}}{\text{ft}^2}$$

$$a = 4\ \text{ft}$$



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Solution:

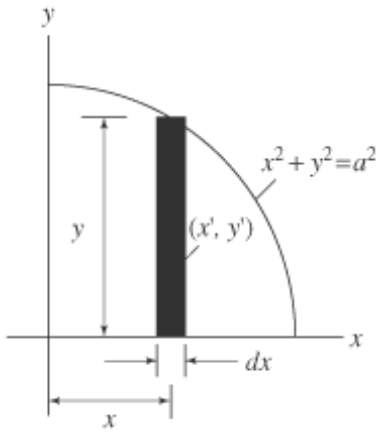
$$x^2 + y^2 = a^2$$

$$y = \sqrt{a^2 - x^2}$$

$$A = \pi\frac{a^2}{4} \qquad W = A\gamma \qquad W = 62.832\,\text{lb}$$

$$x_C = \frac{1}{A}\int_0^a x\sqrt{a^2 - x^2}\,dx \qquad x_C = 1.698\,\text{ft}$$

$$y_C = \frac{1}{A}\int_0^a \frac{1}{2}\left(a^2 - x^2\right)dx \qquad y_C = 1.698\,\text{ft}$$



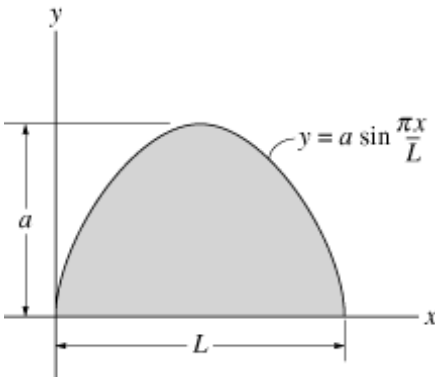
Guesses $T_A = 1\text{lb}$ $T_B = 1\text{lb}$

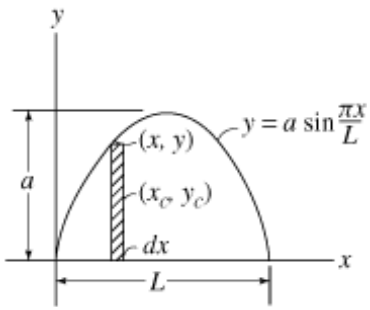
Given $T_A + T_B - W = 0$ $T_B a - W x_C = 0$

$$\begin{pmatrix} T_A \\ T_B \end{pmatrix} = \text{Find}\left(T_A, T_B\right) \qquad \begin{pmatrix} T_A \\ T_B \end{pmatrix} = \begin{pmatrix} 36.2 \\ 26.7 \end{pmatrix}\text{lb}$$

Problem *9-12

Locate the centroid of the shaded area.





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Solution:

$$dA = y\,dx \qquad x_C = x \qquad y_C = \frac{y}{2}$$

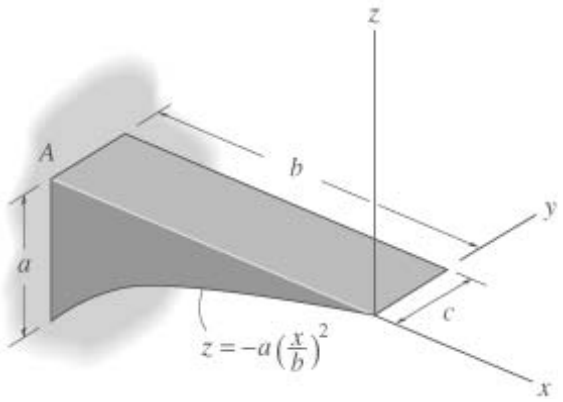
$$A = \int_0^L a \sin\left(\frac{\pi x}{L}\right) dx = \frac{2}{\pi} L\, a$$

$$x_C = \frac{\pi}{2L\,a} \int_0^L x\, a \sin\left(\frac{\pi x}{L}\right) dx = \frac{1}{2}\, L \qquad x_C = \frac{1}{2}L$$

$$y_C = \frac{\pi}{2L\,a} \int_0^L \frac{1}{2}\left(a \sin\left(\frac{\pi x}{L}\right)\right)^2 dx = \frac{1}{8}\, \pi a \qquad y_C = \frac{1}{8}\pi a$$

Problem 9-13

Locate the center of gravity of the homogeneous cantilever beam and determine the reactions at the fixed support.The material has a density of ρ.



Units Used: $\text{Mg} = 10^3\ \text{kg}$ $\text{kN} = 10^3\ \text{N}$

Given: $\rho = 8\ \frac{\text{Mg}}{\text{m}^3}$ $a = 1\ \text{m}$

$b = 4\ \text{m}$

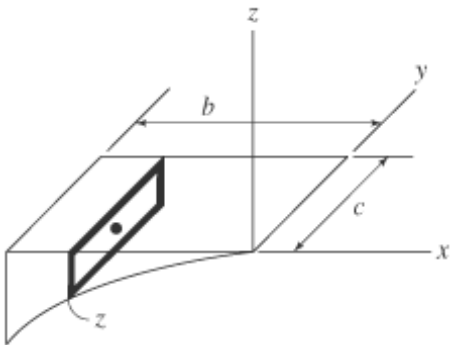
$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $c = 0.5\ \text{m}$

Solution:

$$V = \int_{-b}^0 c\, a \left(\frac{x}{b}\right)^2 dx \qquad W = \rho g V$$

$$x_C = \frac{1}{V} \int_{-b}^0 x\, c\, a \left(\frac{x}{b}\right)^2 dx$$

$$y_C = \frac{1}{V} \int_{-b}^0 \frac{c}{2} c\, a \left(\frac{x}{b}\right)^2 dx$$



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$$z_C = \frac{1}{V}\int_{-b}^{0} \frac{-c}{2}\left[a\left(\frac{x}{b}\right)^2\right]^2 \, dx \qquad \begin{pmatrix} x_C \\ y_C \\ z_C \end{pmatrix} = \begin{pmatrix} -3.00 \\ 0.25 \\ -0.30 \end{pmatrix} \text{m}$$

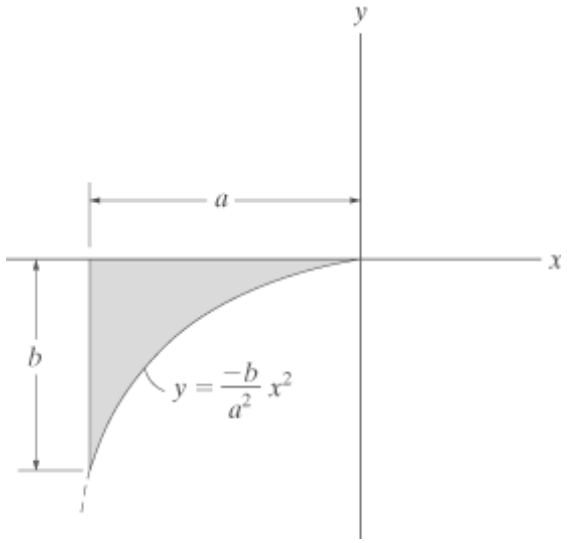
Guesses $A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $A_z = 1\ \text{N}$ $M_A = 1\ \text{N}\cdot\text{m}$

Given $A_x = 0$ $A_y = 0$ $A_z - W = 0$ $M_A - W(b + x_C) = 0$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \\ M_A \end{pmatrix} = \text{Find}(A_x, A_y, A_z, M_A) \qquad \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 0.00 \\ 0.00 \\ 52.32 \end{pmatrix} \text{kN} \qquad M_A = 52.32\ \text{kN}\cdot\text{m}$$

Problem 9-14

Locate the centroid (x_c, y_c) of the exparabolic segment of area.



Solution:

$$A = \int_{-a}^{0} \frac{b}{a^2}x^2 \, dx = \frac{1}{3}\,a\,b$$

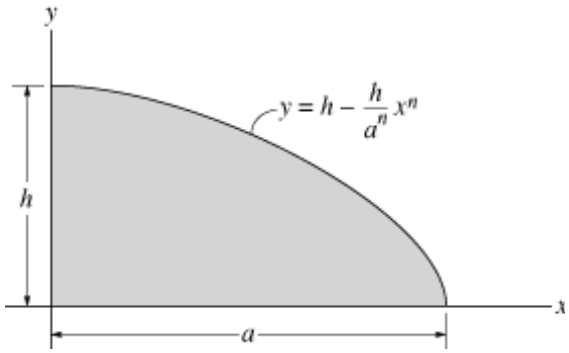
$$x_C = \frac{3}{a\,b}\int_{-a}^{0} x\frac{b}{a^2}x^2 \, dx = \frac{-3}{4}\,a \qquad x_C = \frac{-3}{4}a$$

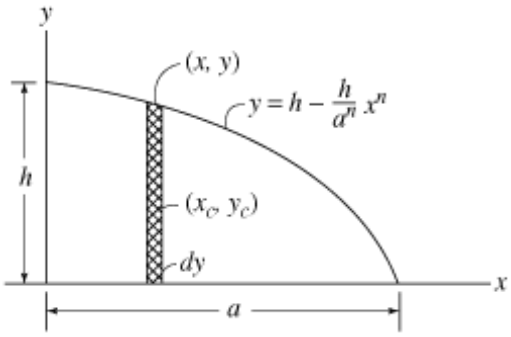
$$y_C = \frac{3}{a\,b}\int_{-a}^{0} -\frac{1}{2}\left(\frac{b}{a^2}x^2\right)^2 dx = \frac{-3}{10}b \qquad y_C = \frac{-3}{10}b$$

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Problem 9-15

Locate the centroid of the shaded area.





Solution:

$$x_C = \frac{\displaystyle\int_0^a x\left(h - \frac{h}{a^n}x^n\right)\,dx}{\displaystyle\int_0^a \left(h - \frac{h}{a^n}x^n\right)\,dx}$$

$$x_C = \frac{n+1}{2(n+2)}a \quad \text{provided that } n \neq -2$$

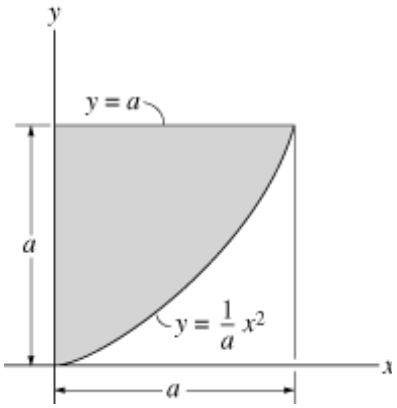
$$y_C = \frac{\dfrac{1}{2}\displaystyle\int_0^a \left(h - \frac{h}{a^n}x^n\right)^2\,dx}{\displaystyle\int_0^a \left(h - \frac{h}{a^n}x^n\right)\,dx}$$

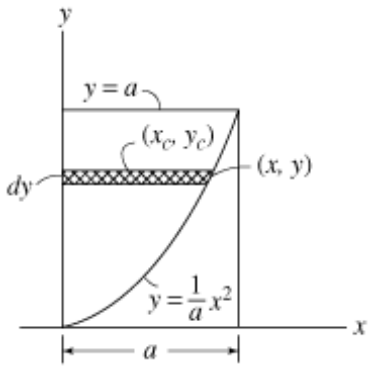
$$y_C = \frac{n}{2n+1}h \quad \text{provided that } n \neq \frac{-1}{2}$$

Problem *9-16

Locate the centroid of the shaded area bounded by the parabola and the line $y = a$.

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Solution:

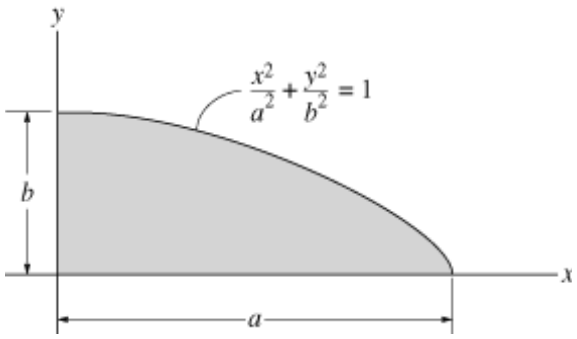
$$A = \int_0^a \sqrt{a\,y}\,\mathrm{d}y = \frac{2}{3}\,\frac{\left(a^2\right)^{\frac{3}{2}}}{a} \qquad A = \frac{2a^2}{3}$$

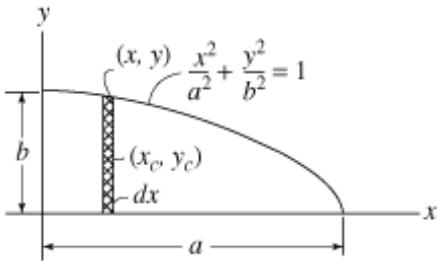
$$x_C = \frac{3}{2a^2}\int_0^a \frac{1}{2}a\,y\,\mathrm{d}y = \frac{3}{8}\,a \qquad x_C = \frac{3}{8}a$$

$$y_C = \frac{3}{2a^2}\int_0^a y\sqrt{a\,y}\,\mathrm{d}y = \frac{3}{5\,a^4}\left(a^2\right)^{\frac{5}{2}} \qquad y_C = \frac{3}{5}a$$

Problem 9-17

Locate the centroid of the quarter elliptical area.





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Solution:

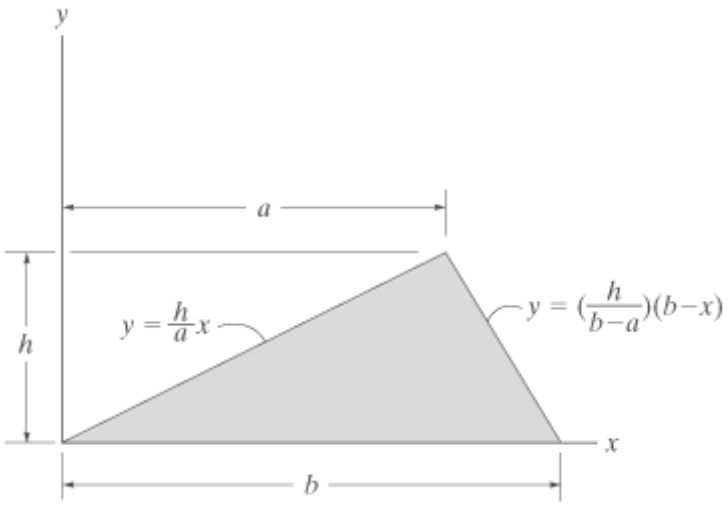
$$A = \int_0^a b\sqrt{1-\left(\frac{x}{a}\right)^2}\,dx \qquad A = \frac{\pi a b}{4}$$

$$x_C = \frac{4}{\pi a b}\int_0^a x b\sqrt{1-\left(\frac{x}{a}\right)^2}\,dx = \frac{4}{3\ \pi}\ a \qquad x_C = \frac{4}{3\pi}a$$

$$y_C = \frac{4}{\pi a b}\int_0^a \frac{1}{2}\left[b\sqrt{1-\left(\frac{x}{a}\right)^2}\right]^2 dx = \frac{4}{3\ \pi}\ b \qquad y_C = \frac{4}{3\pi}b$$

Problem 9-18

Locate the centroid x_c of the triangular area.



Solution:

$$A = \frac{b\,h}{2}$$

$$x_C = \frac{2}{b\,h}\left[\int_0^a x\frac{h}{a}x\,dx + \int_a^b x\frac{h}{b-a}(b-x)\,dx\right] \qquad x_C = \frac{a+b}{3}$$

Problem 9-19

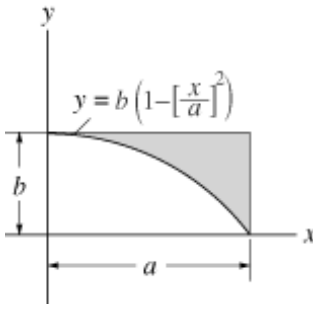
Locate the centroid of the shaded area.

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Given:

$a = 8$ m

$b = 4$ m



Solution:

$$A = \int_0^a \left[b - b\left[1 - \left(\frac{x}{a}\right)^2\right]\right] dx \qquad A = 10.667 \text{ m}^2$$

$$x_C = \frac{1}{A}\int_0^a x\left[b - b\left[1 - \left(\frac{x}{a}\right)^2\right]\right] dx \qquad x_C = 6 \text{ m}$$

$$y_C = \frac{1}{A}\int_0^a \frac{1}{2}\left[b + b\left[1 - \left(\frac{x}{a}\right)^2\right]\right]\left[b - b\left[1 - \left(\frac{x}{a}\right)^2\right]\right] dx \qquad y_C = 2.8 \text{ m}$$

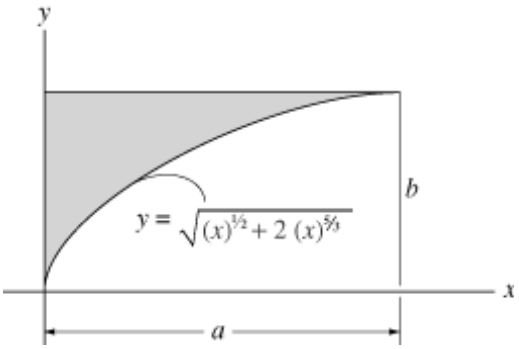
Problem 9-20

Locate the centroid x_c of the shaded area. Solve the problem by evaluating the integrals using Simpson's rule.

Given:

$a = 2$ ft

$b = \sqrt{a^{\frac{1}{2}} + 2a^{\frac{5}{3}}}$



Solution:

$$A = \int_0^a \left(b - \sqrt{x^{\frac{1}{2}} + 2x^{\frac{5}{3}}}\right) dx \qquad A = 2.177 \text{ ft}^2$$

$$x_C = \frac{1}{A}\int_0^a x\left(b - \sqrt{x^{\frac{1}{2}} + 2x^{\frac{5}{3}}}\right) dx \qquad x_C = 0.649 \text{ ft}$$

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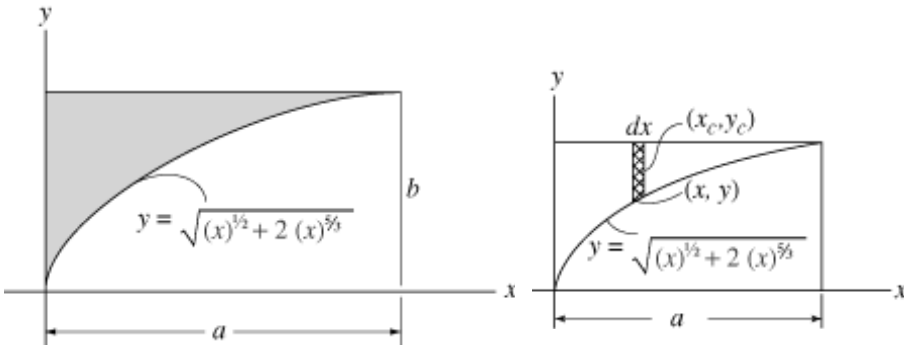
Problem 9-21

Locate the centroid y_c of the shaded area. Solve the problem by evaluating the integrals using Simpson's rule.

Given:

$a = 2$ ft

$b = \sqrt{a^{\frac{1}{2}} + 2a^{\frac{5}{3}}}$



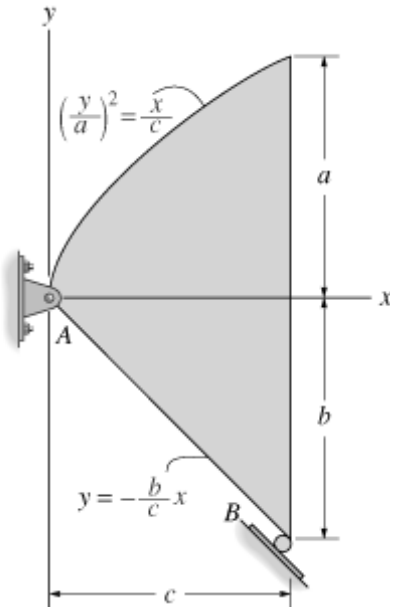
Solution:

$$A = \int_0^a \left(b - \sqrt{x^{\frac{1}{2}} + 2x^{\frac{5}{3}}}\right) dx \qquad A = 2.177 \text{ ft}^2$$

$$y_C = \frac{1}{A}\int_0^a \frac{1}{2}\left(b + \sqrt{x^{\frac{1}{2}} + 2x^{\frac{5}{3}}}\right)\left(b - \sqrt{x^{\frac{1}{2}} + 2x^{\frac{5}{3}}}\right) dx \qquad y_C = 2.04 \text{ ft}$$

Problem 9-22

The steel plate has thickness t and density ρ. Determine the location of its center of mass. Also compute the reactions at the pin and roller support.



Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$t = 0.3$ m $\quad a = 2$ m

$\rho = 7850 \frac{\text{kg}}{\text{m}^3} \quad b = 2$ m

$c = 2$ m

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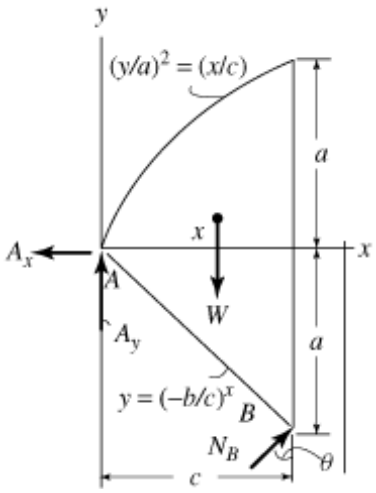
$g = 9.81 \frac{\text{m}}{\text{s}^2}$ $\theta = \text{atan}\left(\frac{b}{c}\right)$

Solution:

$$A = \int_0^c \left(a\sqrt{\frac{x}{c}} + \frac{bx}{c}\right) dx \qquad A = 4.667\ \text{m}^2$$

$$x_c = \frac{1}{A}\int_0^c x\left(a\sqrt{\frac{x}{c}} + \frac{bx}{c}\right) dx \qquad x_c = 1.257\ \text{m}$$

$$y_c = \frac{1}{A}\int_0^c \frac{1}{2}\left(a\sqrt{\frac{x}{c}} + \frac{bx}{c}\right)\left(a\sqrt{\frac{x}{c}} - \frac{bx}{c}\right) dx \qquad y_c = 0.143\ \text{m}$$



Equilibrium $W = \rho A t g$

Guesses $A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $N_B = 1\ \text{N}$

Given $A_y - W + N_B\cos(\theta) = 0$ $-A_x + N_B\sin(\theta) = 0$ $N_B\sqrt{b^2 + c^2} - W x_c = 0$

$$\begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \text{Find}(A_x, A_y, N_B) \qquad \begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \begin{pmatrix} 33.9 \\ 73.9 \\ 47.9 \end{pmatrix} \text{kN}$$

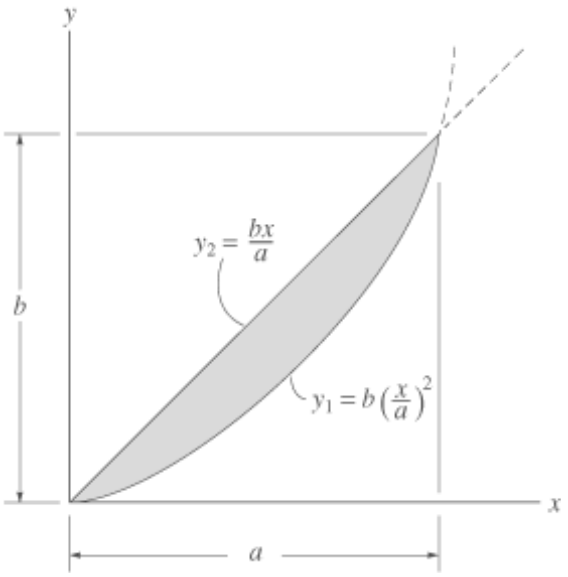
Problem 9-23

Locate the centroid x_c of the shaded area.

Given:

$a = 4\ \text{ft}$

$b = 4\ \text{ft}$



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Solution:

$$A = \int_0^a \left[\frac{b\,x}{a} - b\left(\frac{x}{a}\right)^2\right] \mathrm{d}x$$

$$x_C = \frac{1}{A}\int_0^a x\left[\frac{b\,x}{a} - b\left(\frac{x}{a}\right)^2\right] \mathrm{d}x \qquad x_C = 2.00\ \text{ft}$$

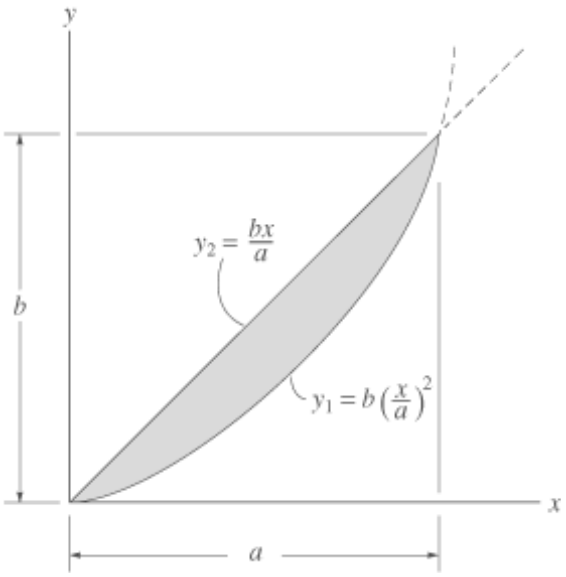
Problem 9-24

Locate the centroid y_c of the shaded area.

Given:

$a = 4$ ft

$b = 4$ ft



Solution:

$$A = \int_0^a \left[\frac{b\,x}{a} - b\left(\frac{x}{a}\right)^2\right] \mathrm{d}x$$

$$y_C = \frac{1}{A}\int_0^a \frac{1}{2}\left[b\frac{x}{a} + b\left(\frac{x}{a}\right)^2\right]\left[\frac{b\,x}{a} - b\left(\frac{x}{a}\right)^2\right] \mathrm{d}x \qquad y_C = 1.60\ \text{ft}$$

Problem 9-25

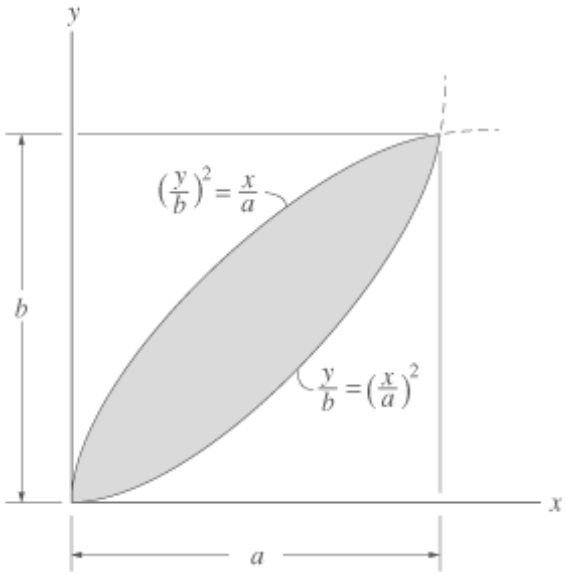
Locate the centroid x_c of the shaded area.

Given:

$a = 4$ m

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$b = 4$ m



Solution:

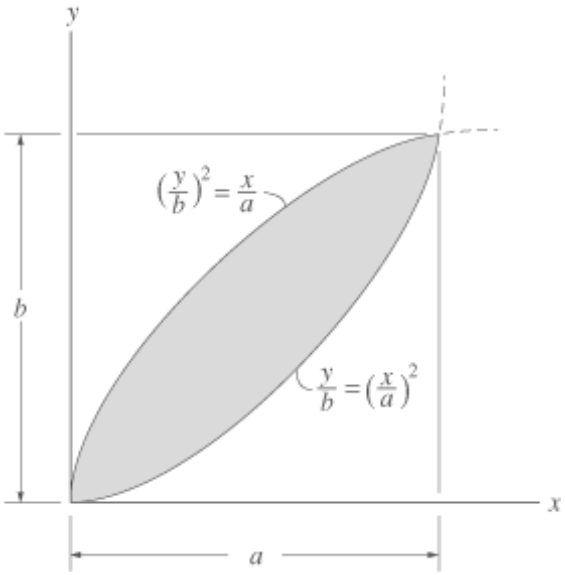
$$A = \int_0^a \left[b\sqrt{\frac{x}{a}} - b\left(\frac{x}{a}\right)^2 \right] dx$$

$$x_C = \frac{1}{A}\int_0^a x\left[b\sqrt{\frac{x}{a}} - b\left(\frac{x}{a}\right)^2 \right] dx$$

$x_C = 1.80$ m

Problem 9-26

Locate the centroid y_c of the shaded area.



Given:

$a = 4$ m

$b = 4$ m

Solution:

$$A = \int_0^a \left[b\sqrt{\frac{x}{a}} - b\left(\frac{x}{a}\right)^2 \right] dx$$

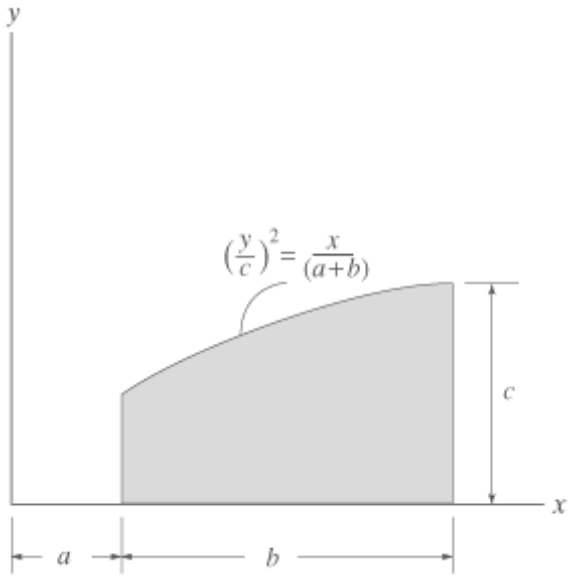
$$y_C = \frac{1}{A}\int_0^a \frac{1}{2}\left[b\sqrt{\frac{x}{a}} + b\left(\frac{x}{a}\right)^2 \right]\left[b\sqrt{\frac{x}{a}} - b\left(\frac{x}{a}\right)^2 \right] dx$$

$y_C = 1.80$ m

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Problem 9-27

Locate the centroid x_c of the shaded area.



Given:

$a = 1$ in

$b = 3$ in

$c = 2$ in

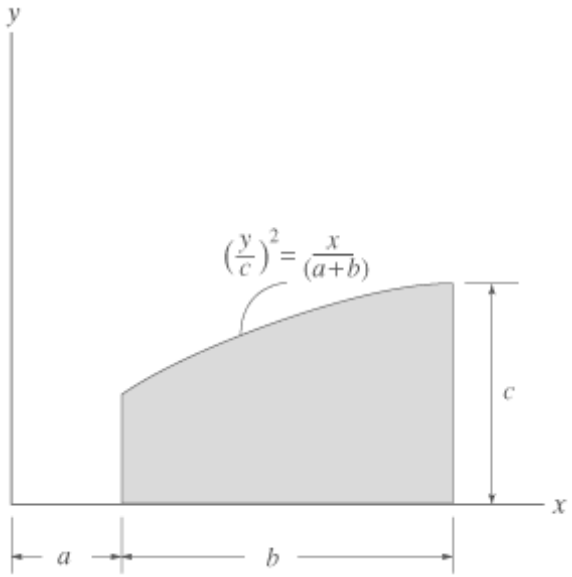
Solution:

$$A = \int_a^{a+b} c\sqrt{\frac{x}{a+b}}\,dx$$

$$x_C = \frac{1}{A}\int_a^{a+b} x\,c\sqrt{\frac{x}{a+b}}\,dx \qquad x_C = 2.66\,\text{in}$$

Problem 9-28

Locate the centroid y_c of the shaded area.



Given:

$a = 1$ in

$b = 3$ in

$c = 2$ in

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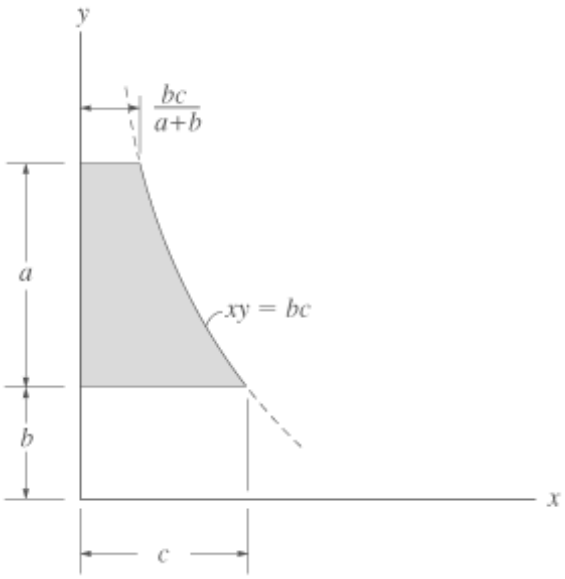
Solution:

$$A = \int_a^{a+b} c\sqrt{\frac{x}{a+b}}\,dx$$

$$y_C = \frac{1}{A}\int_a^{a+b} \frac{1}{2}\left(c\sqrt{\frac{x}{a+b}}\right)^2 dx \qquad y_C = 0.804\,\text{in}$$

Problem 9-29

Locate the centroid x_c of the shaded area.



Given:

$a = 4$ in

$b = 2$ in

$c = 3$ in

Solution:

$$A = \int_b^{a+b} \frac{b\,c}{y}\,dy \qquad A = 6.592\,\text{in}^2$$

$$x_C = \frac{1}{A}\int_b^{a+b} \frac{1}{2}\left(\frac{b\,c}{y}\right)^2 dy \qquad x_C = 0.910\,\text{in}$$

Problem 9-30

Locate the centroid y_c of the shaded area.

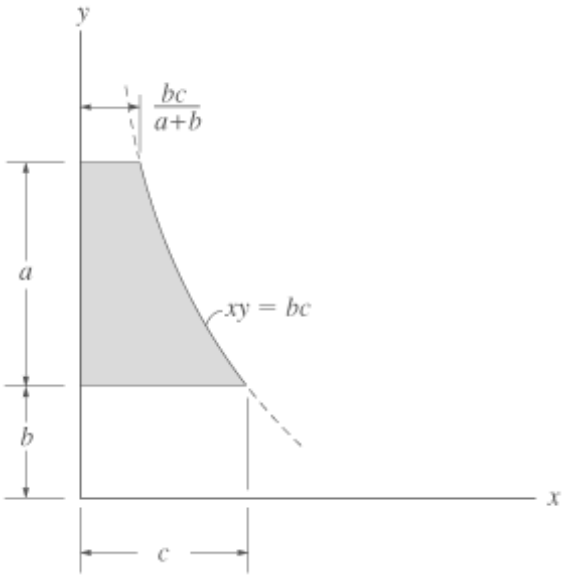
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Given:

$a = 4$ in

$b = 2$ in

$c = 3$ in



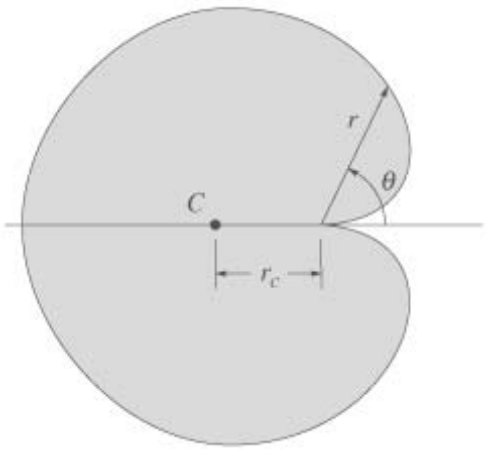
Solution:

$$A = \int_b^{a+b} \frac{b\,c}{y}\,dy \qquad A = 6.592\,\text{in}^2$$

$$y_C = \frac{1}{A}\int_b^{a+b} y\left(\frac{b\,c}{y}\right) dy \qquad y_C = 3.64\,\text{in}$$

Problem 9-31

Determine the location r_c of the centroid C of the cardioid, $r = a(1 - \cos\theta)$.



Solution:

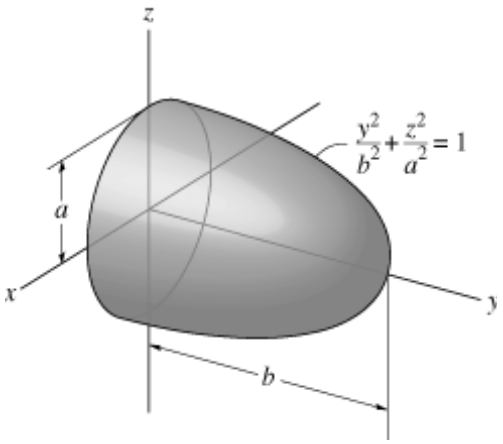
$$A = \int_0^{2\pi} \int_0^{a(1-\cos(\theta))} r\,dr\,d\theta = \frac{3}{2}\,a^2\,\pi$$

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$$x_C = \frac{2}{3a^2\pi}\int_0^{2\pi}\int_0^{a(1-\cos(\theta))} r\cos(\theta) r\,dr\,d\theta = \frac{-5}{6}\,a \qquad r_C = \frac{5a}{6}$$

Problem 9-32

Locate the centroid of the ellipsoid of revolution.



Solution: $dV = \pi z^2\,dy \qquad z^2 = a^2\left(1-\frac{y^2}{b^2}\right)$

$$V = \int_0^b \pi a^2\left(1-\frac{y^2}{b^2}\right)dy = \frac{1}{3}\,b\,\frac{3\,b^2-b^2}{b^2}\,a^2\,\pi$$

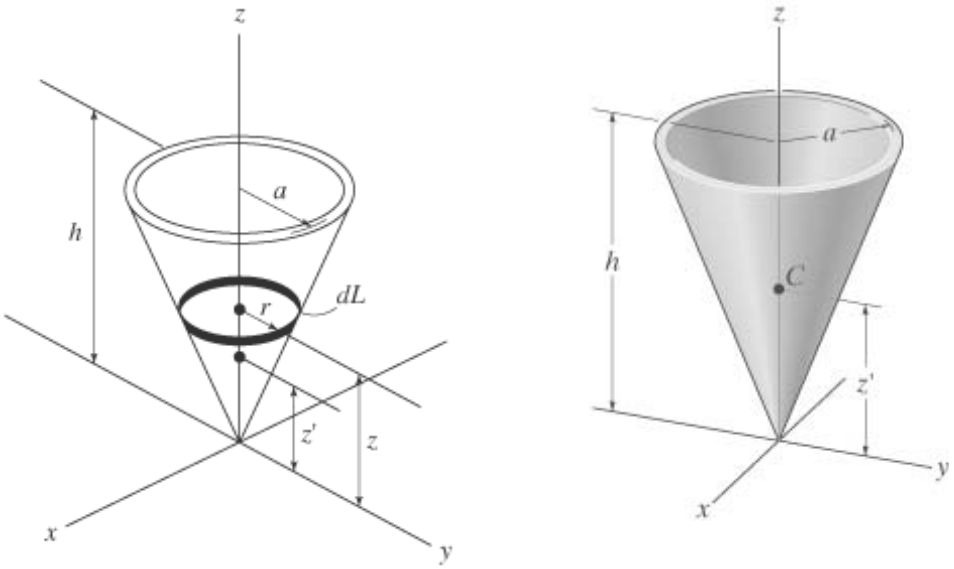
$$y_C = \frac{3}{2b\,a^2\pi}\int_0^b y\pi a^2\left(1-\frac{y^2}{b^2}\right)dy = \frac{3}{8\,b}\,b^2 \qquad y_C = \frac{3b}{8}$$

By symmetry $x_C = z_C = 0$

Problem 9-33

Locate the centroid z_c of the very thin conical shell. *Hint:* Use thin ring elements having a center at (0, 0, *z*), radius *y*, and width $dL = \sqrt{(dy)^2 + (dz)^2}$

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Solution:

$$dL = \sqrt{dy^2 + dz^2} = \sqrt{1 + \left(\frac{dy}{dz}\right)^2}\, dz = \sqrt{1 + \left(\frac{a}{h}\right)^2}\, dz \qquad r = \frac{a z}{h}$$

$$A = \int_0^h 2\pi \frac{a z}{h} \sqrt{1 + \left(\frac{a}{h}\right)^2}\, dz = h^2\ \pi\ \frac{a}{h} \left(\sqrt{\frac{h^2 + a^2}{h^2}}\right)$$

$$z_C = \frac{1}{\pi a \sqrt{a^2 + h^2}} \int_0^h z 2\pi \frac{a z}{h} \sqrt{1 + \left(\frac{a}{h}\right)^2}\, dz = \frac{2}{3\ \pi\ a \left(\sqrt{h^2 + a^2}\right)}\ h^2\ \pi\ a \left(\sqrt{\frac{h^2 + a^2}{h^2}}\right)$$

$$z_C = \frac{2h}{3}$$

Problem 9-34

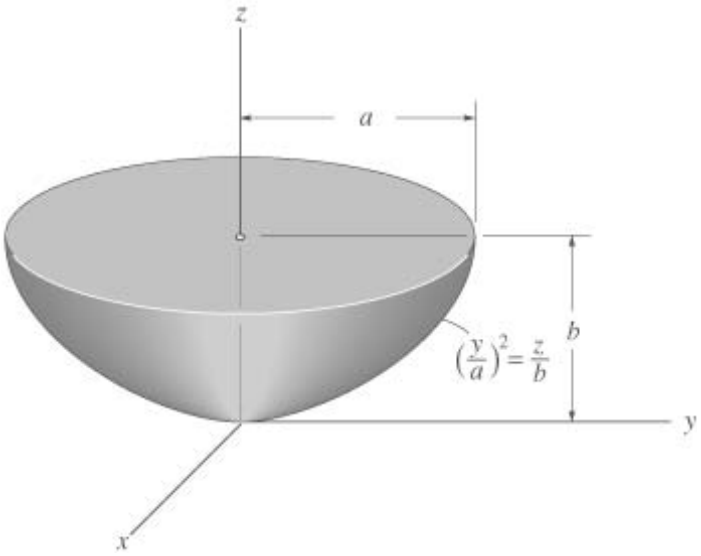
Locate the centroid z_c of the volume.

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Given:

$a = 2$ ft

$b = 2$ ft



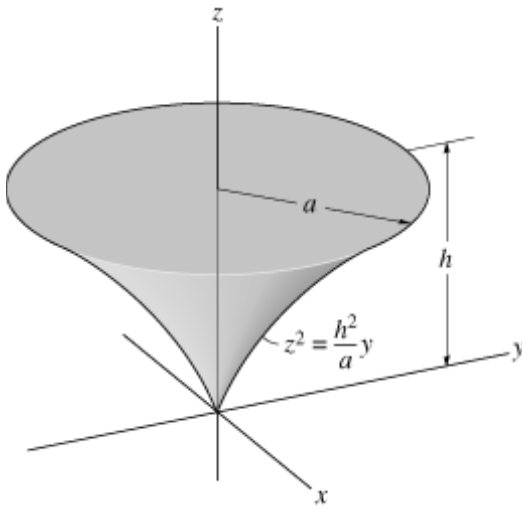
Solution:

$$V = \int_0^b \pi \frac{a^2 z}{b}\, dz \qquad V = 12.566\,\text{ft}^3$$

$$z_C = \frac{1}{V}\int_0^b z\pi \frac{a^2 z}{b}\, dz \qquad z_C = 1.333\,\text{ft}$$

Problem 9-35

Locate the centroid of the solid.



Solution:

$$z^2 = \frac{h^2}{a} y \qquad y = a\left(\frac{z}{h}\right)^2$$

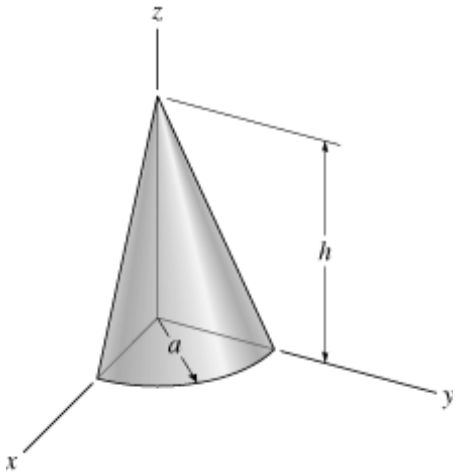
$x_C = y_C = 0$ By symmetry

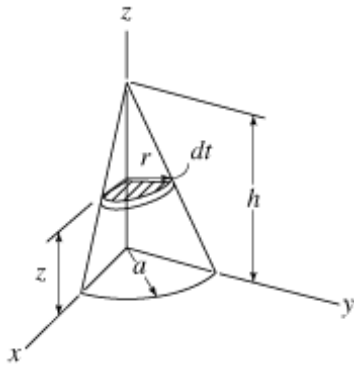
$$z_C = \frac{\displaystyle\int_0^h z\pi\left[a\left(\frac{z}{h}\right)^2\right]^2 dz}{\displaystyle\int_0^h \pi\left[a\left(\frac{z}{h}\right)^2\right]^2 dz} = \frac{5}{6}\,\frac{h^6}{h^5} \qquad z_C = \frac{5h}{6}$$

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Problem 9-36

Locate the centroid of the quarter-cone.





Solution:

$$r = \frac{a}{h}(h - z) \qquad z_C = z \qquad x_C = y_C = \frac{4r}{3\pi}$$

$$V = \int_0^h \frac{\pi}{4}\left[\frac{a}{h}(h - z)\right]^2 \mathrm{dz} = \frac{1}{12}\, h\, a^2\, \pi$$

$$z_C = \frac{12}{h a^2 \pi}\int_0^h z\frac{\pi}{4}\left[\frac{a}{h}(h - z)\right]^2 \mathrm{dz} = \frac{1}{4}\, h$$

$$x_C = \frac{12}{h a^2 \pi}\left[\int_0^h \frac{4}{3\pi}\left[\frac{a}{h}(h - z)\right]\frac{\pi}{4}\left[\frac{a}{h}(\mathrm{h} - z)\right]^2 \mathrm{dz}\right] = \frac{a}{\pi}$$

$$x_C = y_C = \frac{a}{\pi} \qquad z_C = \frac{h}{4}$$

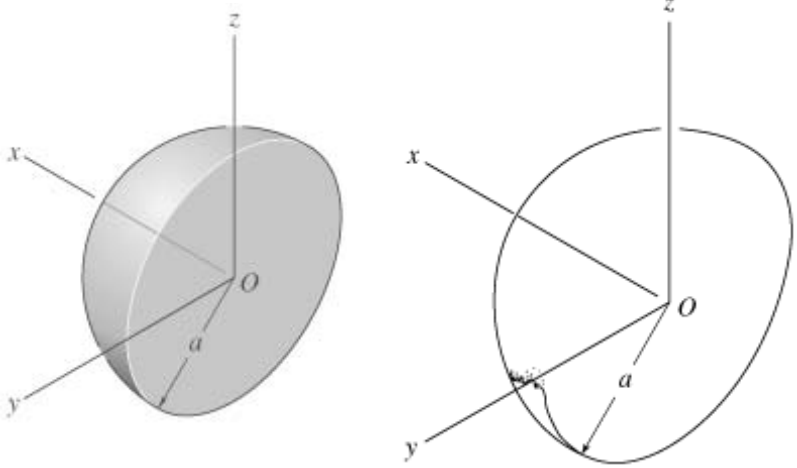
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Problem 9-37

Locate the center of mass x_c of the hemisphere. The density of the material varies linearly from zero at the origin O to ρ_o at the surface. *Hint*: Choose a hemispherical shell element for integration

Solution:

for a spherical shell



$$x_C = \frac{x}{2}$$

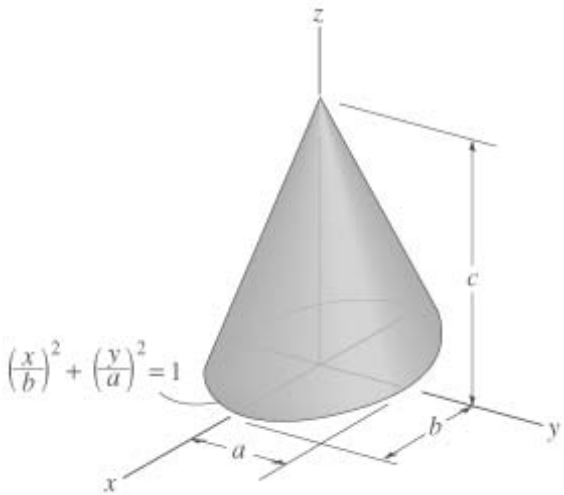
$$\rho = \rho_0\left(\frac{x}{a}\right)$$

$$dV = 2\pi x^2\,dx$$

$$x_C = \frac{\displaystyle\int_0^a \rho_0\left(\frac{x}{a}\right)\frac{x}{2}\,2\pi x^2\,dx}{\displaystyle\int_0^a \rho_0\left(\frac{x}{a}\right)2\pi x^2\,dx} = \frac{2}{5}\cdot a \qquad x_C = \frac{2}{5}a$$

Problem 9-38

Locate the centroid z_c of the right-elliptical cone.



Given:

$a = 3$ ft

$b = 4$ ft

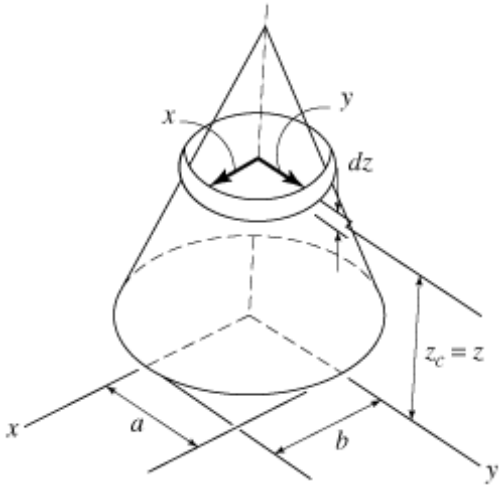
$c = 10$ ft

$$\left(\frac{x}{b}\right)^2 + \left(\frac{y}{a}\right)^2 = 1$$

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Solution:

Volume and Moment Arm : From the geometry,



$$\frac{x}{c-z} = \frac{b}{c} \qquad x = \frac{b}{c}(c-z)$$

$$\frac{y}{c-z} = \frac{a}{c} \qquad y = \frac{a}{c}(c-z)$$

The volume of the thin disk differential element is

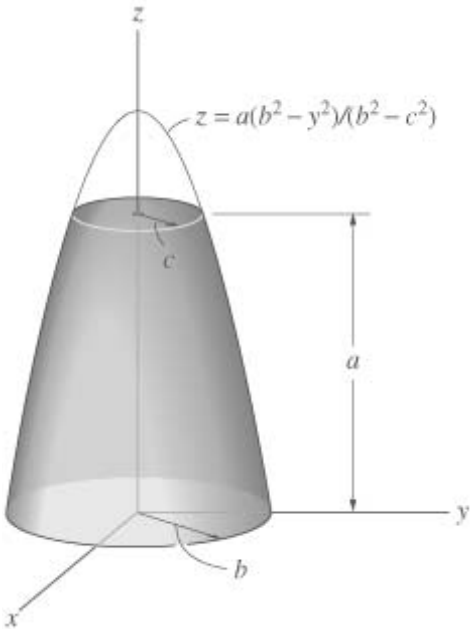
$$dV = \pi \frac{b}{c}(c-z)\frac{a}{c}(c-z)\,dz$$

$$z_C = \frac{\displaystyle\int_0^c z\pi \frac{b}{c}(c-z)\frac{a}{c}(c-z)\,dz}{\displaystyle\int_0^c \pi \frac{b}{c}(c-z)\frac{a}{c}(c-z)\,dz}$$

$z_C = 2.5\ \text{ft}$

Problem 9-39

Locate the center of gravity z_c of the frustum of the paraboloid.The material is homogeneous.



Given:

$a = 1\ \text{m}$

$b = 0.5\ \text{m}$

$c = 0.3\ \text{m}$

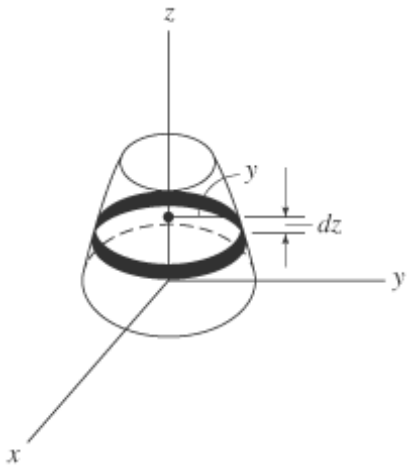
Solution

$$V = \int_0^a \pi\left[b^2 - \frac{z}{a}\left(b^2 - c^2\right)\right] dz$$

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$$z_C = \frac{1}{V}\int_0^a z\pi\left[b^2 - \frac{z}{a}\left(b^2 - c^2\right)\right] \mathrm{d}z$$

$z_C = 0.422$ m



Problem 9-40

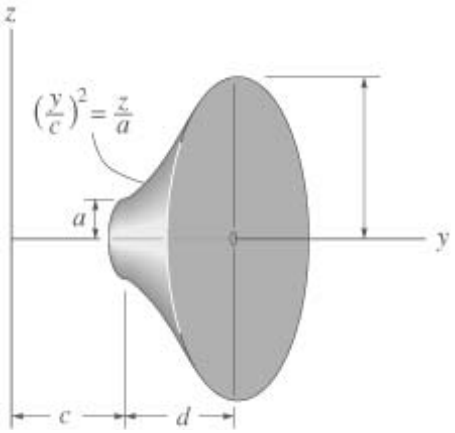
Locate the center of gravity y_c of the volume. The material is homogeneous.

Given:

$a = 25$ mm

$c = 50$ mm

$d = 50$ mm



Solution:

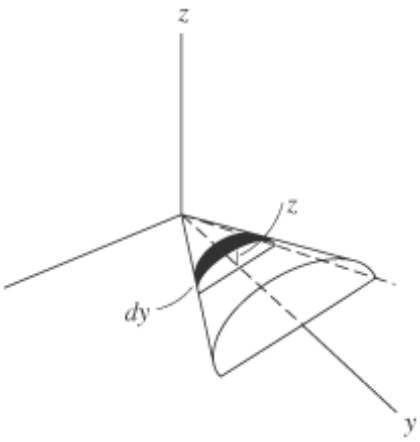
$$V = \int_c^{c+d} \pi\left[a\left(\frac{y}{c}\right)^2\right]^2 \mathrm{d}y$$

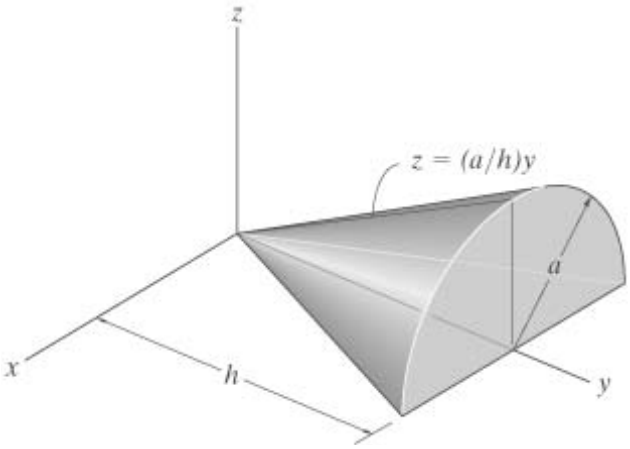
$$y_C = \frac{1}{V}\int_c^{c+d} y\pi\left[a\left(\frac{y}{c}\right)^2\right]^2 \mathrm{d}y \qquad y_C = 84.7\,\mathrm{mm}$$

Problem 9-41

Locate the center of gravity for the homogeneous half-cone.

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Solution:

$$V = \int_0^h \frac{\pi}{2}\left(\frac{a\,y}{h}\right)^2 \mathrm{d}y = \frac{1}{6}\,h\,\pi a^2$$

$$y_C = \frac{6}{h\,a^2\,\pi}\int_0^h y\,\frac{\pi}{2}\left(\frac{a\,y}{h}\right)^2 \mathrm{d}y = \frac{3}{4}h \qquad y_C = \frac{3}{4}h$$

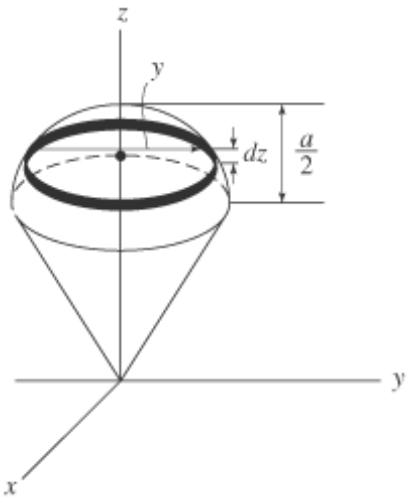
$$z_C = \frac{6}{h\,a^2\,\pi}\int_0^h \left(\frac{4a\,y}{3h\pi}\right)\frac{\pi}{2}\left(\frac{a\,y}{h}\right)^2 \mathrm{d}y = \frac{1}{\pi}a \qquad z_C = \frac{a}{\pi}$$

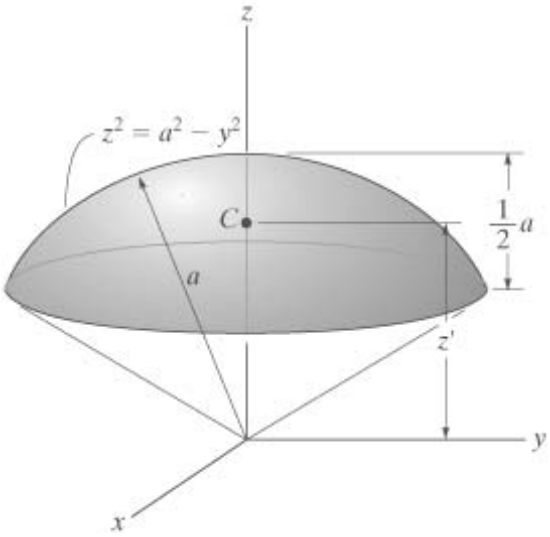
$$x_C = \frac{6}{h\,a^2\,\pi}\int_0^h 0\,\frac{\pi}{2}\left(\frac{a\,y}{h}\right)^2 \mathrm{d}y \qquad x_C = 0$$

Problem 9-42

Locate the centroid z_c of the spherical segment.

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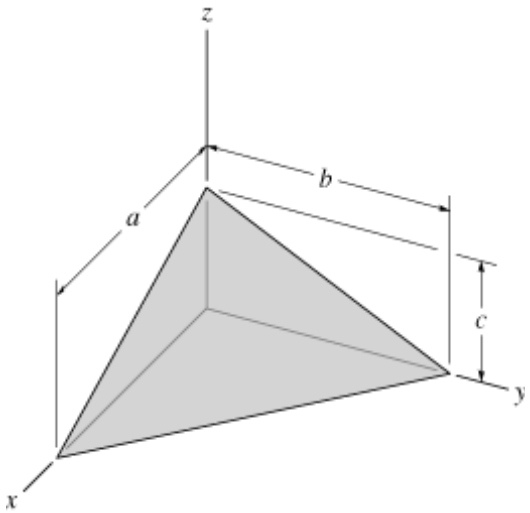
Solution:

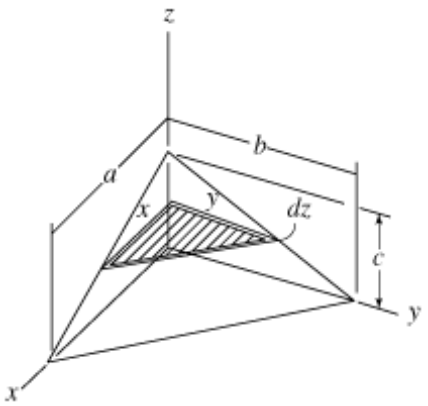
$$V = \int_{\frac{a}{2}}^{a} \pi\left(a^2 - z^2\right) \mathrm{d}z = \frac{5}{24}\, a^3\, \pi$$

$$z_C = \frac{24}{5\pi a^3} \int_{\frac{a}{2}}^{a} z\pi\left(a^2 - z^2\right) \mathrm{d}z = \frac{27}{40}\, a \qquad z_C = \frac{27}{40}\, a$$

Problem 9-43

Determine the location z_c of the centroid for the tetrahedron. *Suggestion:* Use a triangular "plate" element parallel to the *x-y* plane and of thickness d*z*.





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Solution:

$$\frac{c-z}{c}=\frac{x}{a}=\frac{y}{b} \qquad x=\frac{a}{c}(c-z) \qquad y=\frac{b}{c}(c-z)$$

$$z_C=\frac{\int_0^c z\frac{ab}{c^2}(c-z)^2\,dz}{\int_0^c \frac{ab}{c^2}(c-z)^2\,dz}=\frac{1}{4}\,c \qquad z_C=\frac{1}{4}c$$

Problem 9-44

Determine the location (x, y) of the particle M_1 so that the three particles, which lie in the x–y plane, have a center of mass located at the origin O.

Given:

$M_1 = 7$ kg

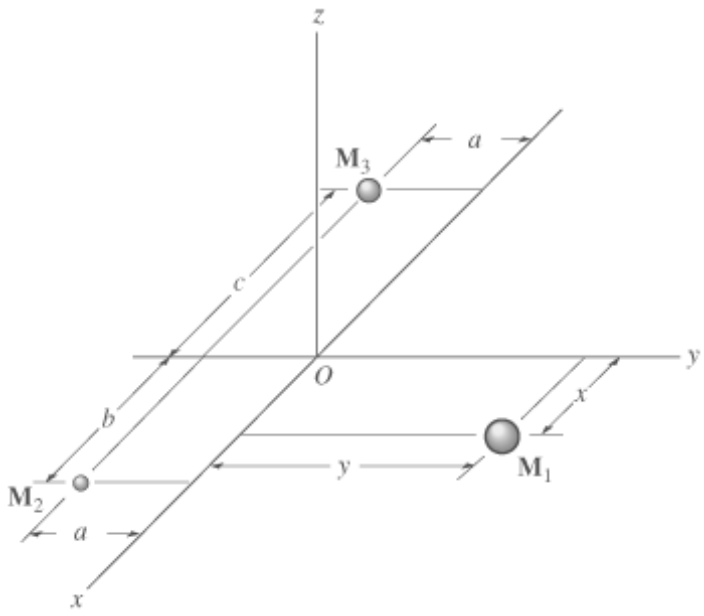
$M_2 = 3$ kg

$M_3 = 5$ kg

$a = 2$ m

$b = 3$ m

$c = 4$ m



Solution:

Guesses $x = 1$ m $y = 1$ m

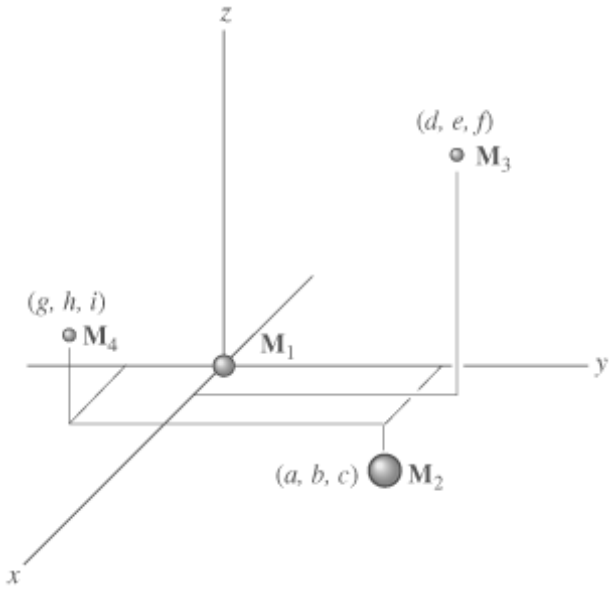
Given $M_1 x + M_2 b - M_3 c = 0$ $M_1 y - M_2 a - M_3 a = 0$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \text{Find}(x, y) \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1.57 \\ 2.29 \end{pmatrix} \text{m}$$

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Problem 9-45

Locate the center of gravity (x_c, y_c, z_c) of the four particles.



Given:

$M_1 = 2$ lb $a = 2$ ft

$M_2 = 3$ lb $b = 3$ ft

$M_3 = 1$ lb $c = -1$ ft

$M_4 = 1$ lb $d = 1$ ft

$f = 4$ ft $e = 4$ ft

$h = -2$ ft $g = 2$ ft

$i = 2$ ft

Solution:

$$x_c = \frac{M_1 0\text{ft} + M_2 a + M_3 d + M_4 g}{M_1 + M_2 + M_3 + M_4} \qquad x_c = 1.29\text{ ft}$$

$$y_c = \frac{M_1 0\text{ft} + M_2 b + M_3 e + M_4 h}{M_1 + M_2 + M_3 + M_4} \qquad y_c = 1.57\text{ ft}$$

$$z_c = \frac{M_1 0\text{ft} + M_2 c + M_3 f + M_4 i}{M_1 + M_2 + M_3 + M_4} \qquad z_c = 0.429\text{ ft}$$

Problem 9-46

A rack is made from roll-formed sheet steel and has the cross section shown. Determine the location (x_c, y_c) of the centroid of the cross section. The dimensions are indicated at the center thickness of each segment.

Given:

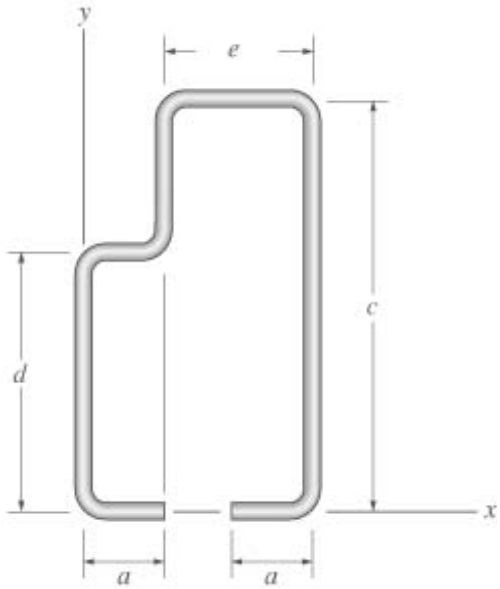
$a = 15$ mm

$c = 80$ mm

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$d = 50\ \text{mm}$

$e = 30\ \text{mm}$



Solution:

$L = 3a + 2c + e$

$$x_C = \frac{2a\frac{a}{2} + a\left(e + \frac{a}{2}\right) + c(a + e) + e\left(a + \frac{e}{2}\right) + (c - d)a}{L} \qquad x_C = 24.4\,\text{mm}$$

$$y_C = \frac{d\frac{d}{2} + c\frac{c}{2} + (c - d)\frac{d + c}{2} + a\,d + e\,c}{L} \qquad y_C = 40.6\,\text{mm}$$

Problem 9-47

The steel and aluminum plate assembly is bolted together and fastened to the wall. Each plate has a constant width w in the z direction and thickness t. If the density of A and B is ρ_s, and the density of C is ρ_{al}, determine the location x_c, the center of mass. Neglect the size of the bolts.

Units Used:

$\text{Mg} = 10^3\ \text{kg}$

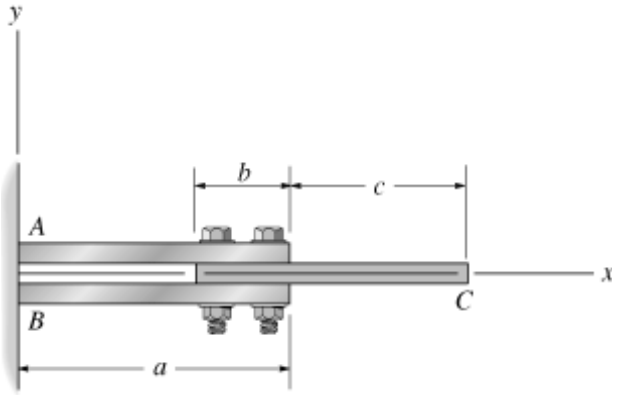
Given:

$w = 200\ \text{mm}$ $a = 300\ \text{mm}$

$t = 20\ \text{mm}$ $b = 100\ \text{mm}$

$\rho_s = 7.85\ \frac{\text{Mg}}{\text{m}^3}$ $c = 200\ \text{mm}$

$\rho_{al} = 2.71\ \frac{\text{Mg}}{\text{m}^3}$



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Solution:

$$x_C = \frac{2\left(\rho_s a\, t\, w\right)\frac{a}{2} + \left[\rho_{al}(b+c)\, t\, w\right]\left(a - b + \frac{b+c}{2}\right)}{2\rho_s a\, t\, w + \rho_{al}(b+c)\, t\, w} \qquad x_C = 179\,\text{mm}$$

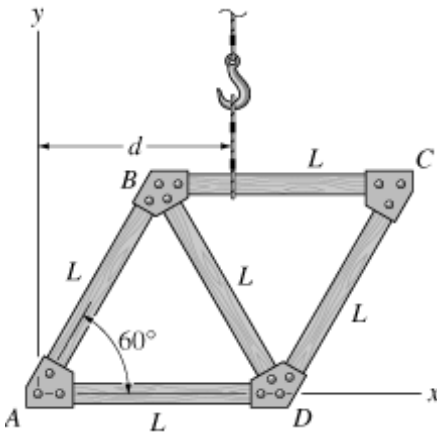
Problem 9-48

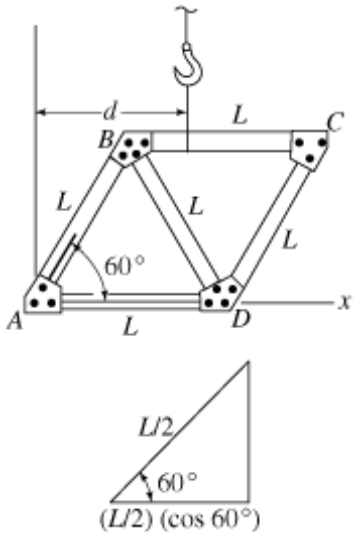
The truss is made from five members, each having a length L and a mass density ρ. If the mass of the gusset plates at the joints and the thickness of the members can be neglected, determine the distance d to where the hoisting cable must be attached, so that the truss does not tip (rotate) when it is lifted.

Given:

$L = 4\ \text{m}$

$\rho = 7\ \frac{\text{kg}}{\text{m}}$





Solution:

$$d = \frac{\rho L\left(\frac{L}{2} + \frac{L}{4} + \frac{3L}{4} + L + \frac{5L}{4}\right)}{5\rho L} \qquad d = 3\ \text{m}$$

Problem 9-49

Locate the center of gravity (x_c, y_c, z_c) of the homogeneous wire.

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Given:

$a = 300$ mm

$b = 400$ mm

Solution:

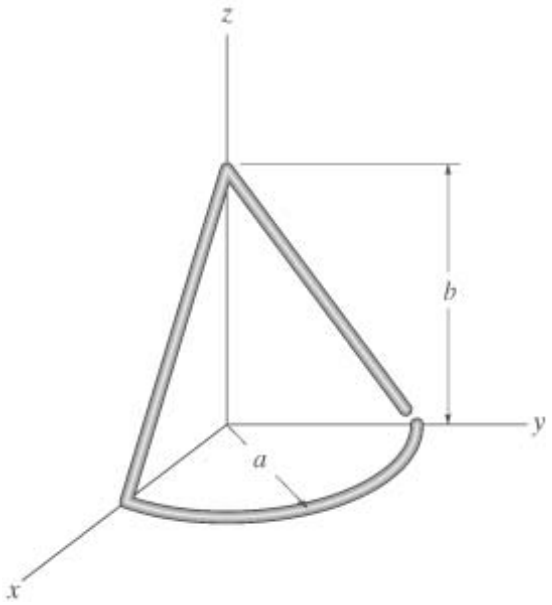
$$L = \frac{\pi a}{2} + 2\sqrt{a^2 + b^2}$$

$$x_C = \frac{1}{L}\left[\sqrt{a^2 + b^2}\,\frac{a}{2} + \frac{\pi a}{2}\left(\frac{2a}{\pi}\right)\right]$$

$$y_C = \frac{1}{L}\left[\sqrt{a^2 + b^2}\,\frac{a}{2} + \frac{\pi a}{2}\left(\frac{2a}{\pi}\right)\right]$$

$$z_C = \frac{1}{L}\left[2\sqrt{a^2 + b^2}\left(\frac{b}{2}\right)\right]$$

$$\begin{pmatrix} x_C \\ y_C \\ z_C \end{pmatrix} = \begin{pmatrix} 112.2 \\ 112.2 \\ 135.9 \end{pmatrix} \text{mm}$$



Problem 9-50

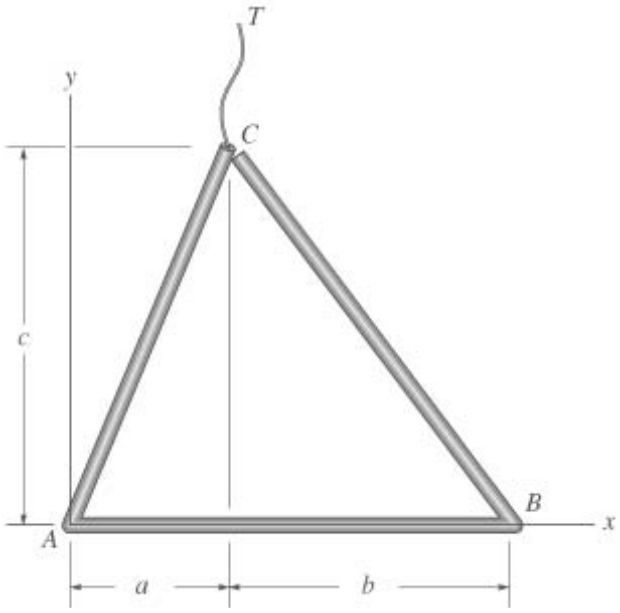
Determine the location (x_c, y_c) of the center of gravity of the homogeneous wire bent in the form of a triangle. Neglect any slight bends at the corners. If the wire is suspended using a thread *T* attached to it at *C*, determine the angle of tilt *AB* makes with the horizontal when the wire is in equilibrium.

Given:

$a = 5$ in

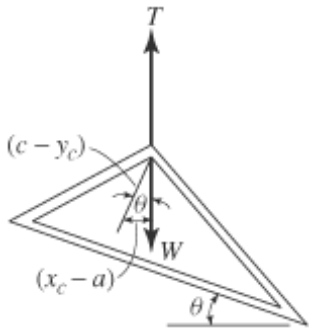
$b = 9$ in

$c = 12$ in



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Solution:



$$L = a + b + \sqrt{a^2 + c^2} + \sqrt{b^2 + c^2}$$

$$x_C = \frac{1}{L}\left[(a+b)\frac{a+b}{2} + \sqrt{a^2+c^2}\,\frac{a}{2} + \sqrt{b^2+c^2}\left(a + \frac{b}{2}\right)\right] \qquad x_C = 6.50\,\text{in}$$

$$y_C = \frac{1}{L}\left(\sqrt{a^2+c^2}\,\frac{c}{2} + \sqrt{b^2+c^2}\,\frac{c}{2}\right) \qquad y_C = 4.00\,\text{in}$$

$$\theta = \text{atan}\left(\frac{x_C - a}{c - y_C}\right) \qquad \theta = 10.6\,\text{deg}$$

Problem 9-51

The three members of the frame each have weight density γ. Locate the position (x_c, y_c) of the center of gravity. Neglect the size of the pins at the joints and the thickness of the members. Also, calculate the reactions at the fixed support A.

Given:

$\gamma = 4\ \frac{\text{lb}}{\text{ft}}$

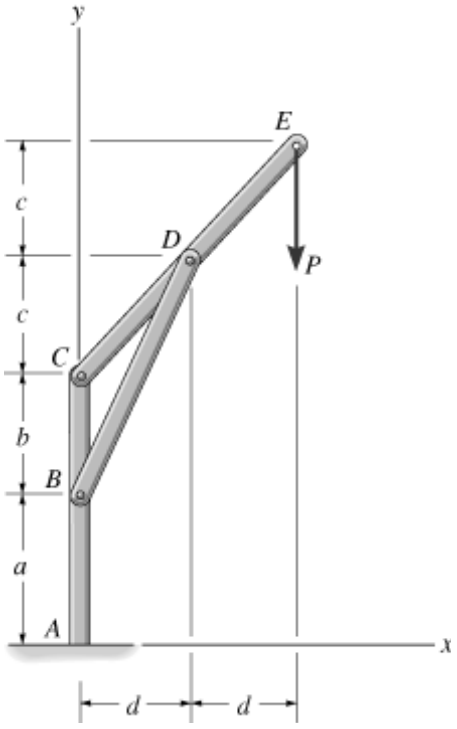
$P = 60$ lb

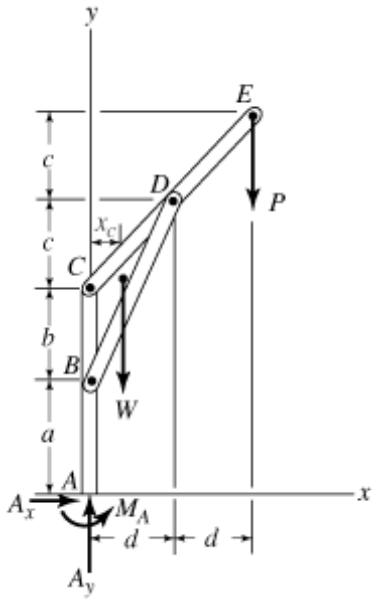
$a = 4$ ft

$b = 3$ ft

$c = 3$ ft

$d = 3$ ft





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Solution:

$$W = \gamma\sqrt{d^2 + (b+c)^2} + \gamma 2\sqrt{d^2 + c^2} + \gamma(a+b) \qquad W = 88.774\,\text{lb}$$

$$x_C = \frac{\gamma\sqrt{d^2 + (b+c)^2}\,\frac{d}{2} + \gamma 2\sqrt{d^2 + c^2}\,d}{W} \qquad x_C = 1.6\,\text{ft}$$

$$y_C = \frac{\gamma(a+b)\left(\frac{a+b}{2}\right) + \gamma\sqrt{d^2 + (b+c)^2}\left(a + \frac{b+c}{2}\right) + \gamma 2\sqrt{d^2 + c^2}(a+b+c)}{W}$$

$$y_C = 7.043\,\text{ft}$$

Equilibrium

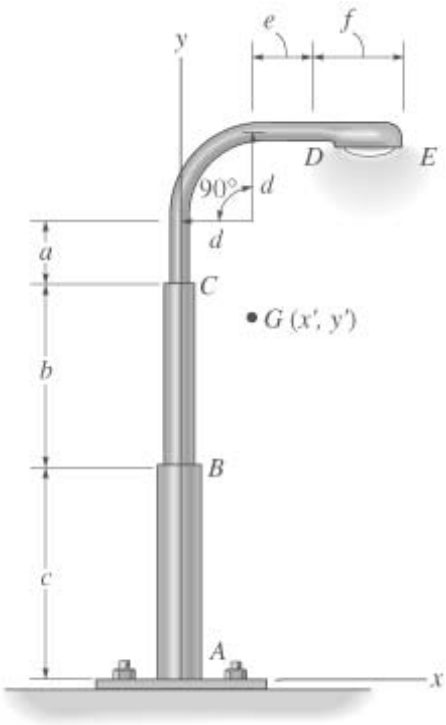
$$A_x = 0 \qquad A_x = 0\text{lb} \qquad A_x = 0\,\text{lb}$$

$$A_y - W - P = 0 \qquad A_y = W + P \qquad A_y = 148.8\,\text{lb}$$

$$M_A - W x_C - P 2d = 0 \qquad M_A = W x_C + P 2d \qquad M_A = 502\,\text{lb ft}$$

Problem 9-52

Locate the center of gravity G(x_c, y_c) of the streetlight. Neglect the thickness of each segment. The mass per unit length of each segment is given.



Given:

$a = 1\ \text{m}$ $\quad \rho_{AB} = 12\ \frac{\text{kg}}{\text{m}}$

$b = 3\ \text{m}$ $\quad \rho_{BC} = 8\ \frac{\text{kg}}{\text{m}}$

$c = 4\ \text{m}$ $\quad \rho_{CD} = 5\ \frac{\text{kg}}{\text{m}}$

$d = 1\ \text{m}$ $\quad \rho_{DE} = 2\ \frac{\text{kg}}{\text{m}}$

$e = 1\ \text{m}$ $\quad f = 1.5\ \text{m}$

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Solution:

$$M = \rho_{AB} c + \rho_{BC} b + \rho_{CD}\left(a + e + \frac{\pi d}{2}\right) + \rho_{DE} f$$

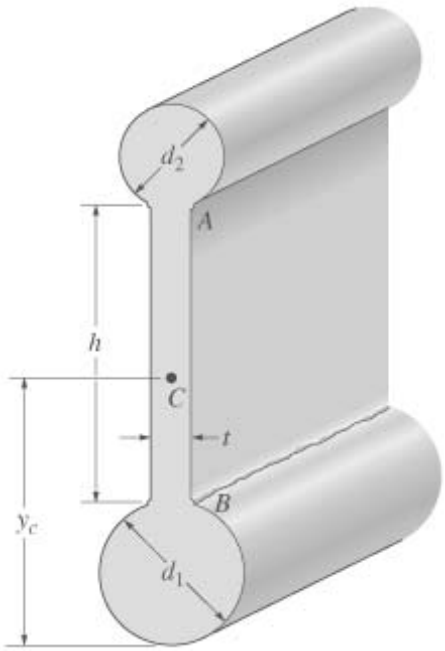
$$x_C = \frac{1}{M}\left[\rho_{CD}\frac{\pi d}{2}\left(d - \frac{2d}{\pi}\right) + \rho_{CD} e\left(d + \frac{e}{2}\right) + \rho_{DE} f\left(d + e + \frac{f}{2}\right)\right]$$

$$y_C = \frac{1}{M}\left[\begin{array}{l}\rho_{CD}\left[a\left(c + b + \frac{a}{2}\right) + \frac{\pi d}{2}\left(c + b + a + \frac{2d}{\pi}\right) + e(c + b + a + d)\right] \ldots \\ + \rho_{DE} f(c + b + a + d) + \rho_{BC} b\left(c + \frac{b}{2}\right) + \rho_{AB} c\frac{c}{2}\end{array}\right]$$

$$\begin{pmatrix} x_C \\ y_C \end{pmatrix} = \begin{pmatrix} 0.200 \\ 4.365 \end{pmatrix} \text{ m}$$

Problem 9-53

Determine the location y_c of the centroid of the beam's cross-sectional area. Neglect the size of the corner welds at *A* and *B* for the calculation.



Given:

$d_1 = 50$ mm

$d_2 = 35$ mm

$h = 110$ mm

$t = 15$ mm

Solution:

$$y_C = \frac{\pi\left(\frac{d_1}{2}\right)^2 \frac{d_1}{2} + h t\left(d_1 + \frac{h}{2}\right) + \pi\left(\frac{d_2}{2}\right)^2\left(d_1 + h + \frac{d_2}{2}\right)}{\pi\left(\frac{d_1}{2}\right)^2 + h t + \pi\left(\frac{d_2}{2}\right)^2}$$

$y_C = 85.9\,\text{mm}$

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Problem 9-54

The gravity wall is made of concrete. Determine the location (x_c, y_c) of the center of gravity G for the wall.

Given:

$a = 0.6$ m

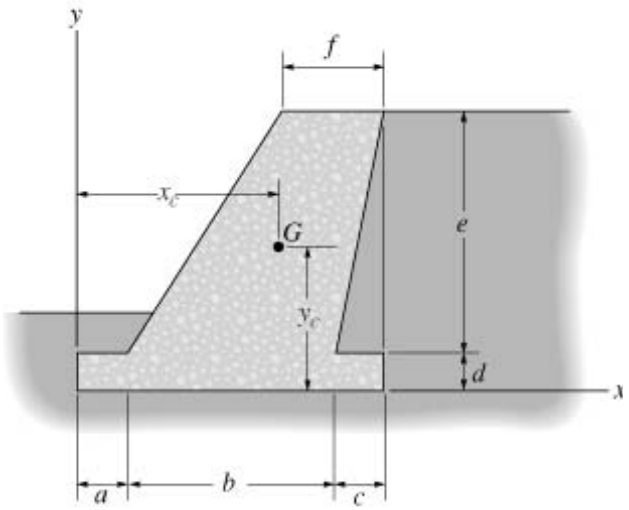
$b = 2.4$ m

$c = 0.6$ m

$d = 0.4$ m

$e = 3$ m

$f = 1.2$ m



Solution:

$$A = (a+b+c)d + (b+c)e - \frac{ce}{2} - (b+c-f)\frac{e}{2} \qquad A = 6.84 \text{ m}^2$$

$$x_C = \frac{1}{A}\left[\begin{array}{l}(a+b+c)d\left(\frac{a+b+c}{2}\right) + (b+c)e\left(a+\frac{b+c}{2}\right) - \frac{ce}{2}\left(a+b+\frac{2c}{3}\right) \ldots \\ + -(b+c-f)\frac{e}{2}\left(a+\frac{b+c-f}{3}\right)\end{array}\right]$$

$$y_C = \frac{1}{A}\left[\begin{array}{l}(a+b+c)d\frac{d}{2} + (b+c)e\left(d+\frac{e}{2}\right) - \frac{ce}{2}\left(d+\frac{e}{3}\right) \ldots \\ + -(b+c-f)\frac{e}{2}\left(d+\frac{2e}{3}\right)\end{array}\right]$$

$$\begin{pmatrix} x_C \\ y_C \end{pmatrix} = \begin{pmatrix} 2.221 \\ 1.411 \end{pmatrix} \text{ m}$$

Problem 9-55

Locate the centroid $(x_c, y_c)$of the shaded area.

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Given:

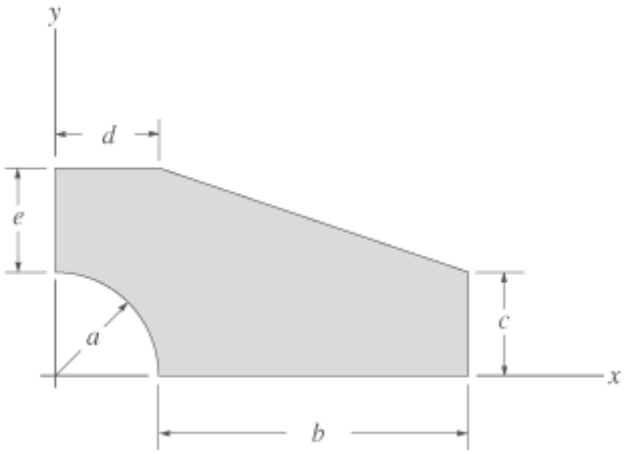
$a = 1$ in

$b = 3$ in

$c = 1$ in

$d = 1$ in

$e = 1$ in



Solution:

$$A = (a+b)(a+e) - \frac{\pi a^2}{4} - \frac{1}{2}(a+b-d)(a+e-c)$$

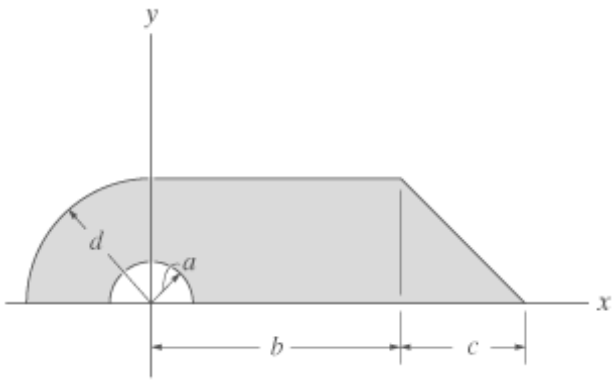
$$x_C = \frac{1}{A}\left[\frac{(a+b)^2}{2}(a+e) - \frac{\pi a^2}{4}\left(\frac{4a}{3\pi}\right) - \frac{1}{2}(a+b-d)(a+e-c)\left(a+b-\frac{a+b-d}{3}\right)\right]$$

$$y_C = \frac{1}{A}\left[(a+b)\frac{(a+e)^2}{2} - \frac{\pi a^2}{4}\left(\frac{4a}{3\pi}\right) - \frac{1}{2}(a+b-d)(a+e-c)\left(a+e-\frac{a+e-c}{3}\right)\right]$$

$$\begin{pmatrix} x_C \\ y_C \end{pmatrix} = \begin{pmatrix} 1.954 \\ 0.904 \end{pmatrix} \text{in}$$

Problem 9-56

Locate the centroid (x_c, y_c) of the shaded area.



Given:

$a = 1$ in

$b = 6$ in

$c = 3$ in

$d = 3$ in

Solution:

$$A = b\,d + \frac{\pi d^2}{4} - \frac{\pi a^2}{2} + \frac{1}{2}(d\,c)$$

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$$x_C = \frac{1}{A}\left[b\,d\,\frac{b}{2} - \frac{\pi d^2}{4}\left(\frac{4d}{3\pi}\right) + \frac{1}{2}d\,c\left(b + \frac{c}{3}\right)\right] \qquad x_C = 2.732\,\text{in}$$

$$y_C = \frac{1}{A}\left[b\,d\left(\frac{d}{2}\right) + \frac{\pi d^2}{4}\left(\frac{4d}{3\pi}\right) - \frac{\pi a^2}{2}\left(\frac{4a}{3\pi}\right) + \frac{1}{2}d\,c\left(\frac{d}{3}\right)\right] \qquad y_C = 1.423\,\text{in}$$

Problem 9-57

Determine the location y_c of the centroidal axis $x_c x_c$ of the beam's cross-sectional area. Neglect the size of the corner welds at *A* and *B* for the calculation.

Given:

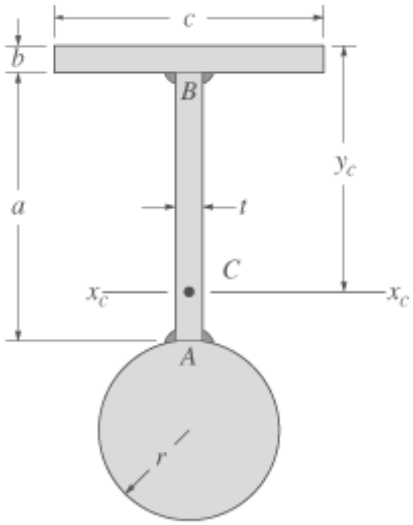
$r = 50$ mm

$t = 15$ mm

$a = 150$ mm

$b = 15$ mm

$c = 150$ mm



Solution:

$$y_C = \frac{b\,c\left(\frac{b}{2}\right) + a\,t\left(b + \frac{a}{2}\right) + \pi r^2(b + a + r)}{b\,c + a\,t + \pi r^2} \qquad y_C = 154.443\,\text{mm}$$

Problem 9-58

Determine the location (x_c, y_c) of the centroid *C* of the area.

Given:

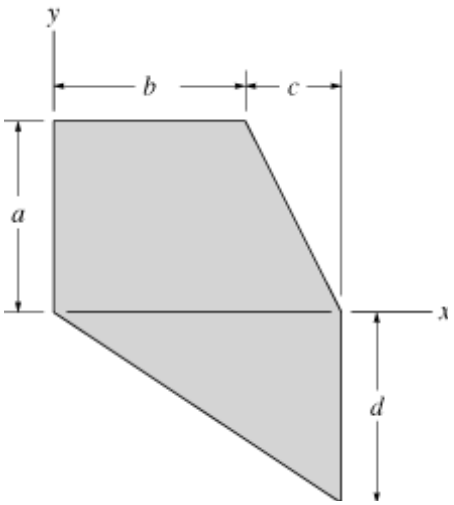
$a = 6$ in

$b = 6$ in

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$c = 3$ in

$d = 6$ in



Solution:

$$x_C = \frac{a\,b\left(\frac{b}{2}\right) + \frac{1}{2}a\,c\left(b + \frac{c}{3}\right) + \frac{1}{2}(b + c)\,d\,\frac{2}{3}(b + c)}{a\,b + \frac{1}{2}c\,a + \frac{1}{2}(b + c)\,d}$$

$x_C = 4.625\,\text{in}$

$$y_C = \frac{a\,b\left(\frac{a}{2}\right) + \frac{1}{2}a\,c\left(\frac{a}{3}\right) - \frac{1}{2}(b + c)\,d\left(\frac{d}{3}\right)}{a\,b + \frac{1}{2}c\,a + \frac{1}{2}(b + c)\,d}$$

$y_C = 1\,\text{in}$

Problem 9-59

Determine the location y_c of the centroid C for a beam having the cross-sectional area shown. The beam is symmetric with respect to the y axis.

Given:

$a = 2$ in

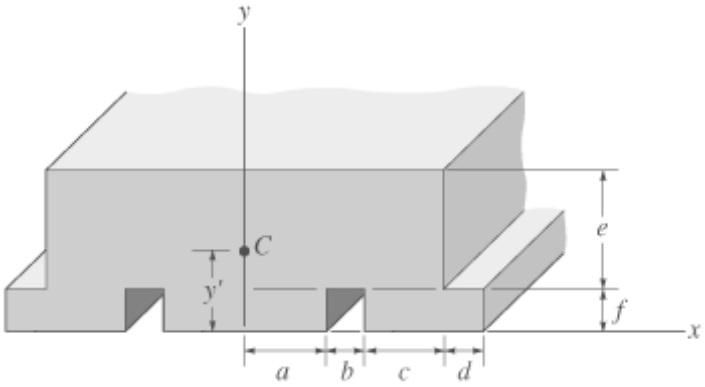
$b = 1$ in

$c = 2$ in

$d = 1$ in

$e = 3$ in

$f = 1$ in



Solution:

$$A = 2[(a + b + c + d)(e + f) - b\,f - d\,e]$$

$A = 40\,\text{in}^2$

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$$y_C = \frac{2}{A}\left[(a+b+c+d)\frac{(e+f)^2}{2} - b\frac{f^2}{2} - de\left(f+\frac{e}{2}\right)\right] \qquad y_C = 2.00\,\text{in}$$

Problem 9-60

The wooden table is made from a square board having weight W. Each of the legs has wieght W_{leg} and length L. Determine how high its center of gravity is from the floor. Also, what is the angle, measured from the horizontal, through which its top surface can be tilted on two of its legs before it begins to overturn? Neglect the thickness of each leg.

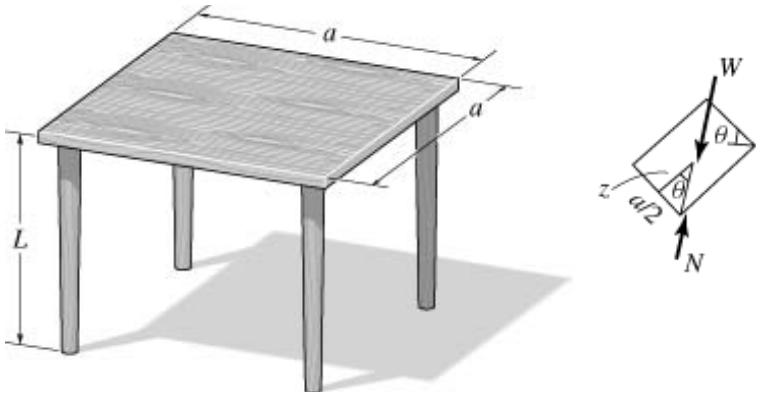
Given:

$W = 15$ lb

$W_{leg} = 2$ lb

$L = 3$ ft

$a = 4$ ft



Solution:

$$z_C = \frac{WL + 4W_{leg}\left(\frac{L}{2}\right)}{W + 4W_{leg}} \qquad z_C = 2.478\,\text{ft}$$

$$\theta = \operatorname{atan}\left(\frac{\frac{a}{2}}{z_C}\right) \qquad \theta = 38.9\,\text{deg}$$

Problem 9-61

Locate the centroid y_c for the beam's cross-sectional area.

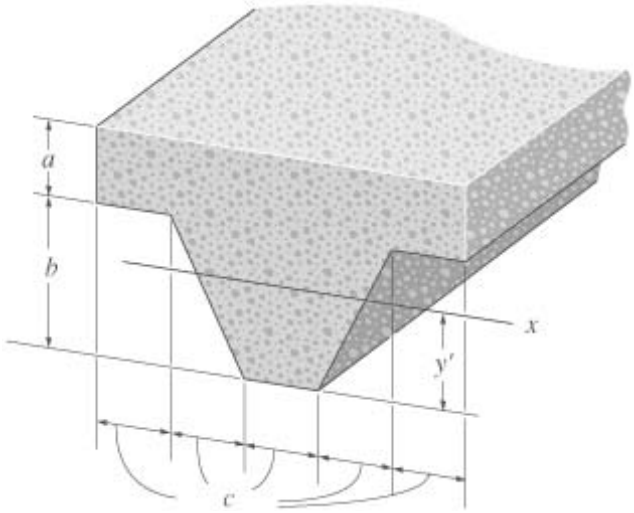
Given:

$a = 120$ mm

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$b = 240$ mm

$c = 120$ mm



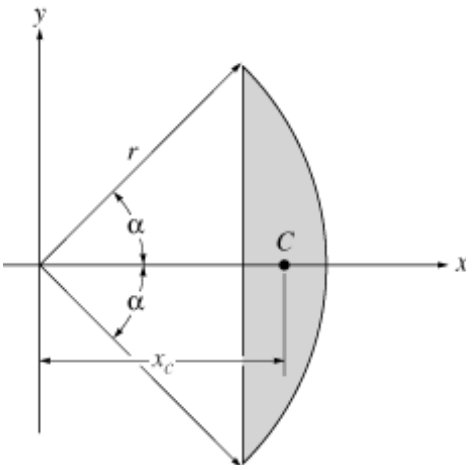
Solution:

$$A = (a+b)5c - 3bc$$

$$y_C = \frac{1}{A}\left[\frac{(a+b)^2}{2}5c - 2bc\left(\frac{b}{2}\right) - bc\left(\frac{b}{3}\right)\right] \qquad y_C = 229\,\text{mm}$$

Problem 9-62

Determine the location x_c of the centroid C of the shaded area which is part of a circle having a radius r.



Solution:

$$A = \alpha r^2 - r^2\sin(\alpha)\cos(\alpha)$$

$$x_C = \frac{1}{A}\left(\alpha r^2\frac{2r\sin(\alpha)}{3\alpha} - r^2\sin(\alpha)\cos(\alpha)\frac{2}{3}r\cos(\alpha)\right)$$

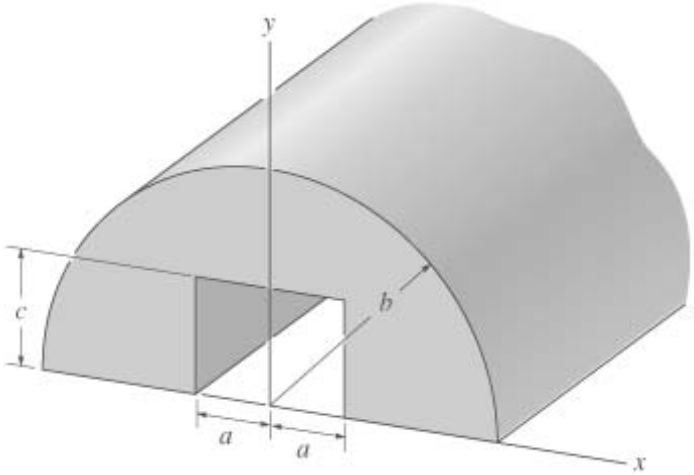
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$$x_C = 2r\frac{\sin(\alpha)}{3}\left(\frac{1-\cos(\alpha)^2}{\alpha-\sin(\alpha)\cos(\alpha)}\right)$$

$$x_C = \frac{2r}{3}\frac{\sin(\alpha)^3}{\alpha-\frac{\sin(2\alpha)}{2}}$$

Problem 9-63

Locate the centroid y_c for the strut's cross-sectional area.



Given:

$a = 40$ mm

$b = 120$ mm

$c = 60$ mm

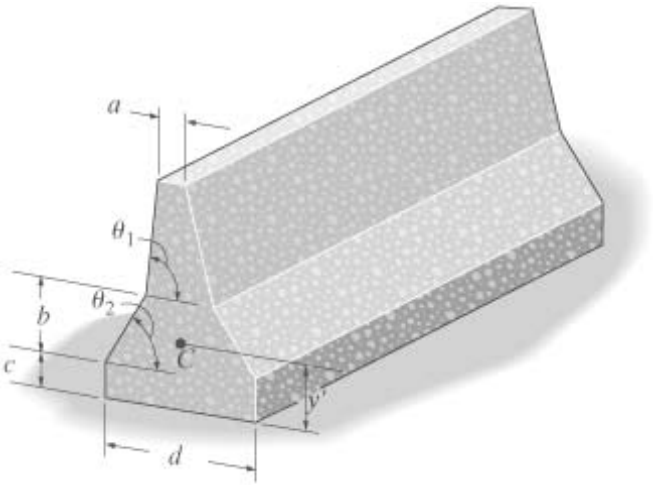
Solution:

$$A = \frac{\pi b^2}{2} - 2ac$$

$$y_C = \frac{1}{A}\left[\frac{\pi b^2}{2}\left(\frac{4b}{3\pi}\right) - 2ac\left(\frac{c}{2}\right)\right] \qquad y_C = 56.6\,\text{mm}$$

Problem 9-64

The "New Jersey" concrete barrier is commonly used during highway construction. Determine the location y_c of its centroid.



Given:

$a = 4$ in

$b = 12$ in

$c = 6$ in

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$d = 24 \text{ in}$

$\theta_1 = 75 \text{ deg}$

$\theta_2 = 55 \text{ deg}$

Solution:

$$e = b\cot(\theta_2) \qquad f = d - 2e \qquad h = \frac{f-a}{2}\tan(\theta_1)$$

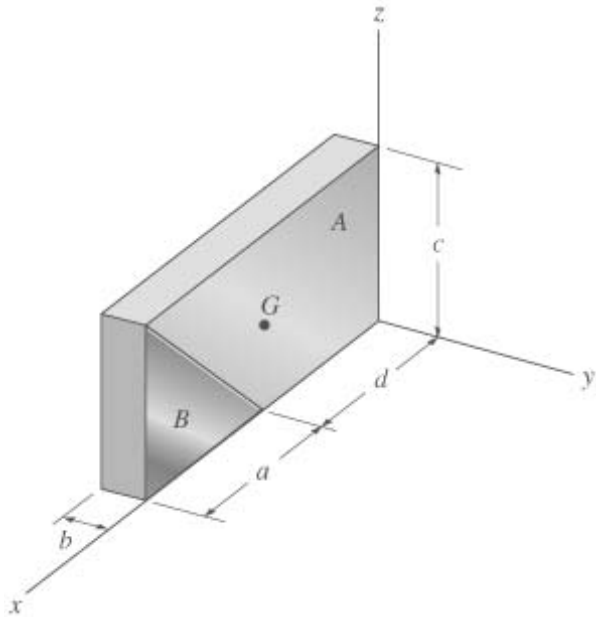
$$A = d(c+b+h) - be - 2he - \frac{1}{2}(f-a)h$$

$$y_C = \frac{1}{A}\left[d\frac{(c+b+h)^2}{2} - be\left(c+\frac{2b}{3}\right) - 2he\left(c+b+\frac{h}{2}\right) - \frac{1}{2}(f-a)h\left(c+b+\frac{2h}{3}\right)\right]$$

$y_C = 8.69 \text{ in}$

Problem 9-65

The composite plate is made from both steel (A) and brass (B) segments. Determine the mass and location (x_c, y_c, z_c) of its mass center G.



Units Used:

$\text{Mg} = 1000 \text{ kg}$

Given:

$\rho_{st} = 7.85 \frac{\text{Mg}}{\text{m}^3}$ $\quad a = 150 \text{ mm}$

$\quad b = 30 \text{ mm}$

$\rho_{br} = 8.74 \frac{\text{Mg}}{\text{m}^3}$ $\quad c = 225 \text{ mm}$

$\quad d = 150 \text{ mm}$

Solution:

$$M = \rho_{st}\left(dbc + \frac{1}{2}abc\right) + \rho_{br}\frac{1}{2}abc$$

$$x_C = \frac{1}{M}\left[\rho_{st}\left[dcb\frac{d}{2} + \frac{1}{2}abc\left(d+\frac{a}{3}\right)\right] + \rho_{br}\frac{1}{2}abc\left(d+\frac{2a}{3}\right)\right]$$

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$$y_C = \frac{-1}{M}\left[\rho_{st}\left(d\,c\,b\,\frac{b}{2} + \frac{1}{2}a\,b\,c\,\frac{b}{2}\right) + \rho_{br}\,\frac{1}{2}a\,b\,c\,\frac{b}{2}\right]$$

$$z_C = \frac{1}{M}\left[\rho_{st}\left(d\,c\,b\,\frac{c}{2} + \frac{1}{2}a\,b\,c\,\frac{2c}{3}\right) + \rho_{br}\,\frac{1}{2}a\,b\,c\,\frac{c}{3}\right]$$

$$\begin{pmatrix} x_C \\ y_C \\ z_C \end{pmatrix} = \begin{pmatrix} 152.8 \\ -15.0 \\ 111.5 \end{pmatrix} \text{mm} \qquad M = 16.347 \text{ kg}$$

Problem 9-66

Locate the centroid y_c of the concrete beam having the tapered cross section shown.

Given:

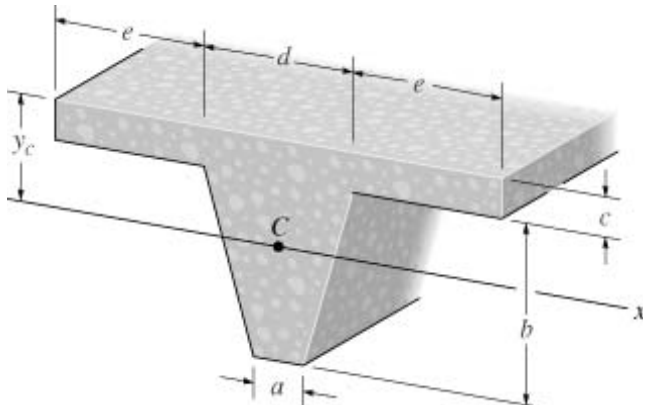
$a = 100$ mm

$b = 360$ mm

$c = 80$ mm

$d = 300$ mm

$e = 300$ mm



Solution:

$$y_C = \frac{(d+2e)c\left(\frac{c}{2}\right) + \frac{1}{2}(d-a)b\left(c+\frac{b}{3}\right) + a\,b\left(c+\frac{b}{2}\right)}{(d+2e)c + \frac{1}{2}(d-a)b + a\,b} \qquad y_C = 135\,\text{mm}$$

Problem 9-67

The anatomical center of gravity G of a person can be determined by using a scale and a rigid board having a uniform weight W_1 and length l. With the person's weight W known, the person lies down on the board and the scale reading **P** is recorded. From this show how to calculate the location x_c of the center of mass. Discuss the best place l_1 for the smooth support at B in order to improve the accuracy of this experiment.

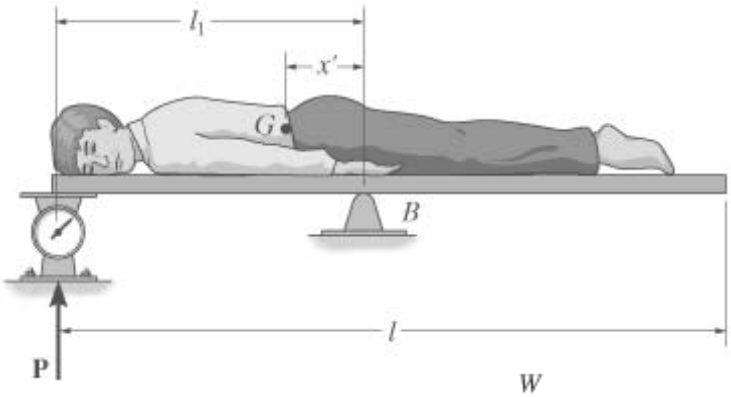
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Given:

$a = 120$ mm

$b = 240$ mm

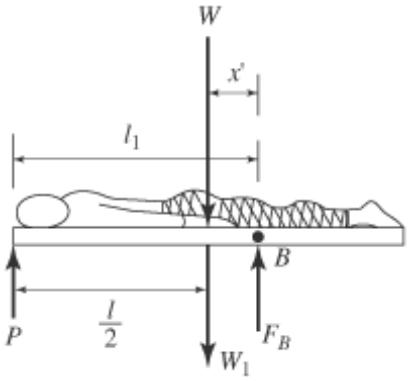
$c = 120$ mm



Solution:

$$\Sigma M_B = 0; \quad W x_C - P l_1 + W_1\left(l_1 - \frac{l}{2}\right) = 0$$

$$x_C = \frac{P l_1 - W_1\left(l_1 - \frac{l}{2}\right)}{W}$$



Put B as close as possible to the center of gravity of the board, i.e., $l_1 = \frac{l}{2}$, then $W_1\left(l_1 - \frac{l}{2}\right) = 0$ and the effect of the board's weight will not be a large factor in the measurement.

Problem 9-68

The tank and compressor have a mass M_T and mass center at G_T and the motor has a mass M_M and a mass center at G_M. Determine the angle of tilt, θ, of the tank so that the unit will be on the verge of tipping over.

Given:

$a = 300$ mm

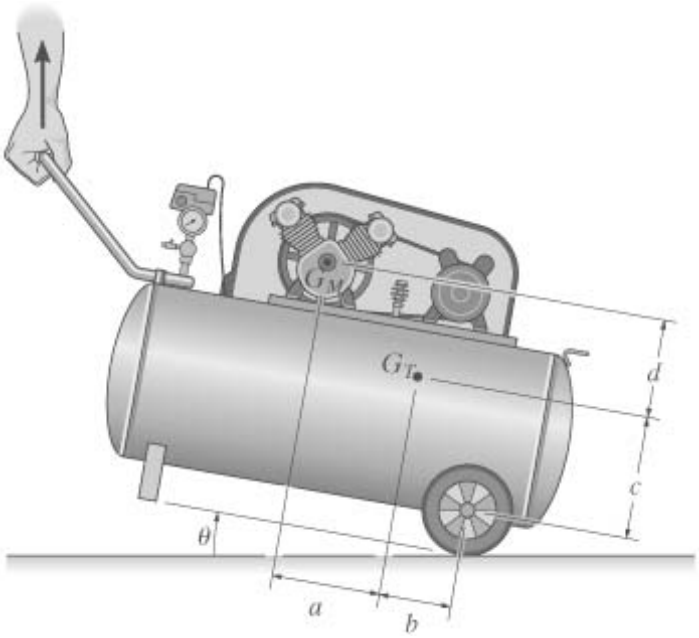
$b = 200$ mm

$c = 350$ mm

$d = 275$ mm

$M_T = 15$ kg

$M_M = 70$ kg



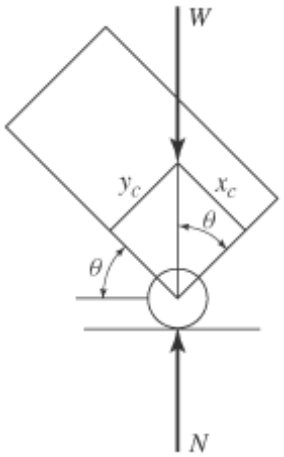
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Solution:

$$x_C = \frac{b M_T + (a+b) M_M}{M_T + M_M} \qquad x_C = 0.4471 \text{ m}$$

$$y_C = \frac{c M_T + (c+d) M_M}{M_T + M_M} \qquad y_C = 0.57647 \text{ m}$$

$$\theta = \operatorname{atan}\left(\frac{x_C}{y_C}\right) \qquad \theta = 37.8 \deg$$



Problem 9-69

Determine the distance h to which a hole of diameter d must be bored into the base of the cone so that the center of mass of the resulting shape is located at z_C. The material has a density ρ.

Given:

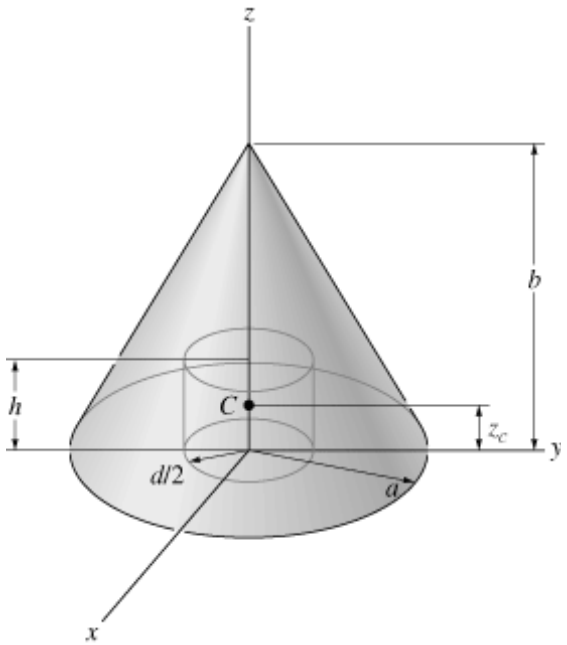
$d = 100$ mm

$z_C = 115$ mm

$\rho = 8 \dfrac{\text{mg}}{\text{m}^3}$

$a = 150$ mm

$b = 500$ mm



Solution:

Guess $h = 200$ mm

Given $$z_C = \frac{\frac{1}{3}\pi a^2 b\left(\frac{b}{4}\right) - \pi\left(\frac{d}{2}\right)^2 h\left(\frac{h}{2}\right)}{\frac{1}{3}\pi a^2 b - \pi\left(\frac{d}{2}\right)^2 h}$$

$h = \text{Find}(h)$ $h = 323\,\text{mm}$

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Problem 9-70

Determine the distance to the centroid of the shape which consists of a cone with a hole of height h bored into its base.

Given:

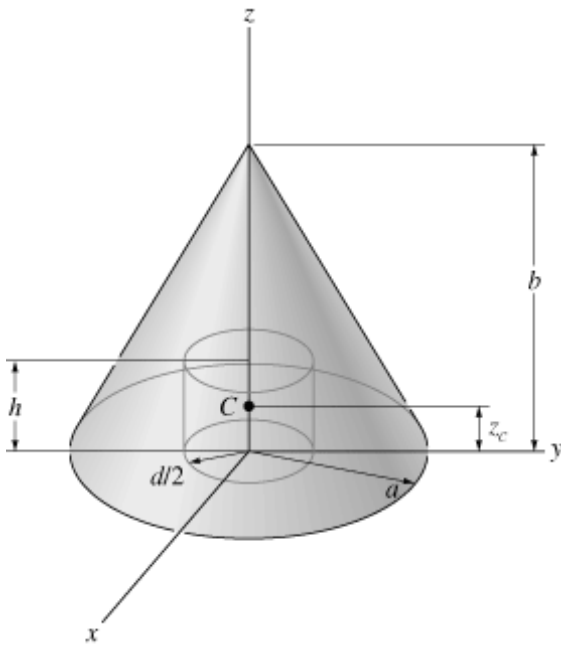
$d = 100$ mm

$h = 50$ mm

$\rho = 8 \frac{\text{mg}}{\text{m}^3}$

$a = 150$ mm

$b = 500$ mm



Solution:

$$z_C = \frac{\frac{1}{3}\pi a^2 b\left(\frac{b}{4}\right) - \pi\left(\frac{d}{2}\right)^2 h\left(\frac{h}{2}\right)}{\frac{1}{3}\pi a^2 b - \pi\left(\frac{d}{2}\right)^2 h} \qquad z_C = 128.4\,\text{mm}$$

Problem 9-71

The sheet metal part has the dimensions shown. Determine the location (x_c, y_c, z_c) of its centroid.

Given:

$a = 3$ in

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$b = 4$ in

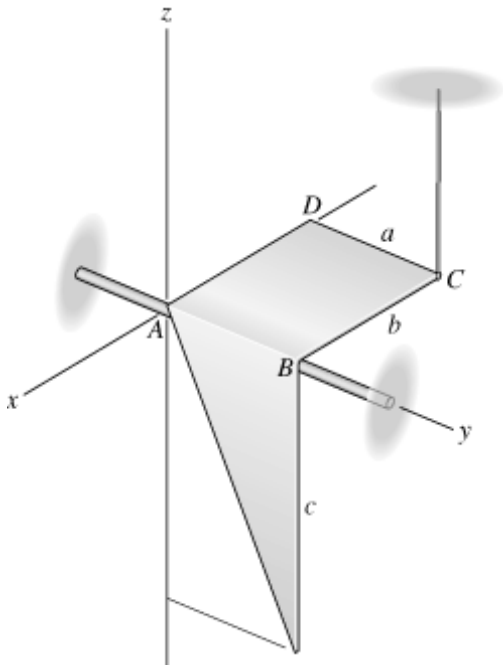
$c = 6$ in

Solution:

$$x_C = \frac{-a\,b\left(\frac{b}{2}\right)}{a\,b + \frac{1}{2}a\,c} \qquad x_C = -1.143\,\text{in}$$

$$y_C = \frac{a\,b\left(\frac{a}{2}\right) + \frac{1}{2}a\,c\left(\frac{2a}{3}\right)}{a\,b + \frac{1}{2}a\,c} \qquad y_C = 1.714\,\text{in}$$

$$z_C = \frac{\frac{-1}{2}a\,c\left(\frac{c}{3}\right)}{a\,b + \frac{1}{2}a\,c} \qquad z_C = -0.857\,\text{in}$$



Problem 9-72

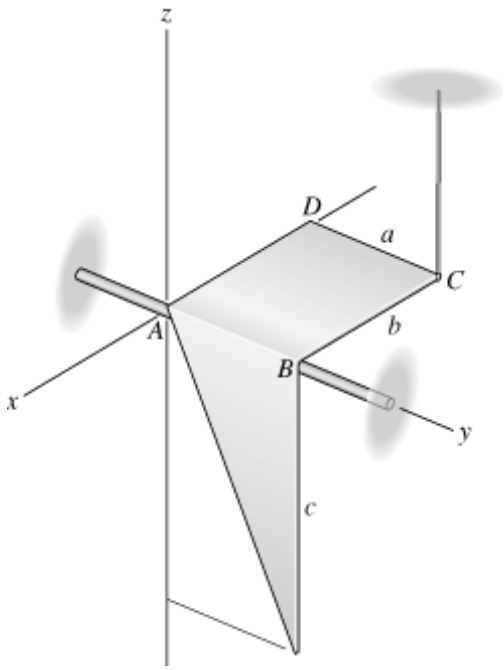
The sheet metal part has a weight per unit area of and is supported by the smooth rod and at *C*. If the cord is cut, the part will rotate about the *y* axis until it reaches equilibrium. Determine the equilibrium angle of tilt, measured downward from the negative *x* axis, that *AD* makes with the -*x* axis.

Given:

$a = 3$ in

$b = 4$ in

$c = 6$ in



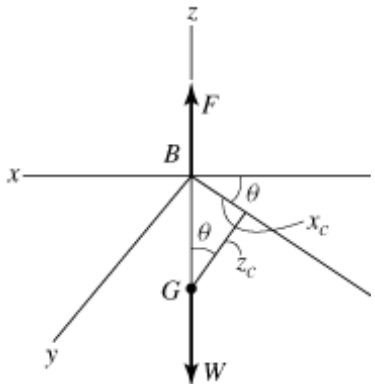
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Solution:

$$x_C = \frac{a\,b\left(\frac{b}{2}\right)}{a\,b + \frac{1}{2}a\,c} \qquad x_C = 1.143\,\text{in}$$

$$z_C = \frac{\frac{1}{2}a\,c\left(\frac{c}{3}\right)}{a\,b + \frac{1}{2}a\,c} \qquad z_C = 0.857\,\text{in}$$

$$\theta = \text{atan}\left(\frac{x_C}{z_C}\right) \qquad \theta = 53.13\,\text{deg}$$



Problem 9-73

A toy skyrocket consists of a solid conical top of density ρ_t, a hollow cylinder of density ρ_c, and a stick having a circular cross section of density ρ_s. Determine the length of the stick, x, so that the center of gravity G of the skyrocket is located along line aa.

Given:

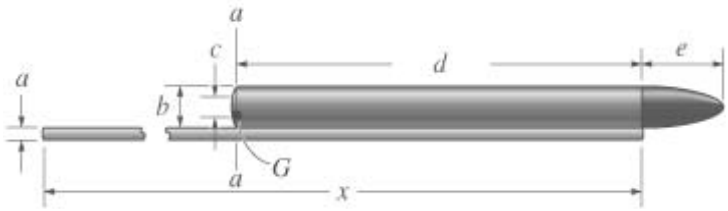
$a = 3\ \text{mm}$ $\quad \rho_t = 600\ \frac{\text{kg}}{\text{m}^3}$

$b = 10\ \text{mm}$

$c = 5\ \text{mm}$ $\quad \rho_c = 400\ \frac{\text{kg}}{\text{m}^3}$

$d = 100\ \text{mm}$

$e = 20\ \text{mm}$ $\quad \rho_s = 300\ \frac{\text{kg}}{\text{m}^3}$



Solution:

Guess $\quad x = 200\ \text{mm}$

Given $\quad \rho_t \pi \left(\frac{b}{2}\right)^2 \frac{e}{3}\left(d + \frac{e}{4}\right) + \rho_c \frac{\pi}{4}\left(b^2 - c^2\right) d\left(\frac{d}{2}\right) + \rho_s \pi \left(\frac{a^2}{4}\right) x\left(d - \frac{x}{2}\right) = 0$

$x = \text{Find}(x) \qquad x = 490\,\text{mm}$

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Problem 9-74

Determine the location (x_c, y_c) of the center of mass of the turbine and compressor assembly. The mass and the center of mass of each of the various components are indicated below.

Given:

$a = 0.75$ m $\quad M_1 = 25$ kg

$b = 1.25$ m $\quad M_2 = 80$ kg

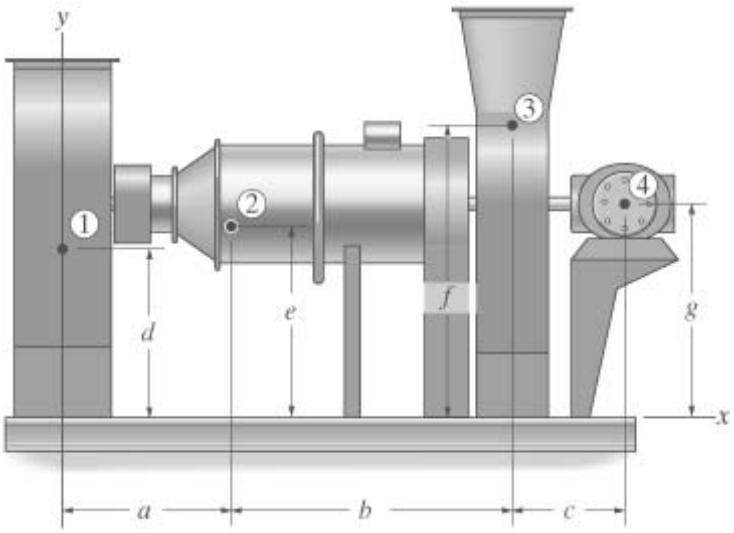
$c = 0.5$ m $\quad M_3 = 30$ kg

$d = 0.75$ m $\quad M_4 = 105$ kg

$e = 0.85$ m

$f = 1.30$ m

$g = 0.95$ m



1. Intake housing $\mathbf{M}_1$
2. Turbine $\mathbf{M}_2$
3. Exhaust housing $\mathbf{M}_3$
4. Compressor $\mathbf{M}_4$

Solution:

$$M = M_1 + M_2 + M_3 + M_4$$

$$x_c = \frac{1}{M}\left[M_2\, a + M_3(a + b) + M_4(a + b + c)\right] \qquad x_c = 1.594 \text{ m}$$

$$y_c = \frac{1}{M}\left(M_1\, d + M_2\, e + M_3\, f + M_4\, g\right) \qquad y_c = 0.940 \text{ m}$$

Problem 9-75

The solid is formed by boring a conical hole into the hemisphere. Determine the distance z_c to the center of gravity.

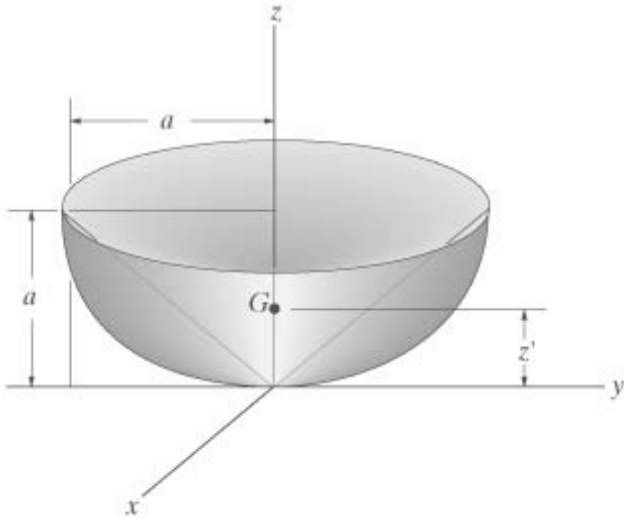
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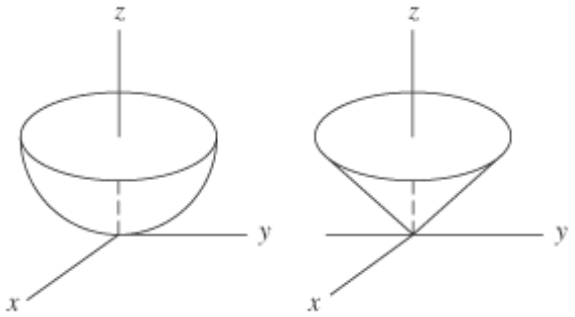
Solution:

$$V = \frac{2}{3}\pi a^3 - \frac{\pi}{3}a^2 a = \frac{\pi}{3}a^3$$

$$z_C = \frac{1}{V}\left[\frac{5a}{8}\left(\frac{2}{3}\pi a^3\right) - \frac{3}{4}a\left(\frac{\pi}{3}a^3\right)\right]$$

$$z_C = \frac{a}{2}$$





Problem 9-76

Determine the location x_c of the centroid of the solid made from a hemisphere, cylinder, and cone.

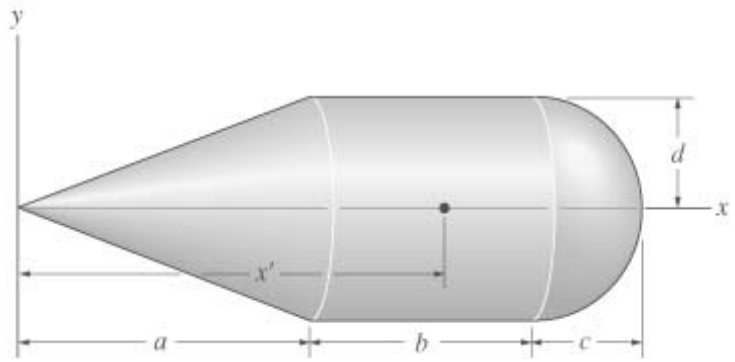
Given:

$a = 80$ mm

$b = 60$ mm

$c = 30$ mm

$d = 30$ mm



Solution:

$$V = \frac{1}{3}\pi d^2 a + \pi d^2 b + \frac{2}{3}\pi d^3$$

$$x_C = \frac{1}{V}\left[\frac{1}{3}\pi d^2 a\left(\frac{3a}{4}\right) + \pi d^2 b\left(a + \frac{b}{2}\right) + \frac{2}{3}\pi d^3\left(a + b + \frac{3c}{8}\right)\right]$$

$$x_C = 105.2\,\text{mm}$$

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Problem 9-77

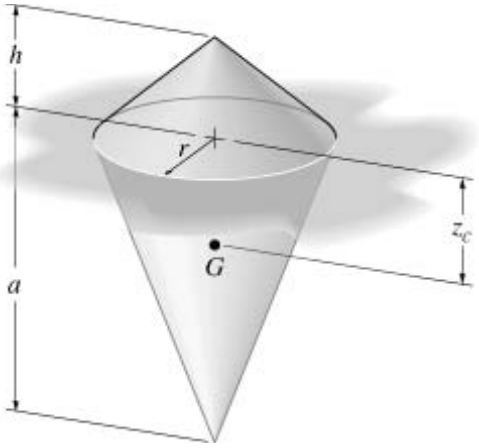
The buoy is made from two homogeneous cones each having radius r. Find the distance z_c to the buoy's center of gravity G.

Given:

$r = 1.5$ ft

$h = 1.2$ ft

$a = 4$ ft



Solution:

$$z_C = \frac{\frac{\pi}{3}r^2 a\left(\frac{a}{4}\right) - \frac{\pi}{3}r^2 h\left(\frac{h}{4}\right)}{\frac{\pi}{3}r^2(a+h)} \qquad z_C = 0.7 \text{ ft}$$

Problem 9-78

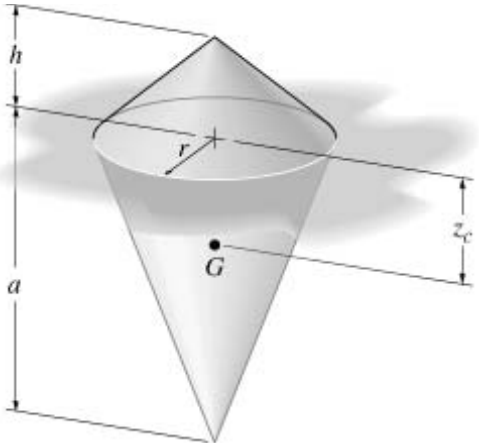
The buoy is made from two homogeneous cones each having radius r. If it is required that the buoy's center of gravity G be located at z_c,determine the height h of the top cone.

Given:

$z_C = 0.5$ ft

$r = 1.5$ ft

$a = 4$ ft



Solution:

Guess $h = 1$ ft

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Given $z_C = \dfrac{\frac{\pi}{3}r^2 a\left(\frac{a}{4}\right) - \frac{\pi}{3}r^2 h\left(\frac{h}{4}\right)}{\frac{\pi}{3}r^2(a+h)}$ $h = \text{Find}(h)$ $h = 2\,\text{ft}$

Problem 9-79

Locate the center of mass z_c of the forked lever, which is made from a homogeneous material and has the dimensions shown.

Given:

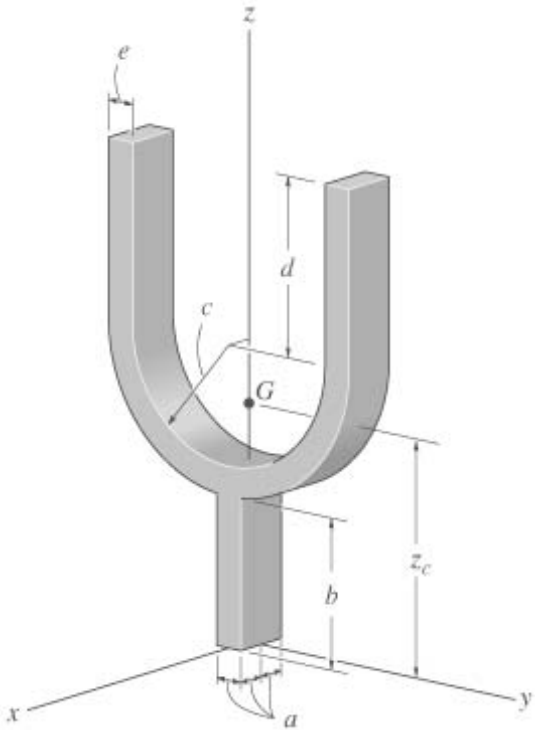
$a = 0.5$ in

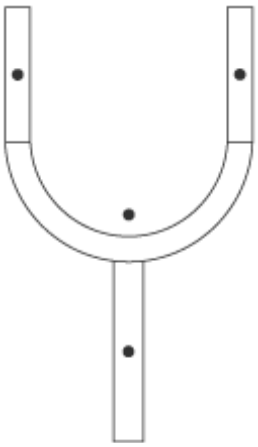
$b = 2.5$ in

$c = 2$ in

$d = 3$ in

$e = 0.5$ in





Solution:

$$V = b\,a^2 + 2e\,a\,d + \frac{\pi}{2}\left[(c+e)^2 - c^2\right]a$$

$$z_C = \frac{1}{V}\left[b\,a^2\left(\frac{b}{2}\right) + 2e\,a\,d\left(b+e+c+\frac{d}{2}\right) + \left(\frac{\pi a}{2}\right)(c+e)^2\left[b+c+e-4\left(\frac{c+e}{3\pi}\right)\right] \ldots \right.$$
$$\left. + \left(\frac{-\pi a}{2}\right)c^2\left[b+c+e-4\left(\frac{c}{3\pi}\right)\right]\right]$$

$z_C = 4.32\,\text{in}$

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Problem 9-80

A triangular plate made of homogeneous material has a constant thickness which is very small. If it is folded over as shown, determine the location y_c of the plate's center of gravity G.

Given:

$a = 6$ in

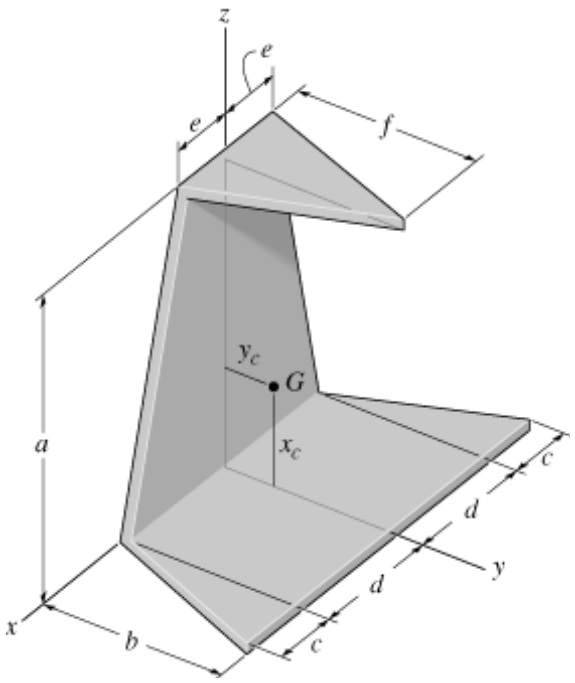
$b = 3$ in

$c = 1$ in

$d = 3$ in

$e = 1$ in

$f = 3$ in



Solution:

$$y_C = \frac{2d\,b\left(\frac{b}{2}\right) + \frac{1}{2}(2c\,b)\left(\frac{2b}{3}\right) + \frac{1}{2}(2e\,f)\left(\frac{f}{3}\right)}{2d\,b + \frac{1}{2}(2c\,b) + \frac{1}{2}(2d)(a + f)} \qquad y_C = 0.75\,\text{in}$$

Problem 9-81

A triangular plate made of homogeneous material has a constant thickness which is very small. If it is folded over as shown, determine the location z_c of the plate's center of gravity G.

Given

$a = 6$ in

$b = 3$ in

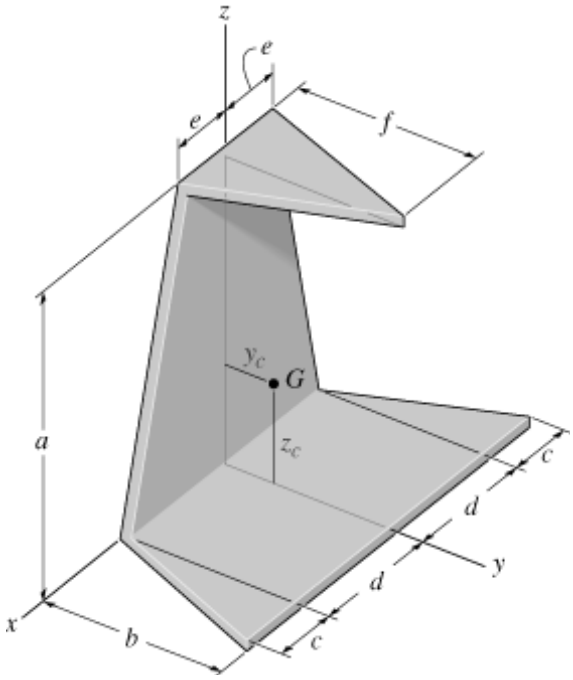
$c = 1$ in

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$d = 3$ in

$e = 1$ in

$f = 3$ in



Solution:

$$z_C = \frac{\frac{1}{2}(2e\,f\,a) + 2e\,a\left(\frac{a}{2}\right) + \frac{1}{2}2(d-e)a\left(\frac{a}{3}\right)}{2d\,b + \frac{1}{2}(2c\,b) + \frac{1}{2}2d(a+f)} \qquad z_C = 1.625\,\text{in}$$

Problem 9-82

Each of the three homogeneous plates welded to the rod has a density ρ and a thickness a. Determine the length l of plate C and the angle of placement, θ, so that the center of mass of the assembly lies on the y axis. Plates A and B lie in the x–y and z–y planes, respectively.

Units Used:

Mg = 1000 kg

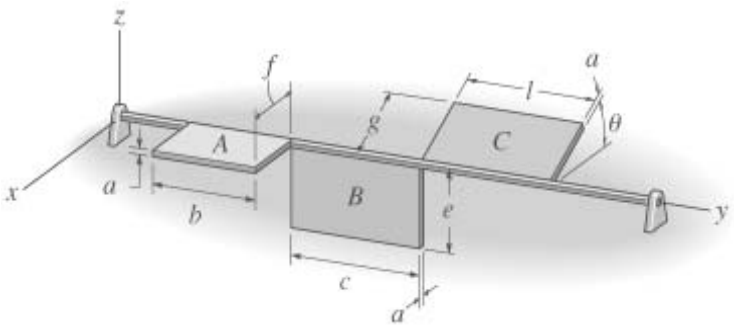
Given:

$a = 10$ mm $\quad f = 100$ mm

$b = 200$ mm $\quad g = 150$ mm

$c = 250$ mm $\quad e = 150$ mm

$\rho = 6\,\frac{\text{Mg}}{\text{m}^3}$



Solution: The thickness and density are uniform

Guesses $\theta = 10$ deg $\quad l = 10$ mm

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Given

$$b\,f\left(\frac{f}{2}\right) - g\,l\left(\frac{g}{2}\right)\cos(\theta) = 0 \qquad -c\,e\left(\frac{e}{2}\right) + g\,l\left(\frac{g}{2}\right)\sin(\theta) = 0$$

$$\begin{pmatrix} l \\ \theta \end{pmatrix} = \text{Find}(l, \theta) \qquad l = 265\,\text{mm} \qquad \theta = 70.4\,\text{deg}$$

Problem 9-83

The assembly consists of a wooden dowel rod of length L and a tight-fitting steel collar. Determine the distance x_c to its center of gravity if the specific weights of the materials are γ_w and γ_{st}. The radii of the dowel and collar are shown.

Given:

$L = 20$ in

$\gamma_w = 150\ \frac{\text{lb}}{\text{ft}^3}$

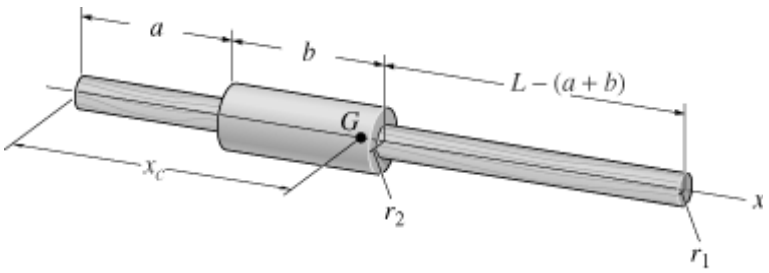
$\gamma_{st} = 490\ \frac{\text{lb}}{\text{ft}^3}$

$a = 5$ in

$b = 5$ in

$r_1 = 1$ in

$r_2 = 2$ in



Solution:

$$x_c = \frac{\gamma_w \pi {r_1}^2 L \frac{L}{2} + \gamma_{st}\pi\left({r_2}^2 - {r_1}^2\right)b\left(a + \frac{b}{2}\right)}{\gamma_w \pi {r_1}^2 L + \gamma_{st}\pi\left({r_2}^2 - {r_1}^2\right)b} \qquad x_c = 8.225\,\text{in}$$

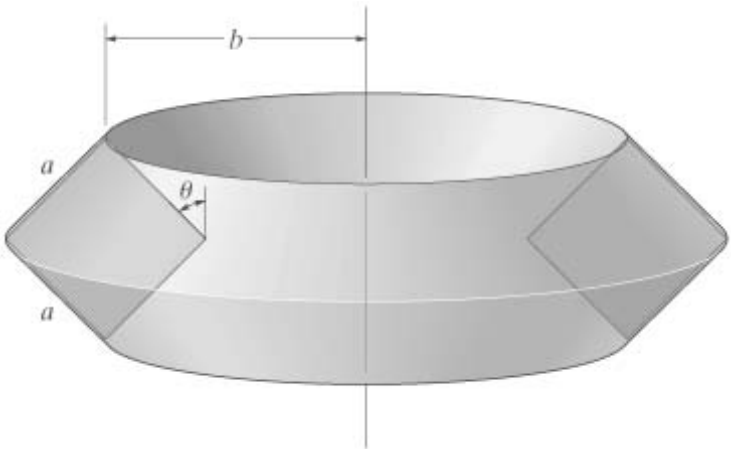
Problem 9-84

Determine the surface area and the volume of the ring formed by rotating the square about the vertical axis.

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Given:

$\theta = 45 \text{ deg}$

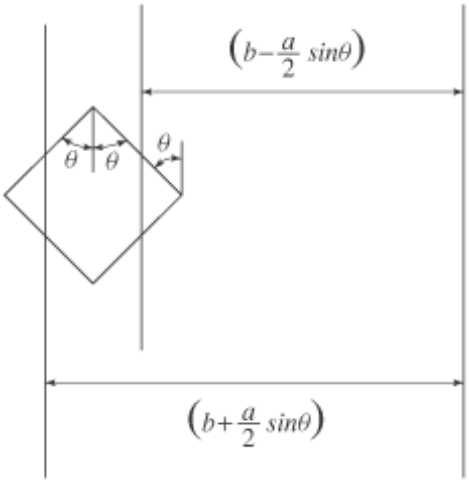


Solution:

$$A = 2\left[2\pi\left(b - \frac{a}{2}\sin(\theta)\right)a\right] \ldots + 2\left[2\pi\left(b + \frac{a}{2}\sin(\theta)\right)a\right]$$

$$A = 8\pi b a$$

$$V = 2\pi b a^2$$



Problem 9-85

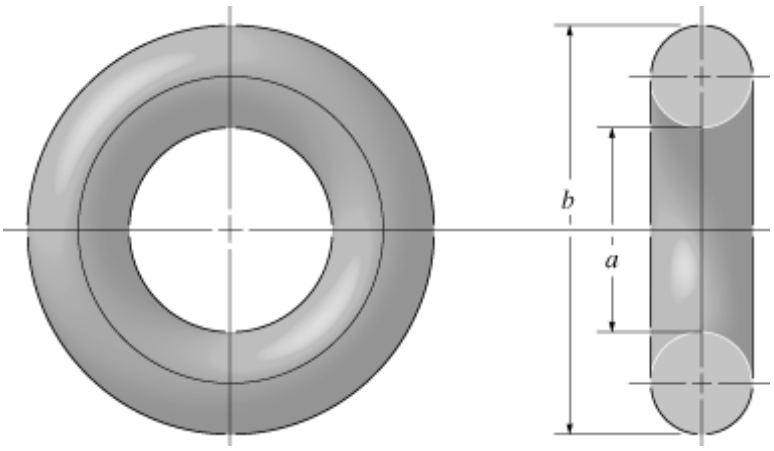
The anchor ring is made of steel having specific weight γ_{st}. Determine the surface area of the ring. The cross section is circular as shown.

Given:

$\gamma_{st} = 490 \dfrac{\text{lb}}{\text{ft}^3}$

$a = 4 \text{ in}$

$b = 8 \text{ in}$



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Solution:

$$A = 2\pi\left(\frac{a}{2} + \frac{b-a}{4}\right)2\pi\left(\frac{b-a}{4}\right) \qquad A = 118\,\text{in}^2$$

Problem 9-86

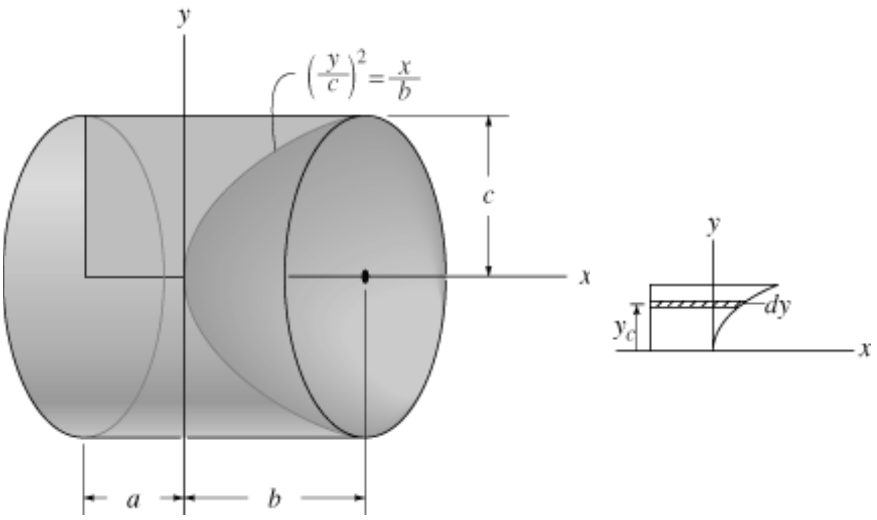
Using integration, determine both the area and the distance y_c to the centroid of the shaded area. Then using the second theorem of PappusGuldinus, determine the volume of the solid generated by revolving the shaded area about the x axis.

Given:

$a = 1$ ft

$b = 2$ ft

$c = 2$ ft



Solution:

$$A = \int_0^c \left[a + \left(\frac{y}{c}\right)^2 b\right] dy \qquad A = 3.333\,\text{ft}^2$$

$$y_C = \frac{1}{A}\int_0^c y\left[a + \left(\frac{y}{c}\right)^2 b\right] dy \qquad y_C = 1.2\,\text{ft}$$

$$V = 2\pi y_C A \qquad V = 25.1\,\text{ft}^3$$

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Problem 9-87

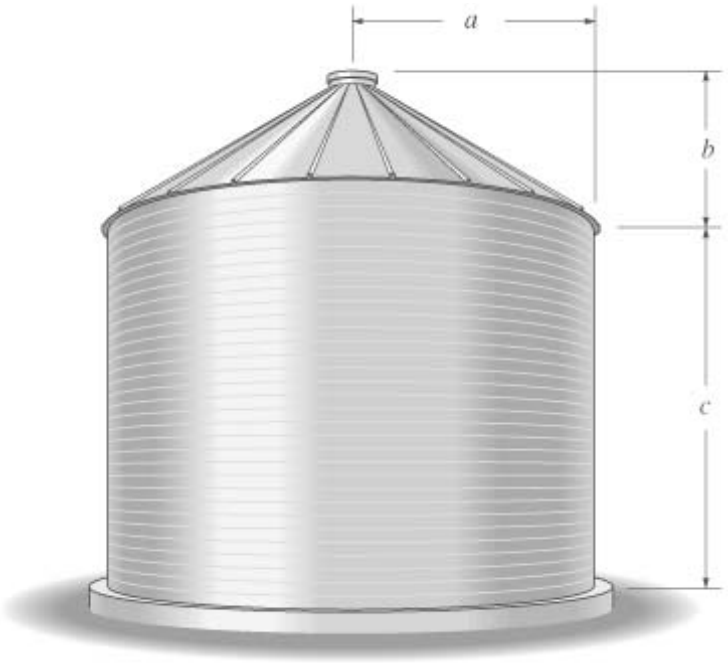
The grain bin of the type shown is manufactured by Grain Systems, Inc. Determine the required square footage of the sheet metal needed to form it, and also the maximum storage capacity (volume) within it.

Given:

$a = 30$ ft

$b = 20$ ft

$c = 45$ ft



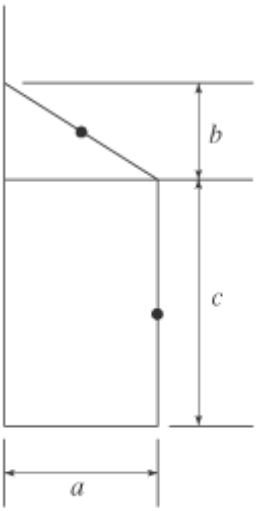
Solution:

$$A = 2\pi a c + 2\pi \frac{a}{2}\sqrt{a^2 + b^2}$$

$$A = 11.9 \times 10^3 \text{ ft}^2$$

$$V = 2\pi a c\left(\frac{a}{2}\right) + 2\pi \frac{a}{3}\left(\frac{1}{2} a b\right)$$

$$V = 146 \times 10^3 \text{ ft}^3$$



Problem 9-88

Determine the surface area and the volume of the conical solid.

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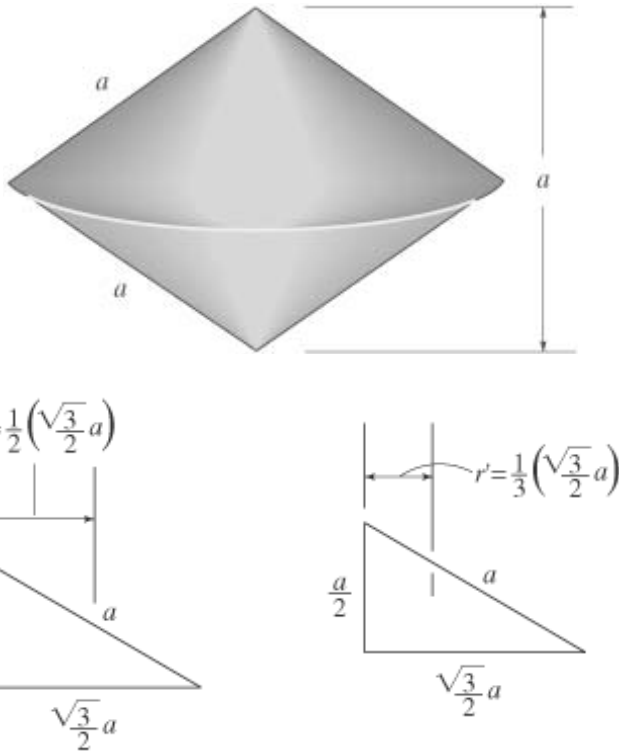
Solution:

$$A = 2a\frac{\sqrt{3}}{2}\frac{a}{2}2\pi$$

$$A = \sqrt{3}\pi a^2$$

$$V = 2\left(\frac{1}{2}\frac{a}{2}\right)\left(\frac{\sqrt{3}}{2}a\right)\left(\frac{\sqrt{3}}{6}a2\pi\right)$$

$$V = \frac{\pi}{4}a^3$$



Problem 9-89

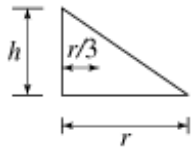
Sand is piled between two walls as shown. Assume the pile to be a quarter section of a cone and that ratio *p* of this volume is voids (air space). Use the second theorem of Pappus-Guldinus to determine the volume of sand.

Given:

$r = 3$ m

$h = 2$ m

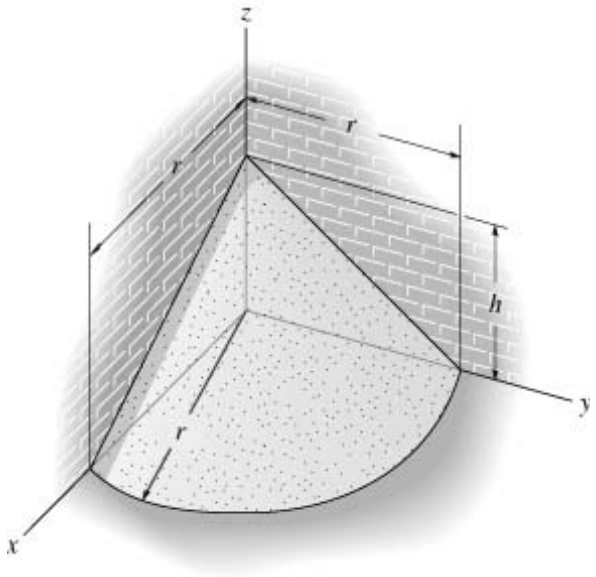
$p = 0.26$



Solution:

$$V = (1 - p)\left(\frac{\pi}{2}\right)\left(\frac{r}{3}\right)\left(\frac{hr}{2}\right)$$

$$V = 3.487\,\text{m}^3$$



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Problem 9-90

The *rim* of a flywheel has the cross section *A-A* shown. Determine the volume of material needed for its construction.

Given:

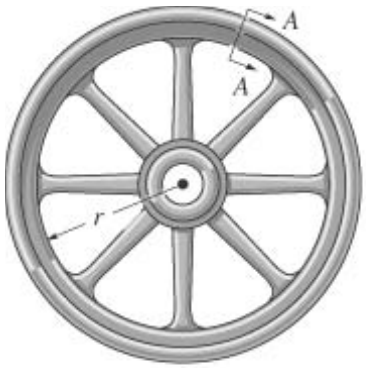
$r = 300\ \text{mm}$

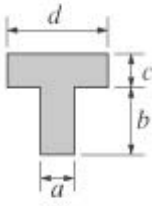
$a = 20\ \text{mm}$

$b = 40\ \text{mm}$

$c = 20\ \text{mm}$

$d = 60\ \text{mm}$



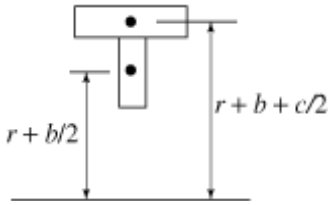


Section A-A

Solution:

$$V = 2\pi\left(r + b + \frac{c}{2}\right)dc + 2\pi\left(r + \frac{b}{2}\right)ba$$

$$V = 4.25 \times 10^6\ \text{mm}^3$$



Problem 9-91

The Gates Manufacturing Co. produces pulley wheels such as the one shown. Determine the weight of the wheel if it is made from steel having a specific weight γ.

Given:

$a = 1\ \text{in}$

$c = 0.5\ \text{in}$

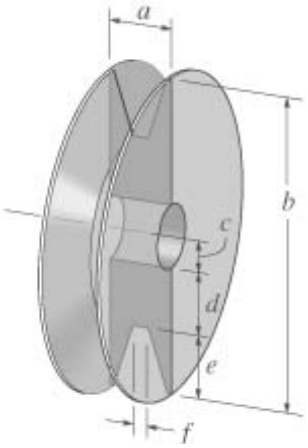
$d = 1\ \text{in}$

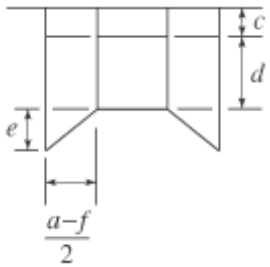
$e = 1\ \text{in}$

$f = 0.25\ \text{in}$

$b = 2(c + d + e)$

$\gamma = 490\ \dfrac{\text{lb}}{\text{ft}^3}$





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Solution:

$$W = \gamma 2\pi\left[d\,a\left(c + \frac{d}{2}\right) + \left(c + d + \frac{a}{3}\right)\left(\frac{a - f}{2}\right)e\right] \qquad W = 3.01\,\text{lb}$$

Problem 9-92

The Gates Manufacturing Co. produces pulley wheels such as the one shown. Determine the total surface area of the wheel in order to estimate the amount of paint needed to protect its surface from rust.

Given:

$a = 1$ in

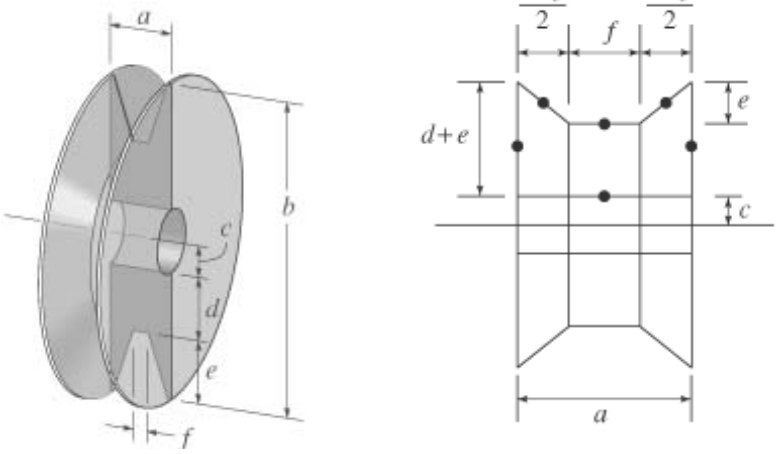
$c = 0.5$ in

$d = 1$ in

$e = 1$ in

$f = 0.25$ in

$b = 2(c + d + e)$



Solution:

$$A = 2\pi\left[f(c + d) + a\,c + 2(d + e)\left(c + \frac{d + e}{2}\right) + 2\sqrt{e^2 + \left(\frac{a - f}{2}\right)^2}\left(c + d + \frac{e}{2}\right)\right]$$

$A = 70\,\text{in}^2$

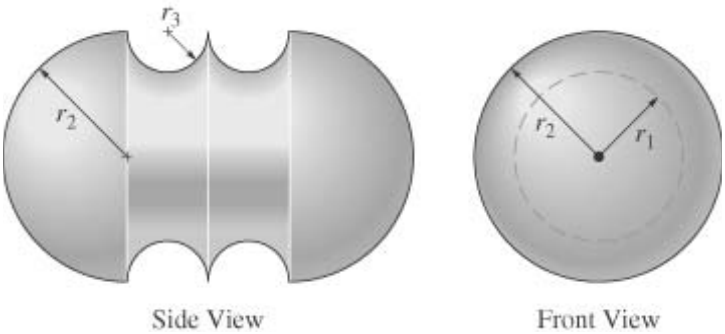
Problem 9-93

Determine the volume of material needed to make the casting.

Given:

$r_1 = 4$ in

$r_2 = 6$ in



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$r_3 = r_2 - r_1$

Solution:

$$V = 2\pi\left[2\left(\frac{\pi}{4}\right)r_2^2\left(\frac{4r_2}{3\pi}\right) + 2r_2(2r_3)\left(\frac{r_2}{2}\right) - 2\left(\frac{\pi}{2}\right)r_3^2\left(r_2 - \frac{4r_3}{3\pi}\right)\right]$$

$$V = 1.40 \times 10^3\ \text{in}^3$$

Problem 9-94

A circular sea wall is made of concrete. Determine the total weight of the wall if the concrete has a specific weight γ_c.

Given:

$\gamma_c = 150\ \frac{\text{lb}}{\text{ft}^3}$

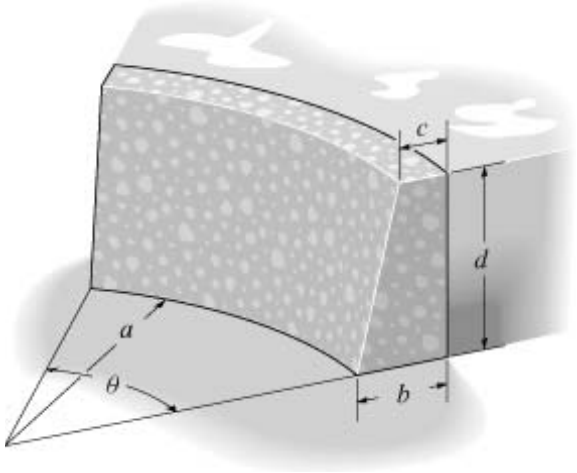
$a = 60$ ft

$b = 15$ ft

$c = 8$ ft

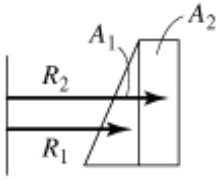
$d = 30$ ft

$\theta = 50$ deg



Solution:

$$W = \gamma_c\ \theta\left[a\left[\frac{1}{2}d(b-c)\right] + \frac{2}{3}(b-c)\left[\frac{1}{2}d(b-c)\right] + \left(a + b - \frac{c}{2}\right)dc\right]$$



$$W = 3.12 \times 10^6\ \text{lb}$$

Problem 9-95

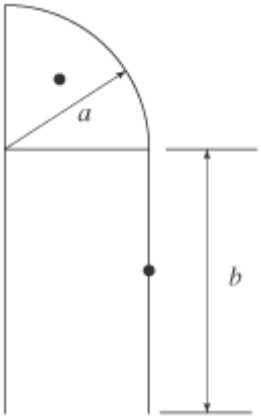
Determine the surface area of the tank, which consists of a cylinder and hemispherical cap.

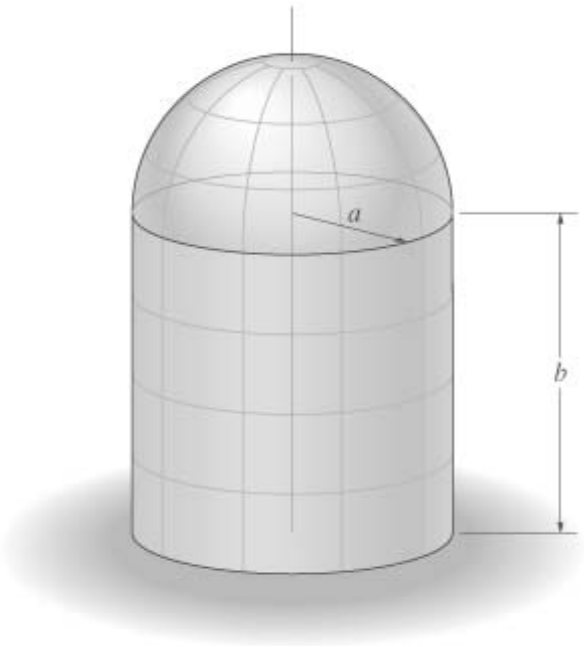
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Given:

$a = 4$ m

$b = 8$ m





Solution:

$$A = 2\pi\left(a\,b + \frac{2a}{\pi}\,\frac{\pi a}{2}\right)$$

$A = 302\ \text{m}^2$

Problem 9-96

Determine the volume of the tank, which consists of a cylinder and hemispherical cap.

Given:

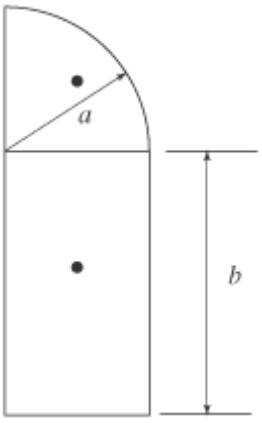
$a = 4$ m

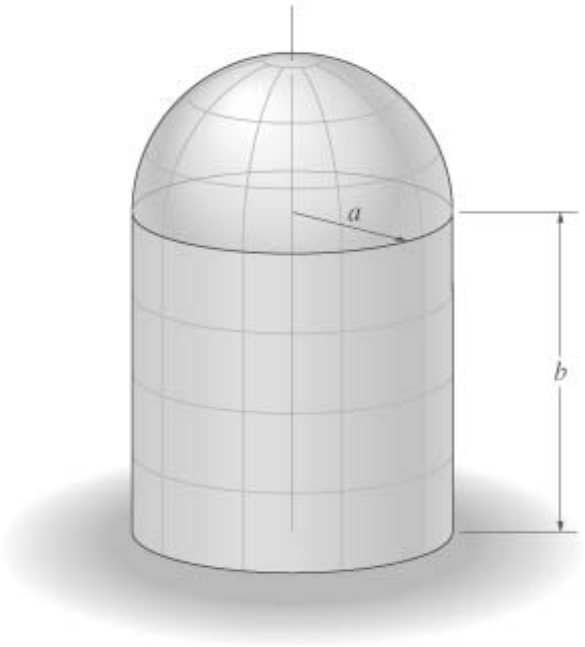
$b = 8$ m

Solution:

$$V = 2\pi\left[\frac{4a}{3\pi}\left(\frac{\pi a^2}{4}\right) + \frac{a}{2}(b\,a)\right]$$

$V = 536\ \text{m}^3$

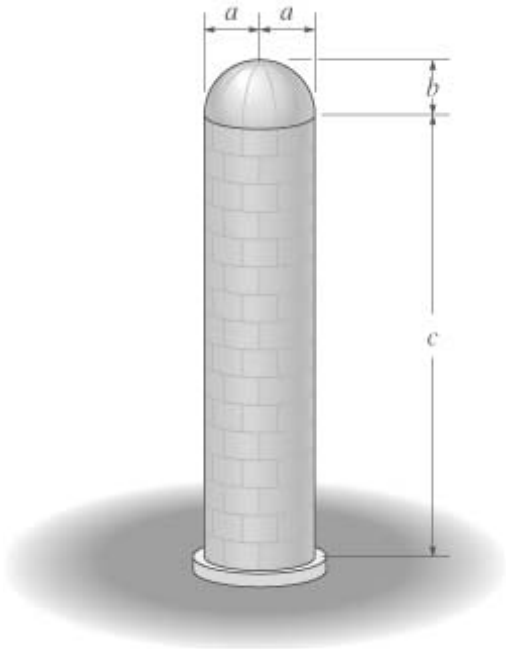




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Problem 9-97

Determine the surface area of the silo which consists of a cylinder and hemispherical cap. Neglect the thickness of the plates.



Given:

$a = 10$ ft

$b = 10$ ft

$c = 80$ ft

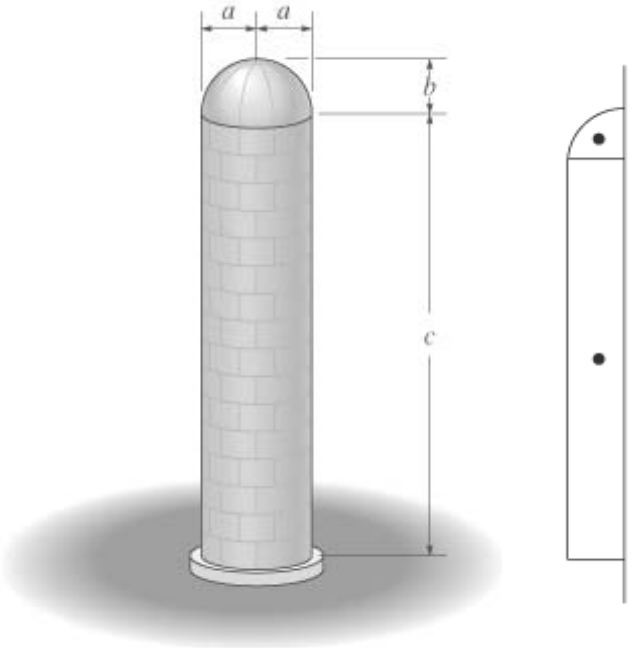
Solution:

$$A = 2\pi\left[\frac{2a}{\pi}\left(\frac{\pi a}{2}\right) + a\,c\right]$$

$$A = 5.65 \times 10^3\ \text{ft}^2$$

Problem 9-98

Determine the volume of the silo which consists of a cylinder and hemispherical cap. Neglect the thickness of the plates.



Given:

$a = 10$ ft

$b = 10$ ft

$c = 80$ ft

Solution:

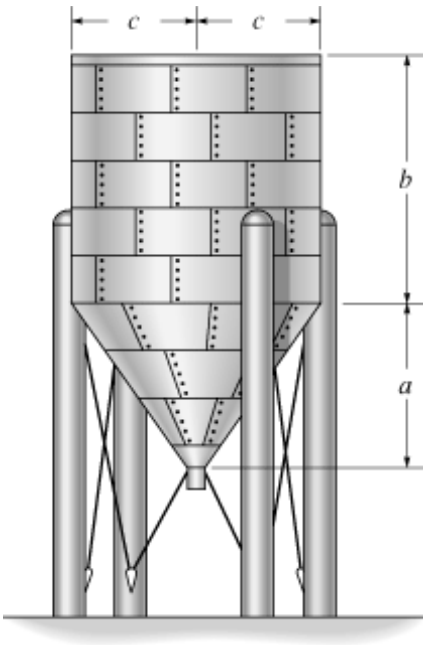
$$V = 2\pi\left[\frac{4a}{3\pi}\left(\frac{\pi a^2}{4}\right) + c\,a\left(\frac{a}{2}\right)\right]$$

$$V = 27.2 \times 10^3\ \text{ft}^3$$

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Problem 9-99

The process tank is used to store liquids during manufacturing. Estimate both the volume of the tank and its surface area. The tank has a flat top and the plates from which the tank is made have negligible thickness.



Given:

$a = 4$ m

$b = 6$ m

$c = 3$ m

Solution:

$$V = 2\pi\left[\frac{c}{3}\left(\frac{c\,a}{2}\right) + \frac{c}{2}(c\,b)\right]$$

$$V = 207\,\mathrm{m}^3$$

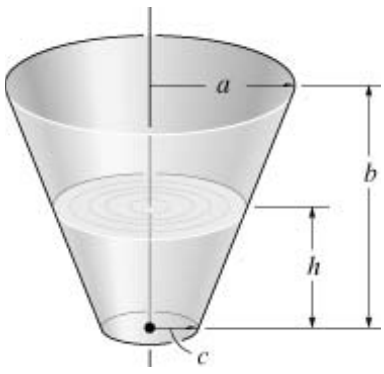
$$A = 2\pi\left(\frac{c}{2}c + c\,b + \frac{c}{2}\sqrt{a^2 + c^2}\right)$$

$$A = 188\ \mathrm{m}^2$$

Problem 9-100

Determine the height h to which liquid should be poured into the cup so that it contacts half the surface area on the inside of the cup. Neglect the cup's thickness for the calculation.

Given:



$a = 30$ mm

$b = 50$ mm

$c = 10$ mm

Solution:

Total area

$$A_{total} = 2\pi\left[c\frac{c}{2} + \frac{a + c}{2}\sqrt{b^2 + (a - c)^2}\right]$$

Guess $h = 1$ mm $e = 1$ mm

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Given

$$\frac{a-c}{b} = \frac{e-c}{h}$$

$$\frac{A_{total}}{2} = 2\pi\left[c\frac{c}{2} + \frac{e+c}{2}\sqrt{h^2 + (e-c)^2}\right]$$

$$\begin{pmatrix} e \\ h \end{pmatrix} = \text{Find}(e, h) \qquad e = 21.942\,\text{mm} \qquad h = 29.9\,\text{mm}$$

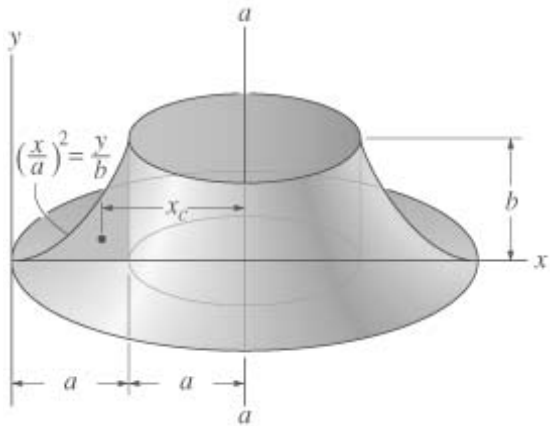
Problem 9-101

Using integration, compute both the area and the centroidal distance x_c of the shaded region. Then, using the second theorem of Pappus–Guldinus, compute the volume of the solid generated by revolving the shaded area about the *aa* axis.

Given:

$a = 8$ in

$b = 8$ in



Solution:

$$A = \int_0^a b\left(\frac{x}{a}\right)^2 dx$$

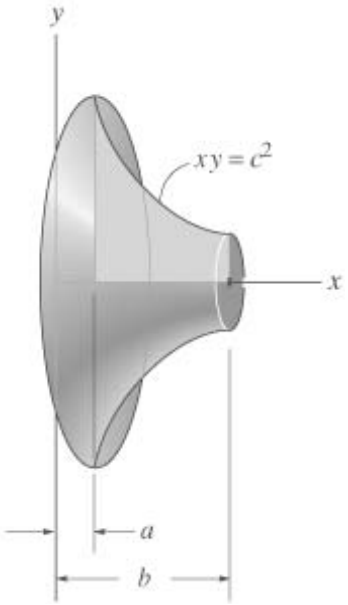
$$x_C = 2a - \frac{1}{A}\int_0^a x\,b\left(\frac{x}{a}\right)^2 dx \qquad A = 21.333\,\text{in}^2 \qquad x_C = 10\,\text{in}$$

$$V = 2\pi A x_C \qquad V = 1.34 \times 10^3\,\text{in}^3$$

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Problem 9-102

Using integration, determine the area and the centroidal distance y_c of the shaded area. Then, using the second theorem of Pappus–Guldinus, determine the volume of a solid formed by revolving the area about the x axis.



Given:

$a = 0.5$ ft

$b = 2$ ft

$c = 1$ ft

Solution:

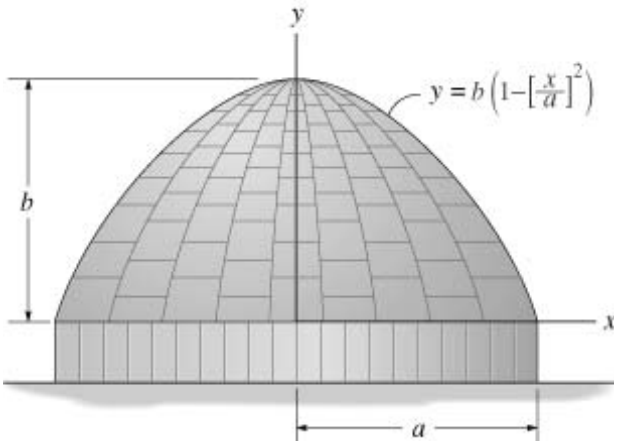
$$A = \int_a^b \frac{c^2}{x}\,dx \qquad A = 1.386\,\text{ft}^2$$

$$y_C = \frac{1}{A}\int_a^b \frac{1}{2}\left(\frac{c^2}{x}\right)^2 dx \qquad y_C = 0.541\,\text{ft}$$

$$V = 2\pi A\, y_C \qquad V = 4.71\,\text{ft}^3$$

Problem 9-103

Determine the surface area of the roof of the structure if it is formed by rotating the parabola about the y axis.



Given:

$a = 16$ m

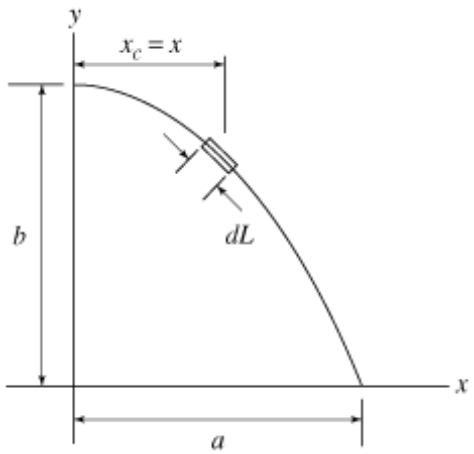
$b = 16$ m

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Solution:

Centroid : The length of the differential element is

$$dL = \sqrt{dx^2 + dy^2} = \left[\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\right] dx$$



and its centroid is $x_c = x$

Here, $\frac{dy}{dx} = -2\frac{bx}{a^2}$

Evaluating the integrals, we have

$$L = \int_0^a \sqrt{1 + \frac{4b^2 x^2}{a^4}}\, dx \qquad L = 23.663 \text{ m}$$

$$x_C = \frac{1}{L}\int_0^a x\sqrt{1 + \frac{4b^2 x^2}{a^4}}\, dx \qquad x_C = 9.178 \text{ m}$$

$$A = 2\pi x_C L \qquad A = 1.365 \times 10^3 \text{ m}^2$$

Problem 9-104

The suspension bunker is made from plates which are curved to the natural shape which a completely flexible membrane would take if subjected to a full load of coal.This curve may be approximated by a parabola, $y/b = (x/a)^2$. Determine the weight of coal which the bunker would contain when completely filled. Coal has a specific weight of γ, and assume there is a fraction loss p in volume due to air voids. Solve the problem by integration to determine the cross-sectional area of *ABC*; then use the second theorem of Pappus–Guldinus to find the volume.

Units Used:

kip $= 10^3$ lb

Given:

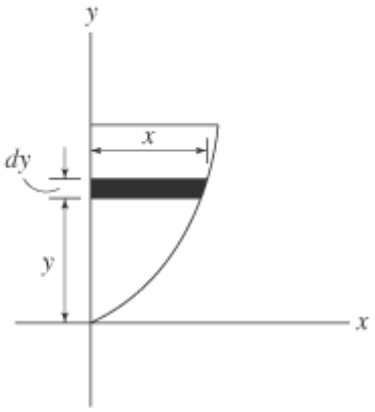
$a = 10$ ft

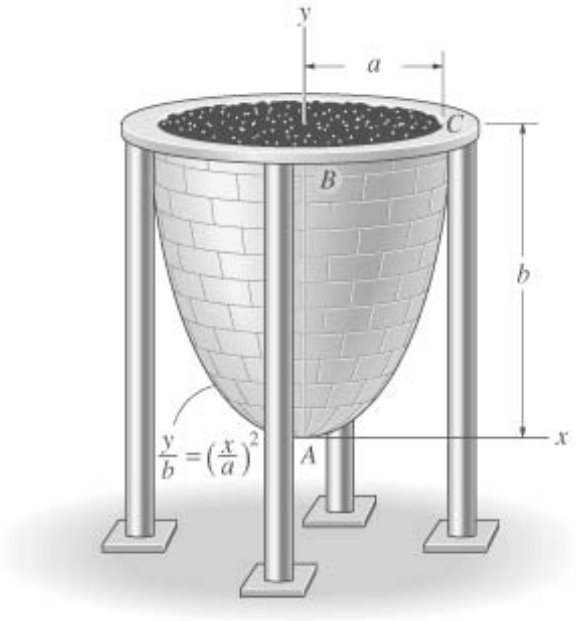
$b = 20$ ft

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$\gamma = 50 \dfrac{\text{lb}}{\text{ft}^3}$

$p = 0.2$





Solution:

$$A = \int_0^b a\sqrt{\frac{y}{b}}\,dy \qquad A = 133.3\,\text{ft}^2$$

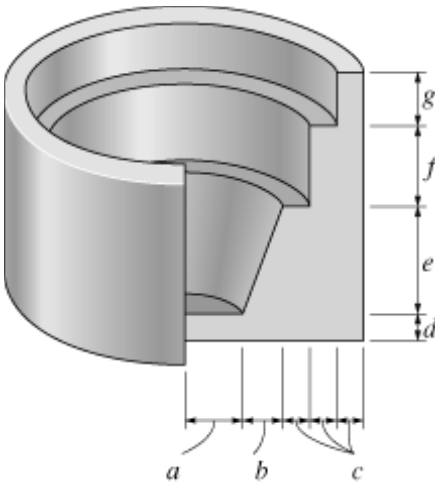
$$x_C = \frac{1}{A}\int_0^b \frac{1}{2}\left(a\sqrt{\frac{y}{b}}\right)^2 dy \qquad x_C = 3.75\,\text{ft}$$

$$V = 2\pi A x_C \qquad V = 3.142 \times 10^3\,\text{ft}^3$$

$$W = (1 - p)\gamma V \qquad W = 125.7\,\text{kip}$$

Problem 9-105

Determine the interior surface area of the brake piston. It consists of a full circular part. Its cross section is shown in the figure.



Given:

$a = 40$ mm

$b = 30$ mm

$c = 20$ mm

$d = 20$ mm

$e = 80$ mm

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$f = 60\ \text{mm}$

$g = 40\ \text{mm}$

Solution:

$$A = 2\pi\left[\frac{a}{2}a + \left(a + \frac{b}{2}\right)\sqrt{b^2 + e^2} + c\left(a + b + \frac{c}{2}\right) + (a + b + c)f + \left(a + b + \frac{3c}{2}\right)c \ldots + (a + b + 2c)g\right]$$

$A = 119 \times 10^3\ \text{mm}^2$

Problem 9-106

Determine the magnitude of the resultant hydrostatic force acting on the dam and its location H, measured from the top surface of the water. The width of the dam is w; the mass density is ρ_w.

Units Used:

$\text{Mg} = 10^3\ \text{kg}$

$\text{MN} = 10^6\ \text{N}$

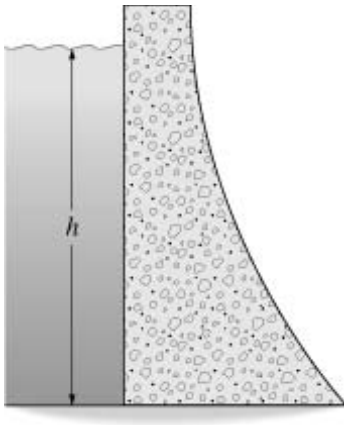
Given:

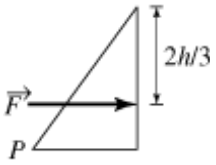
$w = 8\ \text{m}$

$\rho_w = 1\ \frac{\text{Mg}}{\text{m}^3}$

$h = 6\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$





Solution:

$p = h\rho_w g$ $\quad p = 58860\ \frac{\text{N}}{\text{m}^2}$

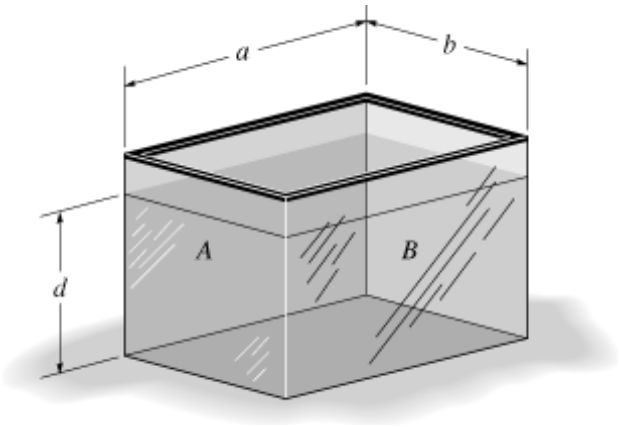
$F = \frac{1}{2}hwp$ $\quad F = 1.41\ \text{MN}$

$H = \left(\frac{2}{3}\right)h$ $\quad H = 4\ \text{m}$

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Problem 9-107

The tank is filled with water to a depth d. Determine the resultant force the water exerts on side A and side B of the tank. If oil instead of water is placed in the tank, to what depth d should it reach so that it creates the same resultant forces? The densities are ρ_0 and ρ_w.



Given: $\text{kN} = 10^3\ \text{N}$

$d = 4\ \text{m}$

$a = 3\ \text{m}$

$b = 2\ \text{m}$

$\rho_o = 900\ \frac{\text{kg}}{\text{m}^3}$

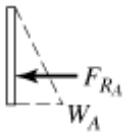
$\rho_w = 1000\ \frac{\text{kg}}{\text{m}^3}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

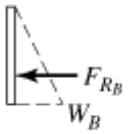
Solution:

For water

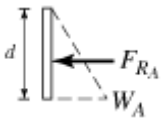
At side A: $W_A = b\ \rho_w\ g\ d \qquad W_A = 78480\ \frac{\text{N}}{\text{m}}$



$F_{RA} = \frac{1}{2} W_A d \qquad F_{RA} = 157\,\text{kN}$



At side B: $W_B = a\rho_w g d \qquad W_B = 117720\ \frac{\text{N}}{\text{m}}$



$F_{RB} = \frac{1}{2} W_B d \qquad F_{RB} = 235\,\text{kN}$

For oil

At side A: $F_{RA} = \frac{1}{2} b \rho_o g\, d_1\, d_1$

$d_1 = \sqrt{\frac{2F_{RA}}{b\rho_o g}} \qquad d_1 = 4.216\ \text{m}$

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Problem 9-108

The factor of safety for tipping of the concrete dam is defined as the ratio of the stabilizing moment about O due to the dam's weight divided by the overturning moment about O due to the water pressure. Determine this factor if the concrete has specific weight γ_{conc} and water has specific weight γ_w.

Given:

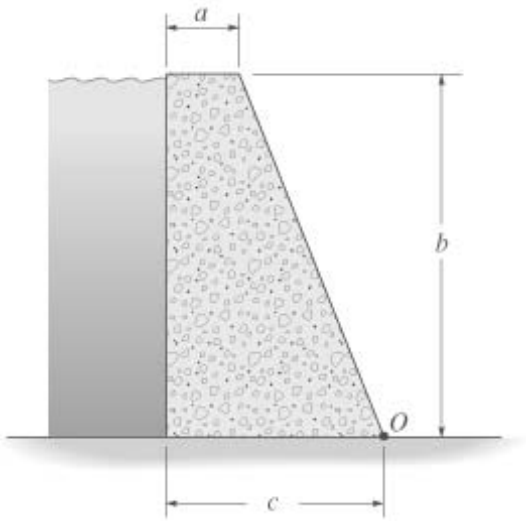
$a = 3$ ft

$b = 15$ ft

$c = 9$ ft

$\gamma_w = 62.4 \frac{\text{lb}}{\text{ft}^3}$

$\gamma_{conc} = 150 \frac{\text{lb}}{\text{ft}^3}$



Solution:

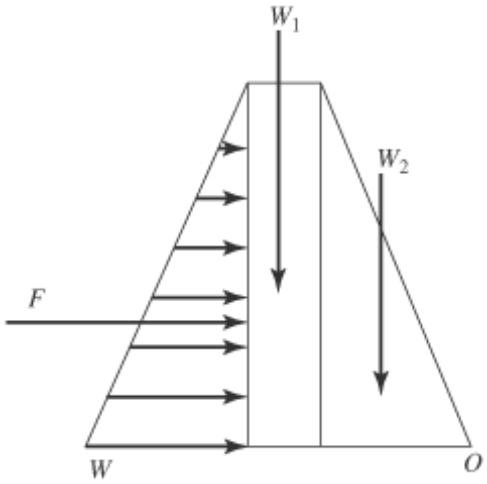
For a 1-ft thick section:

$W = \gamma_w b(1\text{ft})$ $\quad W = 936 \frac{\text{lb}}{\text{ft}}$

$F = \frac{1}{2} W b$ $\quad F = 7020\,\text{lb}$

$W_1 = \gamma_{conc}(1\text{ft}) a\, b$ $\quad W_1 = 6750\,\text{lb}$

$W_2 = \gamma_{conc} \frac{1}{2}(c - a) b(1\text{ft})$ $\quad W_2 = 6750\,\text{lb}$



Moment to overturn:

$M_O = F \frac{1}{3} b$ $\quad M_O = 35100\,\text{lb ft}$

Moment to stabilize:

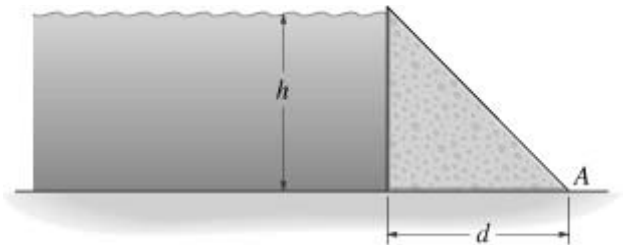
$M_S = W_1\left[(c - a) + \frac{a}{2}\right] + W_2\left[\frac{2}{3}(c - a)\right]$ $\quad M_S = 77625\,\text{lb}\cdot\text{ft}$

$F_s = \frac{M_S}{M_O}$ $\quad F_s = 2.21$

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Problem 9-109

The concrete "gravity" dam is held in place by its own weight. If the density of concrete is ρ_c and water has a density ρ_w, determine the smallest dimension d that will prevent the dam from overturning about its end A.



Units Used:

$$\text{Mg} = 10^3 \text{ kg}$$

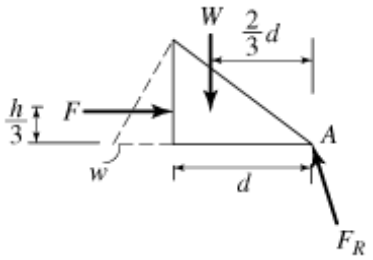
Given:

$$\rho_c = 2.5 \frac{\text{Mg}}{\text{m}^3}$$

$$\rho_w = 1.0 \frac{\text{Mg}}{\text{m}^3}$$

$$h = 6 \text{ m}$$

$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$



Solution:

Consider a dam of width $a = 1$ m.

$$w = \rho_w g h a \qquad w = 58860 \frac{\text{N}}{\text{m}} \qquad F = \frac{1}{2} w h \qquad F = 176580 \text{ N}$$

$$W = \frac{1}{2} \rho_c g d h a$$

Equilibrium $\qquad W \frac{2d}{3} - F \frac{h}{3} = 0$

$$\frac{1}{2} \rho_c g d h a \frac{2d}{3} = \frac{F h}{3}$$

$$d = \sqrt{\frac{F}{\rho_c g a}} \qquad d = 2.683 \text{ m}$$

Problem 9-110

The concrete dam is designed so that its face AB has a gradual slope into the water as shown. Because of this, the frictional force at the base BD of the dam is increased due to the hydrostatic force of the water acting on the dam. Calculate the hydrostatic force acting on the face AB of the dam. The dam has width w, the water density is γ_w.

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Units Used:

$\text{kip} = 10^3\ \text{lb}$

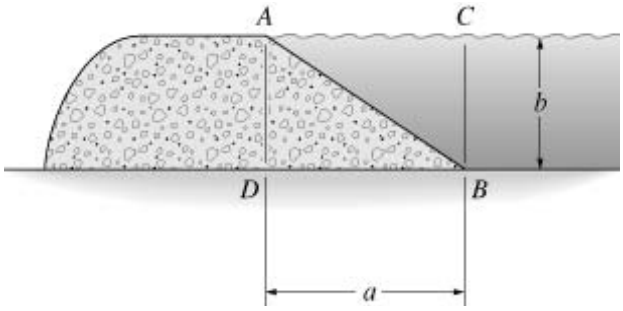
Given:

$w = 60\ \text{ft}$

$\gamma_w = 62.4\ \frac{\text{lb}}{\text{ft}^3}$

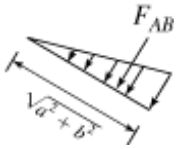
$a = 18\ \text{ft}$

$b = 12\ \text{ft}$



Solution:

$$F_{AB} = \frac{1}{2} w \gamma_w b \sqrt{a^2 + b^2} \qquad F_{AB} = 486\,\text{kip}$$



Problem 9-111

The symmetric concrete "gravity" dam is held in place by its own weight. If the density of concrete is ρ_c and water has a density ρ_w, determine the smallest distance d at its base that will prevent the dam from overturning about its end A.The dam has a width w.

Units Used:

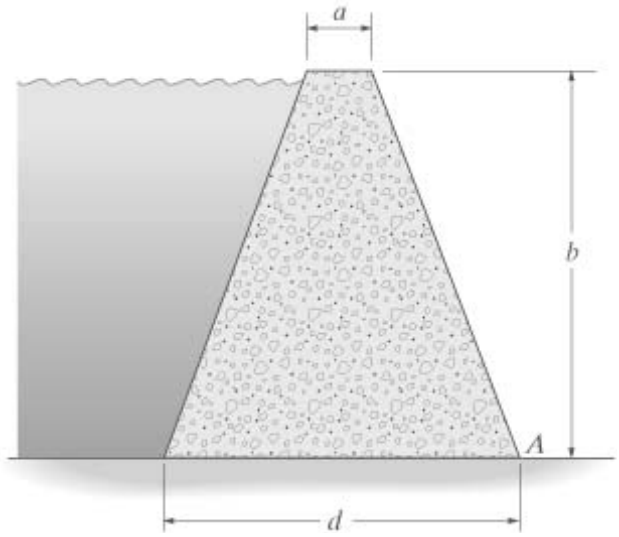
$\text{Mg} = 10^3\ \text{kg} \qquad \text{MN} = 10^6\ \text{N}$

Given:

$a = 1.5\ \text{m} \qquad \rho_c = 2.5\ \frac{\text{Mg}}{\text{m}^3}$

$b = 9\ \text{m}$

$w = 8\ \text{m} \qquad \rho_w = 1.0\ \frac{\text{Mg}}{\text{m}^3}$



Solution:

Guesses

$d = 3\ \text{m} \qquad F_h = 1\ \text{MN}$

$F_v = 1\ \text{MN} \qquad W = 1\ \text{MN}$

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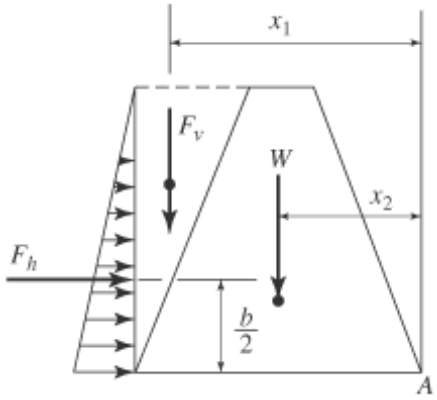
Given

$$F_v = \frac{d-a}{4} b\, w \rho_w g$$

$$F_h = \frac{1}{2} \rho_w g\, b\, w\, b$$

$$W = \rho_c g\, w \left[a\, b + \left(\frac{d-a}{2} \right) b \right]$$

$$W \frac{d}{2} + F_v \left(d - \frac{d-a}{6} \right) - F_h \frac{b}{3} = 0$$



$$\begin{pmatrix} F_v \\ F_h \\ W \\ d \end{pmatrix} = \text{Find}\left(F_v, F_h, W, d\right) \qquad \begin{pmatrix} F_v \\ F_h \\ W \end{pmatrix} = \begin{pmatrix} 0.379 \\ 3.178 \\ 4.545 \end{pmatrix} \text{MN} \qquad d = 3.65 \text{ m}$$

Problem 9-112

The tank is used to store a liquid having a specific weight γ. If it is filled to the top, determine the magnitude of force the liquid exerts on each of its two sides *ABDC* and *BDFE*.

Units used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

$$\gamma = 80 \frac{\text{lb}}{\text{ft}^3}$$

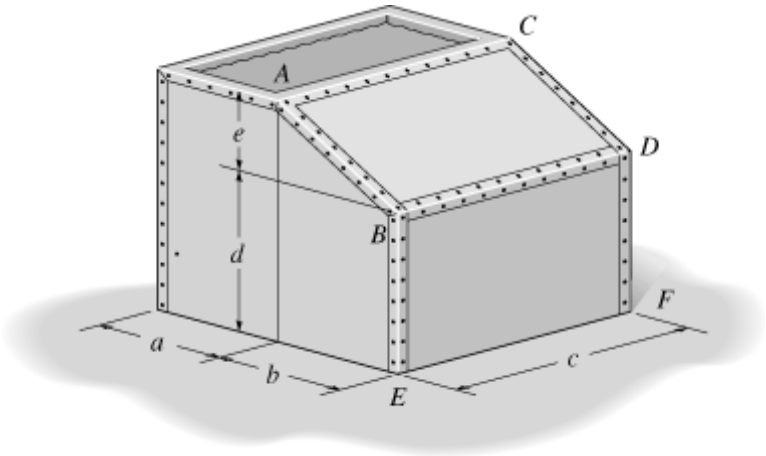
$$a = 6 \text{ ft}$$

$$b = 6 \text{ ft}$$

$$c = 12 \text{ ft}$$

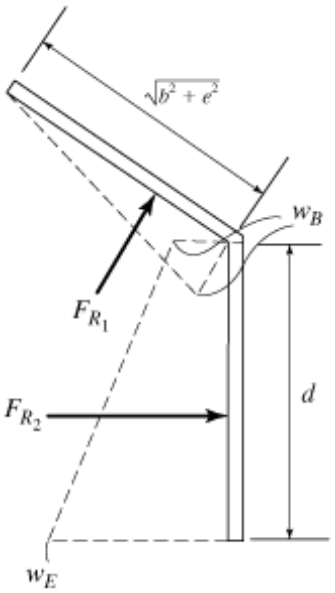
$$d = 8 \text{ ft}$$

$$e = 4 \text{ ft}$$



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Solution:



Fluid Pressure: The fluid pressure at points B and E can be determined using

$$p_B = \gamma e \qquad p_B = 320\,\frac{\text{lb}}{\text{ft}^2}$$

$$p_E = \gamma(e + d) \qquad p_E = 960\,\frac{\text{lb}}{\text{ft}^2}$$

Thus

$$w_B = p_B c \qquad w_B = 3.84\,\frac{\text{kip}}{\text{ft}}$$

$$w_E = p_E c \qquad w_E = 11.52\,\frac{\text{kip}}{\text{ft}}$$

Resultant Forces: The resultant Force acts on surface ABCD is

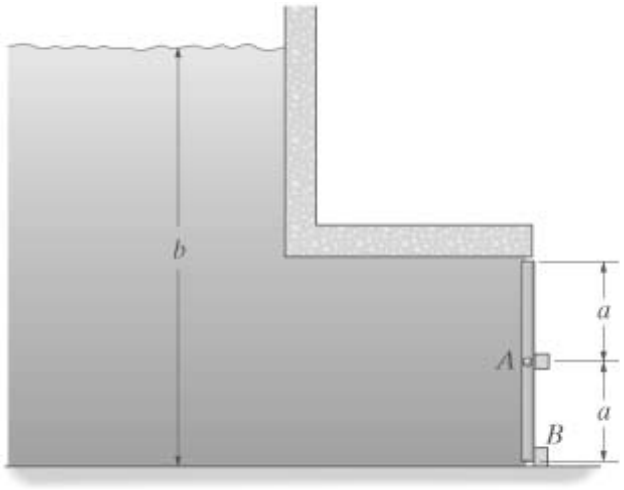
$$F_{R1} = \frac{1}{2} w_B \sqrt{e^2 + b^2} \qquad F_{R1} = 13.8\,\text{kip}$$

and on surface *BDFE* is

$$F_{R2} = \frac{1}{2}\left(w_B + w_E\right) d \qquad F_{R2} = 61.4\,\text{kip}$$

Problem 9-113

The rectangular gate of width w is pinned at its center A and is prevented from rotating by the block at B. Determine the reactions at these supports due to hydrostatic pressure.



Units Used:

$$\text{Mg} = 10^3\ \text{kg} \qquad \text{kN} = 10^3\ \text{N}$$

Given:

$$a = 1.5\ \text{m} \qquad \rho_w = 1.0\,\frac{\text{Mg}}{\text{m}^3}$$

$$b = 6\ \text{m}$$

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$w = 2\ \text{m} \qquad g = 9.81\ \frac{\text{m}}{\text{s}^2}$

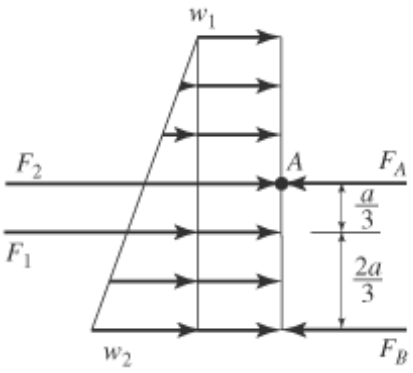
Solution:

$w_1 = \rho_w g(b - 2a)w \qquad w_1 = 59\ \frac{\text{kN}}{\text{m}}$

$w_2 = \rho_w g 2a\, w \qquad w_2 = 59\ \frac{\text{kN}}{\text{m}}$

$F_1 = \frac{1}{2} 2a\, w_1 \qquad F_1 = 88\,\text{kN}$

$F_2 = w_2 2a \qquad F_2 = 177\,\text{kN}$



$\Sigma M_A = 0; \quad F_1 \frac{a}{3} - F_B a = 0 \qquad F_B = \frac{1}{3} F_1 \qquad F_B = 29.4\,\text{kN}$

$\Sigma F_x = 0; \quad F_1 + F_2 - F_B - F_A = 0 \qquad F_A = F_1 + F_2 - F_B \qquad F_A = 235\,\text{kN}$

Problem 9-114

The gate *AB* has width *w*. Determine the horizontal and vertical components of force acting on the pin at *B* and the vertical reaction at the smooth support *A*. The density of water is ρ_w.

Units Used:

$\text{Mg} = 10^3\ \text{kg}$

$\text{kN} = 10^3\ \text{N}$

$\text{MN} = 10^6\ \text{N}$

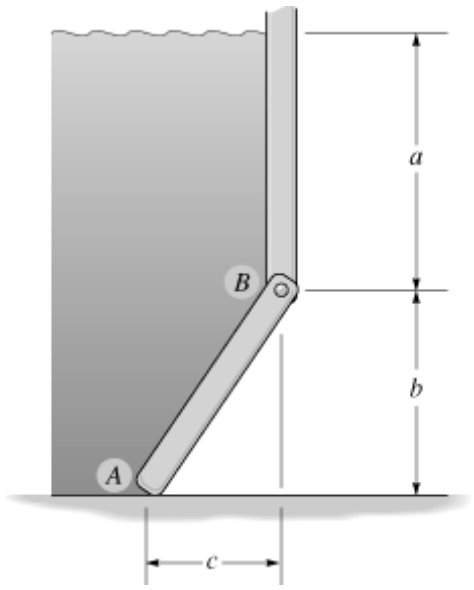
Given:

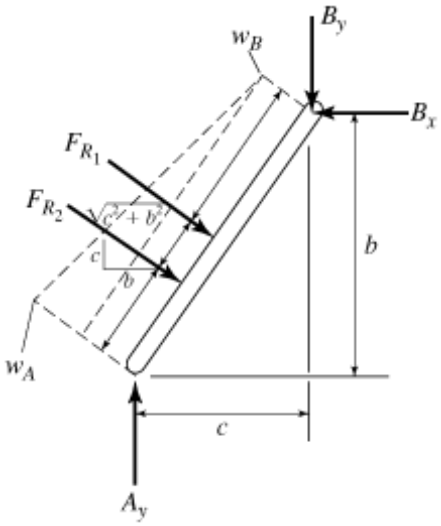
$w = 8\ \text{m}$

$\rho_w = 1.0\ \frac{\text{Mg}}{\text{m}^3}$

$a = 5\ \text{m}$

$b = 4\ \text{m}$





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$c = 3\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

Solution:

Fluid Pressure: The fluid pressure at points *A* and *B* can be determined using Eq. 9-15,

$p_A = \rho_w g(a + b)$ $\quad p_A = 88.29\ \frac{\text{kN}}{\text{m}^2}$ $\quad w_A = p_A w$ $\quad w_A = 706.32\ \frac{\text{kN}}{\text{m}}$

$p_B = \rho_w g\, a$ $\quad p_B = 49.05\ \frac{\text{kN}}{\text{m}^2}$ $\quad w_B = p_B w$ $\quad w_B = 392.4\ \frac{\text{kN}}{\text{m}}$

Equilibrium

$$w_B \frac{b^2 + c^2}{2} + \frac{1}{2}\left(w_A - w_B\right)\frac{b^2 + c^2}{3} - A_y c = 0$$

$$A_y = \frac{w_B \frac{b^2 + c^2}{2} + \frac{1}{2}\left(w_A - w_B\right)\frac{2\left(b^2 + c^2\right)}{3}}{c} \qquad A_y = 2.507\ \text{MN}$$

$$A_y - w_B c - \frac{1}{2}\left(w_A - w_B\right)c - B_y = 0$$

$$B_y = A_y - w_B c - \frac{1}{2}\left(w_A - w_B\right)c \qquad B_y = 858.92\ \text{kN}$$

$$-B_x + w_B b + \frac{1}{2}\left(w_A - w_B\right)b = 0$$

$$B_x = w_B b + \frac{1}{2}\left(w_A - w_B\right)b \qquad B_x = 2.197\ \text{MN}$$

Problem 9-115

The storage tank contains oil having a specific weight γ. If the tank has width *w*, calculate the resultant force acting on the inclined side *BC* of the tank, caused by the oil, and specify its location along *BC*, measured from *B*. Also compute the total resultant force acting on the bottom of the tank.

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Units Used:

$\text{kip} = 10^3\ \text{lb}$

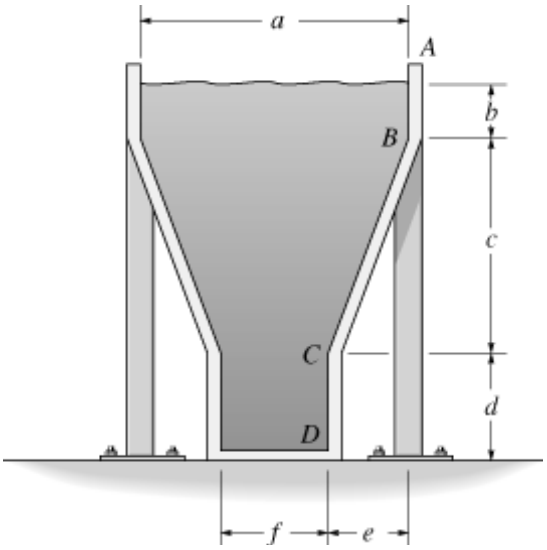
Given:

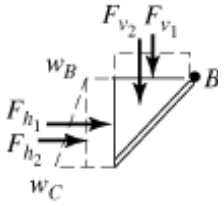
$\gamma = 56\ \frac{\text{lb}}{\text{ft}^3}$ $c = 8\ \text{ft}$

$w = 6\ \text{ft}$ $d = 4\ \text{ft}$

$a = 10\ \text{ft}$ $e = 3\ \text{ft}$

$b = 2\ \text{ft}$ $f = 4\ \text{ft}$





Solution:

$w_B = w\gamma b$ $w_B = 672\ \frac{\text{lb}}{\text{ft}}$ $w_C = w\gamma(b + c)$ $w_C = 3360\ \frac{\text{lb}}{\text{ft}}$

$F_{h1} = w_B c$ $F_{h2} = \frac{1}{2}(w_C - w_B)c$ $F_{v1} = \gamma w b e$ $F_{v2} = \frac{1}{2}\gamma w c e$

The resultant force

$F_{Rx} = F_{h1} + F_{h2}$ $F_{Ry} = F_{v1} + F_{v2}$ $F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$ $F_R = 17.225\ \text{kip}$

The location h measured from point B Guess $h = 1\ \text{ft}$ Given

$$F_{v1}\frac{e}{2} + F_{v2}\frac{2e}{3} + F_{h1}\frac{c}{2} + F_{h2}\frac{2c}{3} = F_{Rx}\frac{c\,h}{\sqrt{c^2 + e^2}} + F_{Ry}\frac{e\,h}{\sqrt{c^2 + e^2}}$$

$h = \text{Find}(h)$ $h = 5.221\ \text{ft}$

On the bottom of the tank

$F_{bot} = \gamma w f(b + c + d)$ $F_{bot} = 18.816\ \text{kip}$

Problem 9-116

The arched surface *AB* is shaped in the form of a quarter circle. If it has a length *L*, determine the horizontal and vertical components of the resultant force caused by the water acting on the surface. The density of water is ρ_w.

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Units Used:

$\text{Mg} = 10^3\ \text{kg}$

$\text{kN} = 10^3\ \text{N}$

Given:

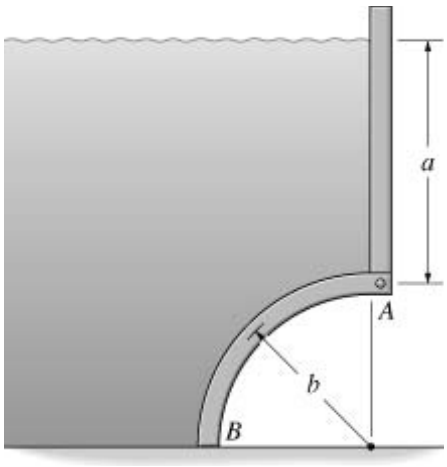
$L = 8\ \text{m}$

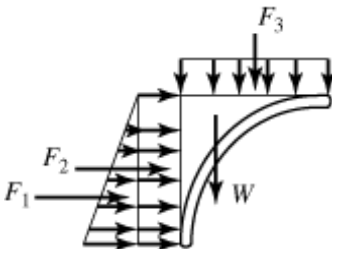
$\rho_w = 1.0\ \frac{\text{Mg}}{\text{m}^3}$

$a = 3\ \text{m}$

$b = 2\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$





Solution:

$F_3 = \rho_w g a b L$ $\quad F_3 = 470.88\,\text{kN}$

$F_2 = \rho_w g a b L$ $\quad F_2 = 470.88\,\text{kN}$

$F_1 = \rho_w g \frac{b}{2} b L$ $\quad F_1 = 156.96\,\text{kN}$

$W = \left(b^2 - \frac{\pi b^2}{4}\right) L \rho_w g$ $\quad W = 67.368\,\text{kN}$

$F_x = F_1 + F_2$ $\quad F_x = 628\,\text{kN}$

$F_y = F_2 + W$ $\quad F_y = 538\,\text{kN}$

Problem 9-117

The rectangular bin is filled with coal, which creates a pressure distribution along wall A that varies as shown, i.e. $p = p_0(z/b)^3$. Determine the resultant force created by the coal and specify its location measured from the top surface of the coal.

Units used:

$\text{kip} = 10^3\ \text{lb}$

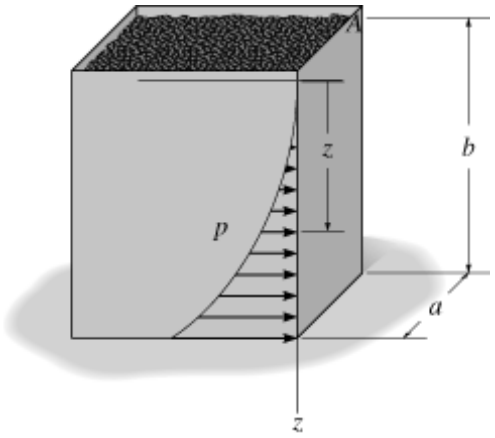
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Given:

$a = 4\ \text{ft}$

$b = 10\ \text{ft}$

$p_0 = 4000\ \frac{\text{lb}}{\text{ft}^2}$



Solution:

Resultant Force and its location:

$$F = \int_0^b p_0 \left(\frac{z}{b}\right)^3 a\ \text{dz} \qquad F = 40\,\text{kip}$$

$$z_C = \frac{1}{F}\int_0^b z\, p_0 \left(\frac{z}{b}\right)^3 a\ \text{dz} \qquad z_C = 8\,\text{ft}$$

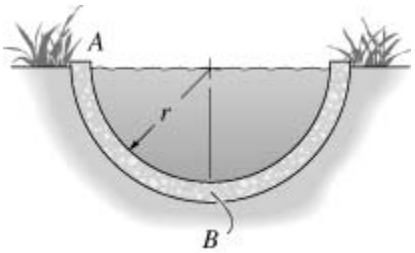
Problem 9-118

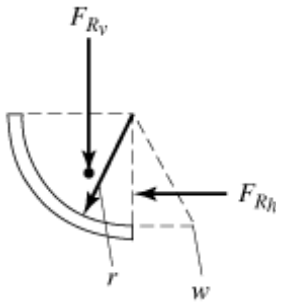
The semicircular drainage pipe is filled with water. Determine the resultant horizontal and vertical force components that the water exerts on the side *AB* of the pipe per foot of pipe length; water has density γ.

Given:

$\gamma = 62.4\ \frac{\text{lb}}{\text{ft}^3}$

$r = 2\ \text{ft}$





Solution:

$w = \gamma r \qquad w = 124.8\ \frac{\text{lb}}{\text{ft}^2}$

Resultant forces (per unit foot):

$$F_{Rh} = \frac{1}{2} w r \qquad F_{Rh} = 124.8\ \frac{\text{lb}}{\text{ft}}$$

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$$F_{Rv} = \gamma \frac{\pi r^2}{4} \qquad F_{Rv} = 196.0 \frac{\text{lb}}{\text{ft}}$$

Problem 9-119

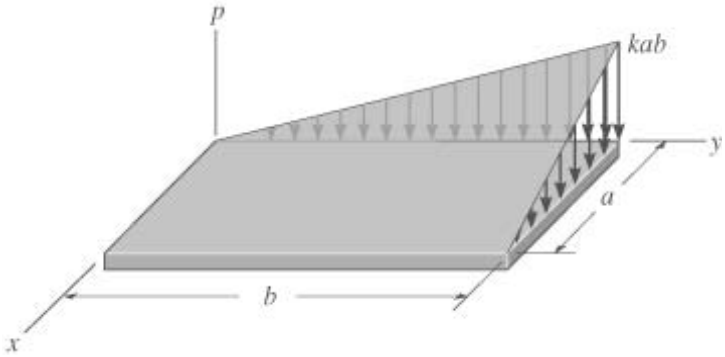
The load over the plate varies linearly along the sides of the plate such that $p = k\,y\,(a\text{-}x)$. Determine the magnitude of the resultant force and the coordinates (x_c, y_c) of the point where the line of action of the force intersects the plate.

Given:

$a = 2$ ft

$b = 6$ ft

$k = 10 \dfrac{\text{lb}}{\text{ft}^4}$



Solution:

$$p(x, y) = k\,y(a - x)$$

$$F_R = \int_0^a \int_0^b p(x, y)\, \mathrm{d}y\, \mathrm{d}x \qquad F_R = 360\,\text{lb}$$

$$x_C = \frac{1}{F_R} \int_0^a \int_0^b x\, p(x, y)\, \mathrm{d}y\, \mathrm{d}x \qquad x_C = 0.667\,\text{ft}$$

$$y_C = \frac{1}{F_R} \int_0^a \int_0^b y\, p(x, y)\, \mathrm{d}y\, \mathrm{d}x \qquad y_C = 4\,\text{ft}$$

Problem 9-120

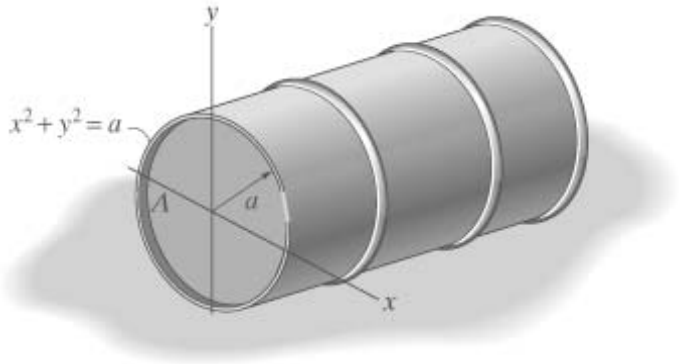
The drum is filled to its top ($y = a$) with oil having a density γ. Determine the resultant force of the oil pressure acting on the flat end of plate A of the drum and specify its location measured from the top of the drum.

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Given:

$a = 1.5 \text{ ft}$

$\gamma = 55 \dfrac{\text{lb}}{\text{ft}^3}$



Solution:

$$F_R = \int_{-a}^{a} \gamma 2\sqrt{a^2 - y^2}(a - y)\, dy \qquad F_R = 583\,\text{lb}$$

$$d = a - \frac{1}{F_R}\int_{-a}^{a} y\gamma 2\sqrt{a^2 - y^2}(a - y)\, dy \qquad d = 1.875\,\text{ft}$$

Problem 9-121

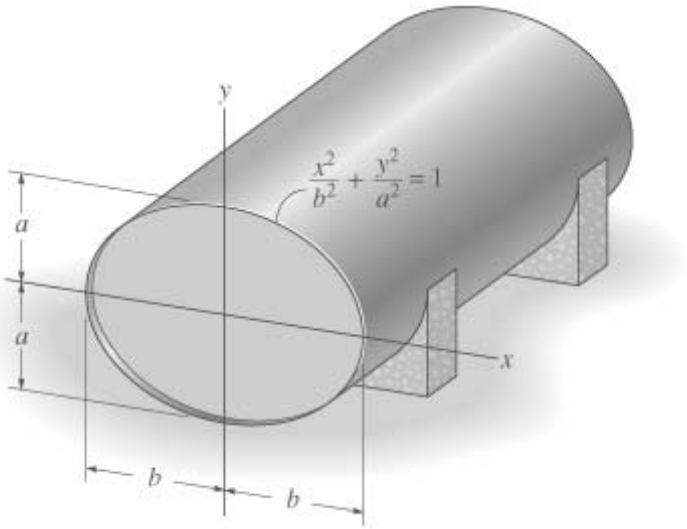
The gasoline tank is constructed with elliptical ends on each side of the tank. Determine the resultant force and its location on these ends if the tank is half full.

Given:

$a = 3 \text{ ft}$

$b = 4 \text{ ft}$

$\gamma = 41 \dfrac{\text{lb}}{\text{ft}^3}$



Solution:

$$F_R = \int_{-a}^{0} -\gamma y 2\left(\frac{b}{a}\sqrt{a^2 - y^2}\right) dy \qquad F_R = 984\,\text{lb}$$

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$$y_C = \frac{1}{F_R}\int_{-a}^{0} y\left[-\gamma y 2\left(\frac{b}{a}\sqrt{a^2 - y^2}\right)\right] dy \qquad y_C = -1.767\,\text{ft} \qquad x_C = 0\ \text{ft}$$

Problem 9-122

The loading acting on a square plate is represented by a parabolic pressure distribution. Determine the magnitude of the resultant force and the coordinates (x_c, y_c) of the point where the line of action of the force intersects the plate. Also, what are the reactions at the rollers B and C and the ball-and-socket joint A? Neglect the weight of the plate.

Units Used:

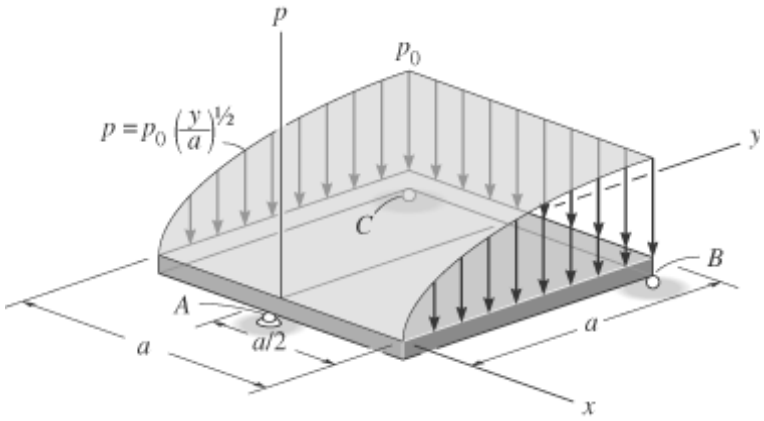
$\text{kPa} = 10^3\ \text{Pa}$

$\text{kN} = 10^3\ \text{N}$

Given:

$a = 4\ \text{m}$

$p_0 = 4\ \text{kPa}$

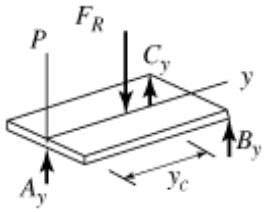


Solution:

Due to symmetry

$$x_C = 0$$

$$F_R = \int_0^a p_0\sqrt{\frac{y}{a}}\,a\,dy \qquad F_R = 42.667\,\text{kN}$$



$$y_C = \frac{1}{F_R}\int_0^a y\,p_0\sqrt{\frac{y}{a}}\,a\,dy \qquad y_C = 2.4\ \text{m}$$

Equilibrium Guesses $A_y = 1\ \text{kN}$ $B_y = 1\ \text{kN}$ $C_y = 1\ \text{kN}$

Given $A_y + B_y + C_y - F_R = 0$

$$(B_y + C_y)a - F_R y_C = 0$$

$$B_y\frac{a}{2} - C_y\frac{a}{2} = 0$$

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$$\begin{pmatrix} A_y \\ B_y \\ C_y \end{pmatrix} = \text{Find}(A_y, B_y, C_y) \qquad \begin{pmatrix} A_y \\ B_y \\ C_y \end{pmatrix} = \begin{pmatrix} 17.067 \\ 12.8 \\ 12.8 \end{pmatrix} \text{kN}$$

Problem 9-123

The tank is filled with a liquid which has density ρ. Determine the resultant force that it exerts on the elliptical end plate, and the location of the center of pressure, measured from the x axis.

Units Used:

$$\text{kN} = 10^3 \text{ N}$$

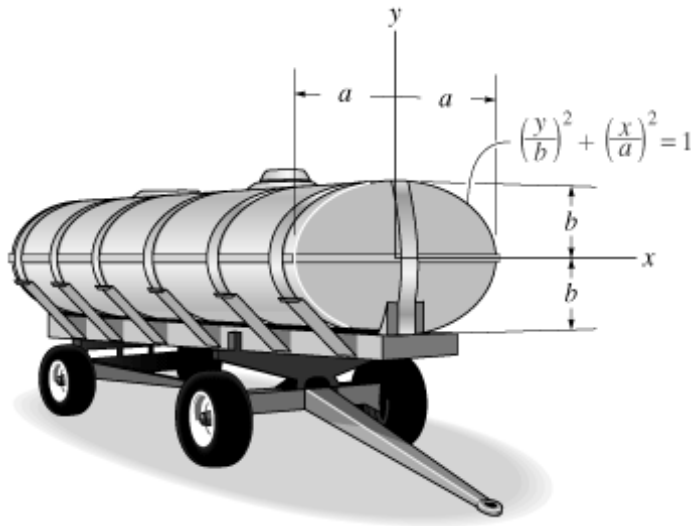
Given:

$$a = 1 \text{ m}$$

$$b = 0.5 \text{ m}$$

$$\rho = 900 \frac{\text{kg}}{\text{m}^3}$$

$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$



Solution:

$$F_R = \int_{-b}^{b} \rho g 2a \sqrt{1 - \left(\frac{y}{b}\right)^2} (b - y)\, dy \qquad F_R = 6.934 \text{ kN}$$

$$y_C = \frac{1}{F_R} \int_{-b}^{b} y \rho g 2a \sqrt{1 - \left(\frac{y}{b}\right)^2} (b - y)\, dy \qquad y_C = -0.125 \text{ m}$$

Problem 9-124

A circular V-belt has an inner radius r and a cross-sectional area as shown. Determine the volume of material required to make the belt.

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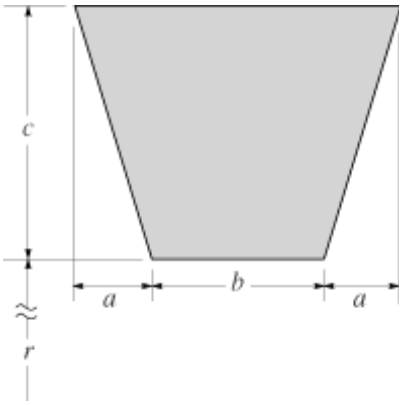
Given:

$r = 600$ mm

$a = 25$ mm

$b = 50$ mm

$c = 75$ mm



Solution:

$$V = 2\pi\left[\left(r + \frac{c}{3}\right)2\left(\frac{1}{2}\right)a\,c + \left(r + \frac{c}{2}\right)b\,c\right] \qquad V = 22.4 \times 10^{-3}\ \text{m}^3$$

Problem 9-125

A circular V-belt has an inner radius r and a cross-sectional area as shown. Determine the surface area of the belt.

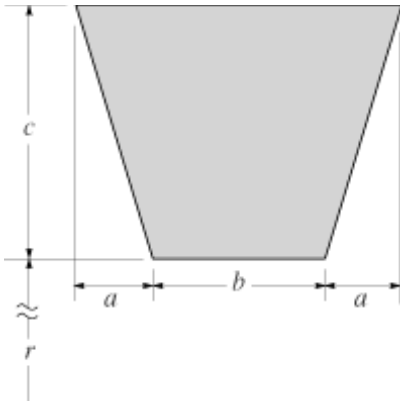
Given:

$r = 600$ mm

$a = 25$ mm

$b = 50$ mm

$c = 75$ mm



Solution:

$$A = 2\pi\left[r\,b + 2\left(r + \frac{c}{2}\right)\sqrt{a^2 + c^2} + (r + c)(b + 2a)\right] \qquad A = 1.246\ \text{m}^2$$

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Problem 9-126

Locate the center of mass of the homogeneous rod.
Given:

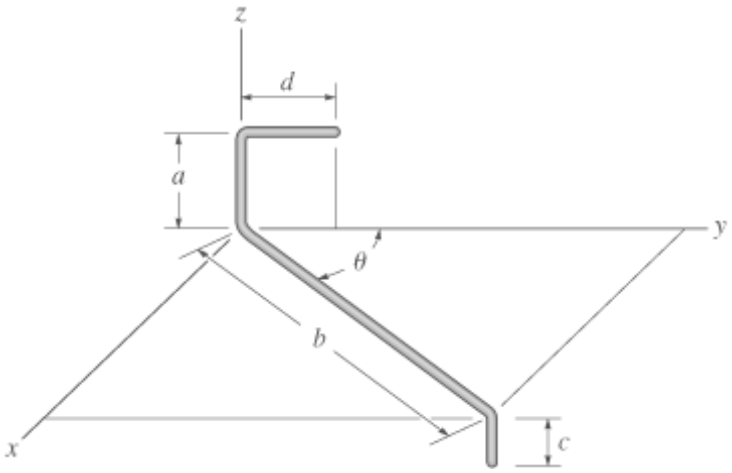
$a = 200$ mm

$b = 600$ mm

$c = 100$ mm

$d = 200$ mm

$\theta = 45$ deg



Solution:

$L = a + b + c + d$

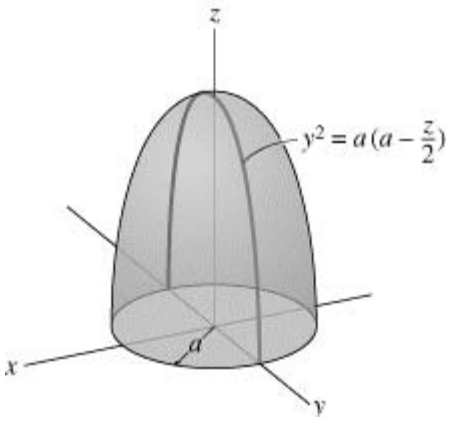
$$x_C = \frac{1}{L}\left(b\frac{b}{2}\sin(\theta) + c\,b\sin(\theta)\right) \qquad x_C = 154.3\,\text{mm}$$

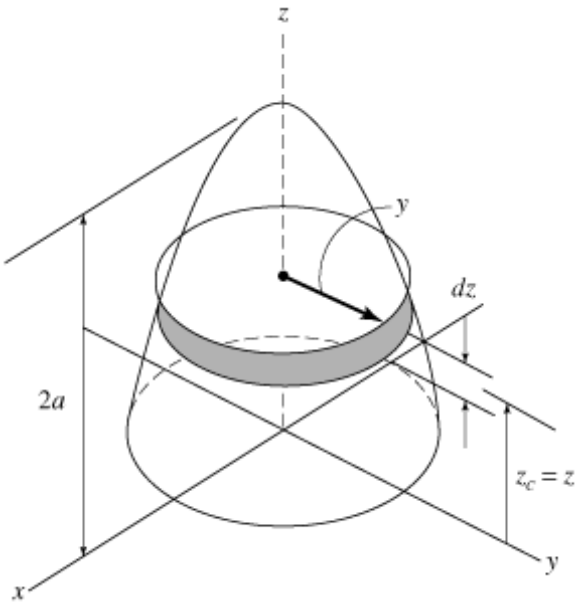
$$y_C = \frac{1}{L}\left(d\frac{d}{2} + b\frac{b}{2}\cos(\theta) + c\,b\cos(\theta)\right) \qquad y_C = 172.5\,\text{mm}$$

$$z_C = \frac{1}{L}\left(a\frac{a}{2} + d\,a - c\frac{c}{2}\right) \qquad z_C = 50.0\,\text{mm}$$

Problem 9-127

Locate the centroid of the solid





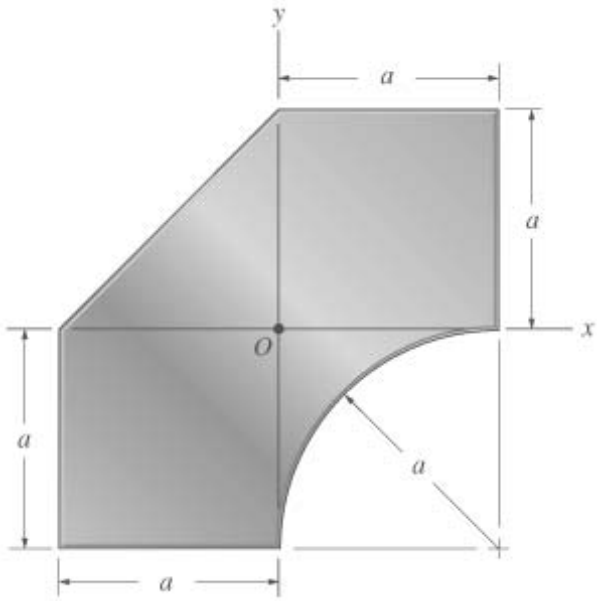
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Solution:

$$y_C = \frac{\int_0^{2a} z\pi a\left(a - \frac{z}{2}\right) dz}{\int_0^{2a} \pi a\left(a - \frac{z}{2}\right) dz} = \frac{2\,a^4\,\pi - a^3\,\pi\,z}{\pi\,a^3} \qquad y_C = \frac{2}{3}\,a$$

Problem 9-128

Locate the centroid (x_c, y_c) of the thin plate.



Given:

$a = 6$ in

Solution:

$$A = 4a^2 - \frac{a^2}{2} - \frac{\pi a^2}{4} \qquad A = 97.7\,\text{in}^2$$

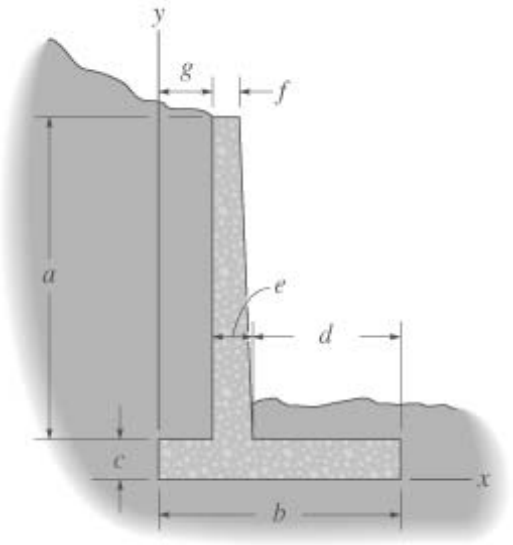
$$x_C = \frac{1}{A}\left[\frac{-a^2}{2}\left(\frac{-2}{3}a\right) - \frac{\pi a^2}{4}\left(a - \frac{4a}{3\pi}\right)\right] \qquad x_C = -0.262\,\text{in}$$

$$y_C = \frac{1}{A}\left[\frac{-a^2}{2}\,\frac{2a}{3} - \frac{\pi a^2}{4}\left(\frac{4a}{3\pi} - a\right)\right] \qquad y_C = 0.262\,\text{in}$$

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Problem 9-129

Determine the weight and location (x_c, y_c) of the center of gravity G of the concrete retaining wall. The wall has a length L, and concrete has a specific gravity of γ.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$a = 12 \text{ ft}$ $f = 1 \text{ ft}$

$b = 9 \text{ ft}$ $g = 2 \text{ ft}$

$c = 1.5 \text{ ft}$ $L = 10 \text{ ft}$

$d = 5.5 \text{ ft}$ $\gamma = 150 \dfrac{\text{lb}}{\text{ft}^3}$

$e = 1.5 \text{ ft}$

Solution:

$$A = b\,c + a\,f + \frac{1}{2}a(e - f) \qquad W = \gamma A L \qquad W = 42.8 \text{ kip}$$

$$x_C = \frac{1}{A}\left[b\,c\frac{b}{2} + a\,f\left(g + \frac{f}{2}\right) + \frac{1}{2}a(e - f)\left(g + f + \frac{e - f}{3}\right)\right] \qquad x_C = 3.52 \text{ ft}$$

$$y_C = \frac{1}{A}\left[b\,c\frac{c}{2} + a\,f\left(c + \frac{a}{2}\right) + \frac{1}{2}a(e - f)\left(c + \frac{a}{3}\right)\right] \qquad y_C = 4.09 \text{ ft}$$

Problem 9-130

The hopper is filled to its top with coal. Determine the volume of coal if the voids (air space) are a fraction p of the volume of the hopper.

Given:

$a = 1.5 \text{ m}$

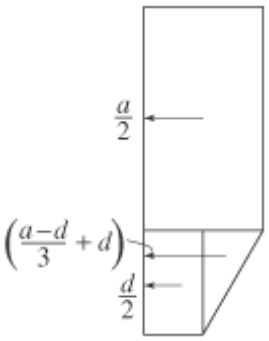
$b = 4 \text{ m}$

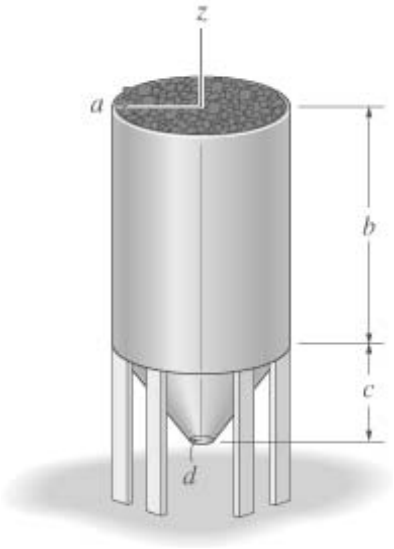
$c = 1.2 \text{ m}$

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$d = 0.2$ m

$p = 0.35$





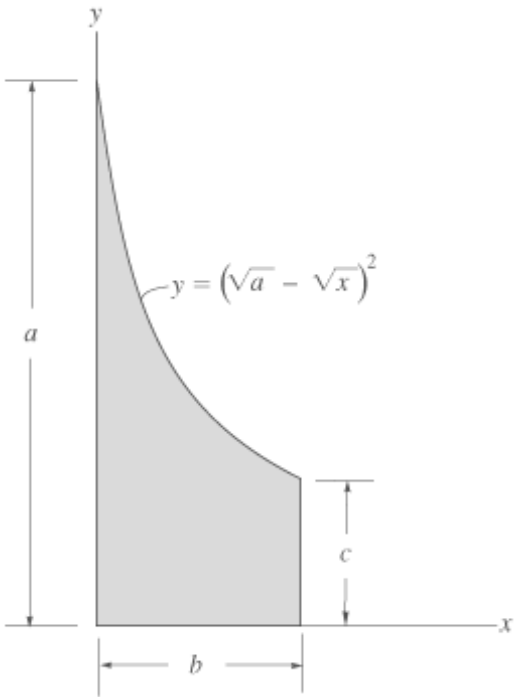
Solution:

$$V = (1 - p)2\pi\left[d\,c\frac{d}{2} + \frac{1}{2}c(a - d)\left(d + \frac{a - d}{3}\right) + a\,b\frac{a}{2}\right]$$

$V = 20.5\ \text{m}^3$

Problem 9-131

Locate the centroid (x_c, y_c) of the shaded area.



Given:

$a = 16$ ft

$b = 4$ ft

$c = \left(\sqrt{a} - \sqrt{b}\right)^2$

Solution:

$$A = \int_0^b \left(\sqrt{a} - \sqrt{x}\right)^2 \mathrm{d}x \qquad A = 29.3\ \text{ft}^2$$

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$$x_C = \frac{1}{A}\int_0^b x\left(\sqrt{a}-\sqrt{x}\right)^2\,dx \qquad x_C = 1.6\,\text{ft}$$

$$y_C = \frac{1}{A}\int_0^b \frac{\left[\left(\sqrt{a}-\sqrt{x}\right)^2\right]^2}{2}\,dx \qquad y_C = 4.15\,\text{ft}$$

Problem 9-132

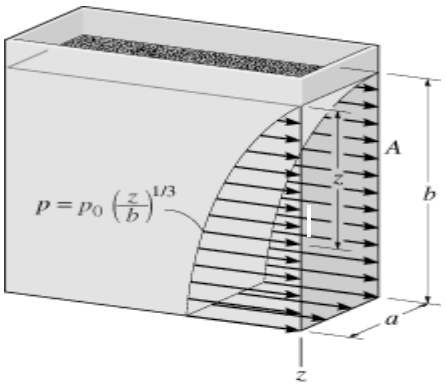
The rectangular bin is filled with coal, which creates a pressure distribution along wall A that varies as shown, i.e., $p = p_0(z/b)^{1/3}$. Compute the resultant force created by the coal, and its location, measured from the top surface of the coal.

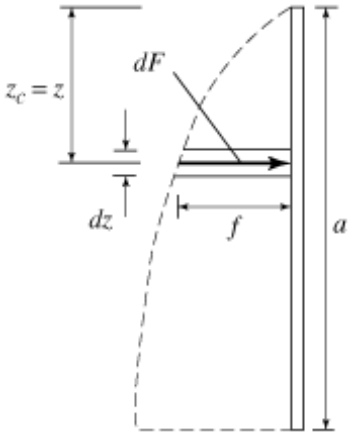
Given:

$p_0 = 8\ \frac{\text{lb}}{\text{ft}^2}$

$a = 3$ ft

$b = 8$ ft





Solution:

$$F = \int_0^b p_0\left(\frac{z}{b}\right)^{\frac{1}{3}} a\,dz \qquad F = 144\,\text{lb}$$

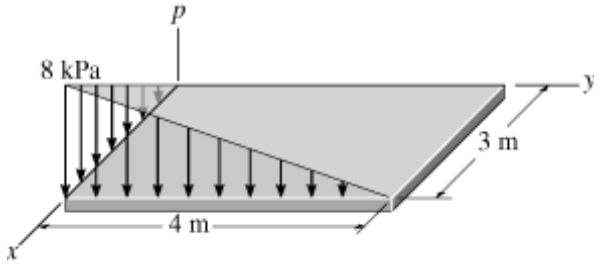
$$z_C = \frac{1}{F}\int_0^b z\,p_0\left(\frac{z}{b}\right)^{\frac{1}{3}} a\,dz \qquad z_C = 4.57\,\text{ft}$$

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Problem 9-133

The load over the plate varies linearly along the sides of the plate such that $p = \frac{2}{3}x(4 - y)$ kPa

Determine the resultant force and its position (x_c, y_c) on the plate.



Solution:

$$F = \int_0^3 \int_0^4 \frac{2}{3}x(4 - y)\, dy\, dx \qquad F = 24 \text{ kN}$$

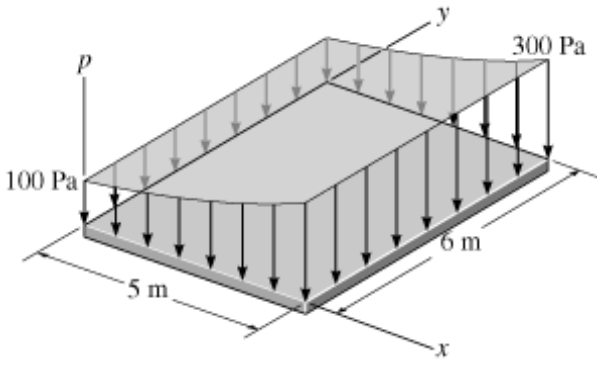
$$x_C = \frac{1}{F}\int_0^3 \int_0^4 x\frac{2}{3}x(4 - y)\, dy\, dx \qquad x_C = 2 \text{ m}$$

$$y_C = \frac{1}{F}\int_0^3 \int_0^4 y\frac{2}{3}x(4 - y)\, dy\, dx \qquad y_C = 1.333 \text{ m}$$

Problem 9-134

The pressure loading on the plate is described by the function $p = \{ -240/(x + 1) + 340 \}$ Pa. Determine the magnitude of the resultant force and coordinates of the point where the line of action of the force intersects the plate.

Solution:



Due to symmetry

$y_C = 3$ m

$$F = \int_0^5 \left(\frac{-240}{x + 1} + 340\right)6\, dx$$

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$$F = 7.62 \times 10^{3}\,\text{N}$$

$$x_C = \frac{1}{F}\int_0^5 x\left(\frac{-240}{x+1} + 340\right) 6\,dx$$

$$x_C = 2.74\,\text{m}$$

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